

IIT JEE / NEET Objective Question Bank

Questions 1 - 1000 (With Answers)

Biology

Q1.

An anti-forest measure is

- (a) Afforestation
- (b) Selective grazing
- (c) Clearing forest
- (d) Selective felling

Answer: C

Chemistry

Q2.

Among the following organic acids, the acid present in rancid butter is:

- (a) pyruvic acid
- (b) lactic acid
- (c) butyric acid
- (d) acetic acid

Answer: C

Physics

Q3.

Which of the following statement regarding transformer is incorrect?

- (a) A transformer makes use of Faradays Law.
- (b) The ferromagnetic properties of an iron core is used by it to efficiently raise or lower AC voltage
- (c) It cannot increase power so that if the voltage is raised, the current is proportionally lowered and vice versa.
- (d) The current induced in primary circuit is of same nature as in secondary circuit with differences in amplitude only.

Answer: D

Biology

Q4.

Fern plants reproduce by

- (a) Seeds
- (b) Spores
- (c) Laying eggs
- (d) Giving birth to young fern plants

Answer: B

Physics

Q5.

Electric current flows through:

- (a) a conductor
- (b) an insulator
- (c) free space
- (d) none of these

Answer: A

Q6.

A particle of mass m is made to move with uniform speed v along the perimeter of a regular hexagon. The magnitude of impulse applied at each corner is:

- (a) $m v$
- (b) $m v \sqrt{3}$
- (c) $(m v)/(2)$
- (d) $(m v)/(\sqrt{3})$

Answer: A

Chemistry

Q7.

A ketone has molar mass 86 . Which of the following cannot be IUPAC name of this ketone?

- (a) 2 -pentanone
- (b) 2-methyl butanone
- (c) 3 -methyl butanone
- (d) 3 -pentanone

Answer: C

Q8.

An ideal gas is compressed in a closed container its U ?

- (a) Increases
- (b) Decreases
- (c) Remains same
- (d) Both (1) & (2)

Answer: D

Physics

Q9.

According to Stoke's law, the viscous drag force, on an oil drop is proportional to:

- (a) $\sqrt{V}$
- (b) V^2
- (c) V^{-1}
- (d) V

Answer: D

Q10.

was the first to discover that lightning is electric discharge.

- (a) Thomas Alva Edison
- (b) Nicolas Tesla
- (c) Benjamin Franklin

(d) Isaac Newton

Answer: C

Q11.

Assertion Newton's law of gravitation resembles Coulomb's law of electrical forces. Reason Coulomb's law has the product of two charges in place of the product of the masses, and the electrostatic constant in place of the gravitational constant.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: A

Chemistry

Q12.

Which of the following will change the value of the equilibrium constant for the reaction? $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{NO}(\text{g})$

- (a) Add more N_2
- (b) Increase of pressure
- (c) Use a smaller reaction vessel
- (d) Increase the temperature

Answer: D

Physics

Q13.

A charged particle is projected in a plane perpendicular to uniform magnetic field. The areal velocity (area swept per unit time) of the particle is : This question has multiple correct options

- (a) directly proportional to kinetic energy of particle
- (b) directly proportional to momentum of the particle
- (c) inversely proportional to magnetic field strength
- (d) inversely proportional to charge on particle

Answer: B,C,D

Q14.

How the gravitational constant will change if a brass plate is introduced between two bodies?

- (a) No change
- (b) Decreases
- (c) Increases
- (d) No sufficient data

Answer: A

Q15.

If the maximum velocity of a particle in SHM is v_0 , then its velocity at half the amplitude from position of rest will be : $A \cdot v_0 / 2$

- (b) v_0
- (c) $v_0 \sqrt{3} / 2$
- (d) $v_0 \sqrt{3} / 4$

Answer: C

Q16.

The product of resistivity and conductivity of a cylindrical conductor depends on

- (a) Temperature
- (b) Material
- (c) Area of cross-section
- (d) None of the above

Answer: D

Chemistry

Q17.

Electronic configuration $[Kr] 4d^{10}$ belongs to:

- (a) Period(4) Group(10) Block(d)
- (b) Period(5) Group(12) Block (d)
- (c) Period(5) Group(10) Block (d)
- (d) Period(4) Group(12) Block(d)

Answer: B

Q18.

Given below are the two reactions of H_2O_2 . Mark the correct statement which follows: (i) $2KMnO_4 + 3H_2SO_4 + 5H_2O_2 \rightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5O_2$ (ii) $2Cr(OH)_3 + 4NaOH + 3H_2O_2 \rightarrow 2Na_2CrO_4 + 8H_2O$

- (a) (i) Shows oxidizing nature of H_2O_2 and (ii) shows reducing nature of H_2O_2
- (b) (i) H_2O_2 acts as a reducing agent and in (ii) it acts as an oxidizing agent
- (c) both (i) and (ii), H_2O_2 acts as an oxidizing agent
- (d) In both (i) and (ii), H_2O_2 acts as a reducing agent

Answer: B

Physics

Q19.

Seema is planning a trip to Kashmir this summer. Which of the following clothes should she definitely pack?

- (a) Denim Jacket
- (b) Leather pants
- (c) Woolen sweater
- (d) cotton shirt

Answer: C

Biology

Q20.

In rabbit and humans, the kidney is

- (a) Metanephric
- (b) Mesonephric
- (c) Pronephric
- (d) Opisthonephric

Answer: A

Physics

Q21.

When forces F_1 , F_2 , F_3 are acting on a particle of mass m such that F_2 and F_3 are mutually perpendicular, then the particle remains stationary. If the force F_1 is now removed then the acceleration of particle the is

- (a) F_1 / m
- (b) $F_2 F_3 / m F_1 \cdot (F_2 - F_3) / m$
- (d) F_2 / m

Answer: A

Biology

Q22.

Snake's venom is

- (a) Alkaline in reaction
- (b) Acidic in reaction
- (c) Alkaline and rich fatty acids
- (d) Acidic and rich fatty acids

Answer: A

Physics

Q23.

The other name for larynx is

- (a) voice processor
- (b) voice modulator
- (c) voice box
- (d) voice generator

Answer: C

Chemistry

Q24.

Internal resistance of a battery cell increases with:

- (a) increase in concentration of electrolyte
- (b) increase in distance between two electrodes
- (c) increase in area of the plates inside the electrolyte
- (d) increase in size of the electrodes

Answer: B

Q25.

Compare the properties of a typical metal and a non-metal on the basis of the following. Conductivity of heat and electricity:

- (a) Metals are good conductor of heat and electricity
- (b) Non metals are good conductor of heat and electricity
- (c) Metals are bad conductor of heat and electricity
- (d) Both B and C

Answer: A

Q26.

0.73 g of organic compound gave 1.32 g CO_2 . The % of carbon in the organic compound is: A · 55.24

- (b) 49.32
- (c) 48.64

(d) 45.24

Answer: A

Physics

Q27.

Mark out the incorrect statement:

- (a) A free neutron can transform itself into photon
- (b) A free proton can transform itself into neutron
- (c) In beta minus decay, the electron originates from nucleus
- (d) All of the above

Answer: D

Chemistry

Q28.

The values of coefficients to balance the following reaction are: $\text{Cr}(\text{OH})_3 + \text{ClO}^- + \text{OH}^- \rightarrow \text{CrO}_4^{2-} + \text{Cl}^- + \text{H}_2\text{O}$?

- A · $\text{Cr}(\text{OH})_3$ -(2), ClO^- -(3), CrO_4^{2-} -(3), Cl^- -(3)
- (b) $\text{Cr}(\text{OH})_3$ -(2), ClO^- -(4), CrO_4^{2-} -(3), Cl^- -(2)
- (c) $\text{Cr}(\text{OH})_3$ -(2), ClO^- -(4), CrO_4^{2-} -(4), Cl^- -(2)
- (d) $\text{Cr}(\text{OH})_3$ -(2), ClO^- -(3), CrO_4^{2-} -(2), Cl^- -(3)

Answer: D

Biology

Q29.

Hydra reproduces by budding. This is an example of

- (a) Asexual reproduction
- (b) Sexual reproduction
- (c) Regeneration
- (d) Parthenocarp

Answer: A

Physics

Q30.

The rays of light, which are not parallel to each other and spread out from a point, are called and a bundle of rays, which are parallel to one another, is called

- (a) parallel beam of light, convergent rays
- (b) divergent rays, convergent rays
- (c) convergent rays, parallel beam of light
- (d) divergent rays, parallel beam of light

Answer: D

Q31.

A steady potential difference of 100 V produces heat at a constant rate in a resistor. The alternating voltage which will produce half the heating effect in the same resistor will be A · 100 v

- (b) 50
- (c) 70.7 v
- (d) 141.4

Answer: C

Q32.

The work done (in Joule) in carrying a charge of 100 coulomb between two points having a potential difference of 10 volt is:

- (a) 0 .
- (b) $10 \text{ C} \cdot 100$
- (d) 1000

Answer: D

Q33.

A body is projected horizontally from a height of 78.4 m with a velocity 10 m s^{-1} . Its velocity after 3 seconds is ($g = 10 \text{ m s}^{-2}$). (Take direction of projection on i and vertically upward direction on j): $A \cdot 10 \text{ i} - 30 \text{ j}$

- (b) $10 \text{ i} + 30 \text{ j}$
- (c) $20 \text{ i} - 30 \text{ j}$
- (d) $10 \text{ i} + 10 \sqrt{30} \text{ j}$

Answer: A

Biology

Q34.

Percentage of water present in oceans and seas is

- (a) 60 %
- (b) $70\% \text{ C} \cdot 80\%$
- (d) 95 %

Answer: D

Chemistry

Q35.

Which of the statements are true about the law of chemical combination? This question has multiple correct options

- (a) Potassium combines with two isotopes of chlorine (^{35}Cl and ^{37}Cl) to form two samples of KCl . Their formation follows the law of definite composition.
- (b) Different proportions of oxygen in the various oxides of sulphur proves the law of multiple proportions. $\text{C} \cdot \text{H}_2\text{O}$ and H_2S contain 11.11 % hydrogen and 5.88 % hydrogen respectively whereas SO_2 contains 50 % sulphur. The above data prove the law of reciprocal proportions.
- (d) In the decomposition of NH_3 ($2 \text{NH}_3 \xrightarrow{\Delta} \text{N}_2 + 3 \text{H}_2$) the ratio of volumes of NH_3 , N_2 and H_2 is 2: 1: 3. The above data proves the Gay Lussac's law.

Answer: A,B,D

Q36.

The plot shows the variation of $-\Delta G_p$ versus temperature for the two reactions. $\text{M(s)} + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{MO(s)}$ and $\text{C(s)} + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{CO(s)}$ Identify the correct statement?

- (a) At $T > 1200 \text{ K}$, carbon will reduce MO(s) to M(s)
- (b) At $T < 1200 \text{ K}$, oxidation of carbon is unfavourable.
- (c) At $T < 1200 \text{ K}$, the reaction $\text{MO(s)} + \text{C(s)} \rightarrow \text{M(s)} + \text{CO(g)}$ is spontaneous
- (d) Oxidation of carbon is favourable at all temperatures

Answer: A,D

Q37.

The number of moles of sulphuric acid present in 24.5 g of H_2SO_4 is:

- (a) 2.5

(b) $0.5 \text{ c} \cdot 4$

(d) 0.25

Answer: D

Q38.

Select the correct statement.

(a) Carbon-carbon bond order in graphite is more than diamond

(b) Carbon-carbon bond order in graphite is less than diamond

(c) Hybridization of each carbon in diamonds is sp^3

(d) Hybridization of each carbon in graphite is sp^2

Answer: A,C

Q39.

$2 \text{Ca}_3(\text{PO}_4)_2 + 6 \text{SiO}_2 + 10 \text{C} \rightarrow \text{P}_4 + \text{CaSiO}_3 + 10 \text{CO}$ Which coefficient balances the reaction given above?

(a) 2

(b) 4

(c) 5

(d) 6

Answer: B

Q40.

Diborane (B_2H_6) reacts independently with O_2 and H_2O to produce :

(a) HBO_2 and H_3BO_3

(b) H_3BO_3 and B_2O_3

(c) B_2O_3 and H_3BO_3

(d) B_2O_3 and $[\text{BH}_4]^-$

Answer: C

Biology

Q41.

If a plant cell is placed in deionised water, the water potential of that cell becomes

(a) More positive because pressure potential becomes more positive

(b) More positive because pressure potential becomes more negative

(c) More negative because pressure potential becomes more negative

(d) Less negative because pressure potential becomes more positive

Answer: D

Chemistry

Q42.

Which one of the following cyano complexes would exhibit the lowest value of paramagnetic behaviour?

(Atomic Numbers: Cr=24 ; Mn= 25 ; Fe=26 ; Co=27) $[\text{Fe}(\text{CN})_6]^{3-}$

(b) $[\text{Co}(\text{CN})_6]^{3-}$

(c) $[\text{Cr}(\text{CN})_6]^{3-}$

(d) $[\text{Mn}(\text{CN})_6]^{3-}$

Answer: B

Physics

Q43.

If for a gas $(R)/(C_v)=0.67$, then the gas is made up of molecules which are :

- (a) Diatomic
- (b) Monoatomic
- (c) Polyatomic
- (d) Mixture of Diatomic & Polyatomic

Answer: A

Q44.

In SHM the net force towards mean position is related to its displacement (x) from mean position by the relation $A \cdot F \propto x$

- (b) $F \propto 1/x$
- (c) $F \propto x^2$
- (d) $F \propto (1)/(x^2)$

Answer: A

Q45.

In hydrogen atom, an electron is revolving in the orbit of radius 0.53 Å with 6.6×10^{15} radiation/s. Magnetic field produced at the centre of the orbit is

- (a) 0.125 Wb / m²
- (b) 1.25 Wb / m²
- (c) 12.5 Wb / m²
- (d) 125 Wb / m²

Answer: C

Q46.

ends so that it lies horizontally and without tension. A weight W is suspended from the middle point of the wire. The vertical depression is (Young's modulus is Y.) $A \cdot \sqrt{((2 T l^2)/(4 A Y) + (T^2 l^2)/(4 A^2 Y^2))}$

- (b) $\sqrt{((2 T l^2)/(4 A Y) - (T^2 l^2)/(4 A^2 Y^2))}$
- (c) $\sqrt{((2 T l^2)/(4 A Y))}$
- (d) $(T l)/(2 A Y)$

Answer: B

Chemistry

Q47.

A beaker contains a saturated solution of copper(I) chloride. Copper(I) chloride is slightly soluble salt with a solubility product of 1.2×10^{-6} . Which of the following salts when added to the solution would precipitate copper(I) chloride?

- (a) Sodium chloride
- (b) Potassium bromide
- (c) Silver(I) nitrate
- (d) Lead(II) acetate E. Magnesium iodide

Answer: A

Physics

Q48.

Higher the refractive index of a medium

- (a) Less it will change the direction of a beam of light passing through it
- (b) More it will change the direction of a beam of light passing through it
- (c) No bending occurs
- (d) All

Answer: B

Chemistry

Q49.

The state of equilibrium refers to:

- (a) State of rest
- (b) Dynamic state
- (c) Stationary state
- (d) State of inertness

Answer: B

Q50.

Internal energy does not include

- (a) vibrational energy
- (b) rotational energy
- (c) energy arising by gravitational pull
- (d) nuclear energy

Answer: C

Q51.

The pH of D_2O and H_2O at 298 K is:

- (a) 7.0, 7.0
- (b) 7.35, 7.0
- (c) 7.0, 6.85
- (d) 6.85, 7.35

Answer: B

Physics

Q52.

The magnetic field generated along the axis of a solenoid is proportional to:

- (a) its length
- (b) square of current flowing in it
- (c) number of turns per unit length in it
- (d) reciprocal of its radius

Answer: C

Q53.

A ball thrown in the air reaches a height of 10 m and drops down to the ground. Find the time taken by the ball to complete this entire journey.

- (a) 2.78 sec
- (b) 3.40 sec
- (c) 1.43 sec
- (d) 2.86 sec

Answer: D

Q54.

A steamer is going downstream overcome a raft point A .2 h r later it turned back and after some time passed the raft at a distance 4 km from point A . The speed of the river is

- (a) 1 km / h
- (b) 2 km / h c .3 km / h

(d) 4 km / h

Answer: B

Q55.

In stationary waves, nodes are the points where there is:

- (a) maximum displacement and minimum pressure change
- (b) minimum displacement and maximum pressure change
- (c) minimum displacement and minimum pressure change
- (d) maximum displacement and maximum pressure change

Answer: B

Q56.

For a metallic wire, the ratio V/i (where, V = applied potential difference and i = current flowing

- (a) is independent of temperature
- (b) increases as the temperature rises
- (c) decreases as the temperature rises
- (d) increases or decreases as temperature rises depending upon the metal.

Answer: B

Chemistry

Q57.

Assertion The change in internal energy and change in heat enthalpy does not depend upon the path by which changes are brought in. Reason Both ΔU and ΔH are path independent as U and H are state functions.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect.
- (d) Assertion is incorrect but Reason is correct

Answer: A

Q58.

What is the ratio of the number of neutrons present in potassium and magnesium atoms with mass number 39 and 24 respectively?

- (a) 19: 12
- (b) 5: 3
- (c) 5: 6
- (d) 4: 3

Answer: B

Physics

Q59.

When a body of mass m_1 moving with uniform velocity 40 m s^{-1} collides with another body of mass m_2 at rest, then the two together begin to move with uniform velocity of 30 m s^{-1} . The ratio of the mass (i.e., $(m_1)/(m_2)$) of the two bodies will be

- (a) 1: 3
- (b) 3: 1
- (c) 1: 1.33
- (d) 1: 0.75

Answer: B

Q60.

1° A is

- (a) 10 nm

- (b) 0.1 n m
- (c) 100 n m
- (d) 1 n m

Answer: B

Q61.

A body is projected vertically upwards. Its momentum is gradually decreasing. In this

- (a) it is a violation of law of conservation of linear momentum
- (b) momentum of the body alone gets conserved
- (c) momentum of the body, earth and air molecules together remains constant
- (d) violates law of conservation of energy.

Answer: C

Chemistry

Q63.

The correct statement(s) among the following is/are: This question has multiple correct options

- (a) beryllium oxide is amphoteric in nature.
- (b) solubility of sulphates of II A group elements decreases down the group.
- (c) reducing power of hydride of alkali metal decreases down the group.
- (d) beryllium has diagonal relationship with aluminum.

Answer: A,B,D

Q64.

What is the hybridisation state of xenon in XeO_4 ? A. $s p^3$

- (b) $s p^3 d$
- (c) $s p^3 d^2$
- (d) $s p^3 d^3$

Answer: C

Q65.

Which of the following is incorrect about polysaccharides? I. These are amorphous solids. II. These are sparingly soluble in water. III. These are sugars

- (a) Only III
- (b) II and II
- (c) I and III
- (d) All of the above

Answer: A

Physics

Q66.

A radioactive nucleus with decay constant $0.5 / s$ is being produced at a constant rate of 100 nuclei/s. If at $t=0$ there were no nuclei, the time when there are 50 nuclei is

- (a) 1 s
- (b) $2 \frac{4}{3} s$
- (c) 2 s
- (d) $\frac{4}{3} s$

Answer: B

Chemistry

Q67.

Which of the following is a true solution?

- (a) Copper in gold
- (b) Sulphur in water
- (c) Milk
- (d) water in sulphur dioxide

Answer: A

Biology

Q68.

Hemoglobin in the RBCs helps in the

- (a) Transport of oxygen
- (b) Transport of nutrients
- (c) Transport of carbon dioxide
- (d) Both A and C

Answer: D

Q69.

Adrenaline is equivalent to which neurotransmitter? A · GABA

- (b) Serotonin
- (c) Epinephrine
- (d) None of the above

Answer: C

Chemistry

Q70.

If the solubility of lithium sodium hexafluoroaluminate, $\text{Li}_3\text{Na}_3[\text{AlF}_6]_2$ is 18.6 g L^{-1} , its solubility product is: (Assume no ionisation of $[\text{AlF}_6]^{3-}$)

- (a) 18 s^3
- (b) $186624 \text{ s}^8 \text{ c} \cdot 1458 \text{ s}^8$
- (d) 2916 s^8

Answer: B

Biology

Q71.

In male grasshoppers and moths, there are two pairs of autosomes and

- (a) x only
- (b) x and
- (c) Y only
- (d) None of the above

Answer: A

Chemistry

Q72.

Liquid benzene burns in oxygen according to: $2 \text{C}_6\text{H}_6 + 15 \text{O}_2 \rightarrow 12 \text{CO}_2 + 6 \text{H}_2\text{O}$ If density of liquid benzene is 0.88 g/cc what volume of O_2 at STP is needed to complete the combustion of 39 cc of liquid benzene?

- (a) 11.2 litre
- (b) 74 litre
- (c) 0.074 m^3
- (d) 37 d m^3

Answer: C

Physics

Q73.

To keep machines running smoothly and to avoid wear and tear of machines, the best and the most commonly used lubricant is:

- (a) Powder
- (b) Sand c · oil
- (d) water

Answer: C

Chemistry

Q74.

Emulsion can be destroyed by: This question has multiple correct options A . the addition of emulsifier which tends to form another emulsion

- (b) electrophoresis with high potential
- (c) freezing
- (d) all

Answer: A,B,C

Biology

Q75.

consider the following statements each with one or two blanks. (i) The ascending limb of loop of Henle is impermeable to (1) but allows transport of (2) (ii) (3) and (4) play a significant role in producing a concentrated urine. (iii) A fall in glomerular blood flow/glomerular blood pressure/GFR can activate the JG cells to release (5) Which one of the following options correctly fills the blanks in any two of the statements?

- (a) (1)-water, (2)-electrolytes, (5)-renin
- (b) (3)-Henle's loop, (4)-vasa recta, (5)-angiotensin
- (c) (1)-electrolytes, (2)-water, (3)- ■■■■, (4)- DCT
- (d) (3)-Henle's loop, (4)-vasa recta, (5)-angiotensinogen

Answer: A

Q76.

Cork is used for insulation and acts as shock absorber because

- (a) It is porous
- (b) It is very hard
- (c) It is light and does not catch fire easily
- (d) The cells of cork are living

Answer: A

Chemistry

Q77.

The number of occupied sub-shell in Cr + is :

- (a) 3
- (b) 4 c · 7
- (d) 6

Answer: C

Q78.

In the atmosphere of industrial smog, copper corrodes to form: This question has multiple correct options

- (a) Basic copper carbonate
- (b) Copper sulphide
- (c) Basic copper sulphate
- (d) copper oxide

Answer: C,D

Biology

Q79.

Assertion Double circulation is incomplete in amphibians and reptiles Reason Unlike in birds and mammals, in amphibian and reptiles, the left atrium receives oxygenated blood and right atrium receives deoxygenated blood.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: B

Q80.

I. Increase in mass II. Differentiation III. Increase in number of individuals. IV. Response to stimuli Which two points are known as the twin characteristics of growth?

- (a) I and II
- (b) II and IV
- (c) II and III
- (d) I and III

Answer: D

Chemistry

Q81.

State in which of the following examples the force of attraction between particles of matter is almost negligible.

- (a) Piece of stone
- (b) Water droplets
- (c) A light gas
- (d) All of the above

Answer: C

Physics

Q82.

Which one of the following statements is true?

- (a) A motor works on the principle of electromagnetic induction

- (b) An electric motor converts mechanical energy into electrical energy
- (c) AC generator has slip rings while DC generator has a commutator
- (d) An electric generator converts electrical energy into mechanical energy

Answer: C

Q83.

Light waves travel in vacuum along the y-axis. Then the wave front is:

- (a) $y = \text{constant}$
- (b) $x = \text{constant}$
- (c) $z = \text{constant}$
- (d) $x + y + z = \text{constant}$

Answer: A

Biology

Q84.

A spinal nerve takes nerve impulses:

- (a) to the CNS
- (b) away from the CNS
- (c) both to and away from the CNS
- (d) only inside the CNS

Answer: C

Chemistry

Q85.

In the conversion of C_2H_4 to C_2H_6 breakage of one π bond accounts for formation of σ bond.

- (a)
- (b) 2
- (c) 3
- (d)

Answer: A

Physics

Q86.

A sphere, a cube and a thin circular disc all made of the same material, have same mass. Their initial temperature is $3 \times 10^3 \text{ }^\circ\text{C}$. The body which cools rapidly is :

- (a) sphere
- (b) cube
- (c) both sphere and cube
- (d) circular plate

Answer: D

Q87.

Braille system is traditionally written with:

- (a) White Paper
- (b) Black Paper
- (c) Embossed Paper
- (d) Typed Paper

Answer: C

Q88.

Consider a long conductor, the middle of which is earthed. If the potential difference across the two ends of the conductor is 220 V , then what is the potential at the ends and the middle point?

- (a) 220 V all over the conductor
- (b) 110 V and -110 V at ends and 0 V at the mid point
- (c) 0 V at ends and 220 V at midpoint
- (d) -220 V all over the conductor

Answer: B

Chemistry

Q89.

True statements for the chemical reaction of the formation of lithium oxide is: $4 \text{Li(s)} + \text{O}_2(\text{g}) \rightarrow 2 \text{Li}_2\text{O(s)}$

- (a) lithium metal is the oxidizing agent
- (b) oxygen gas is the reducing agent
- (c) lithium is oxidized
- (d) oxygen is oxidized E. oxygen loses two electrons to become O^{2-} ion

Answer: C,E

Q90.

Assertion Rate of nitration is $\text{C}_6\text{H}_6 > \text{C}_6\text{D}_6 > \text{C}_6\text{T}_6$ Reason Formation of intermediate is rate determining step in nitration of benzene, not the breaking of C-H or C-D bond

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Assertion is incorrect but Reason is correct

Answer: A

Physics

Q91.

Photodiode is a device

- (a) Which is always operated in reverse bias
- (b) Which is always operated in forward bias
- (c) In which photo current is independent of intensity of incident radiation
- (d) Which may be operated in forward or reverse bias

Answer: A

Biology

Q92.

ICBN was adopted in

- (a) 1951
- (b) 1971
- (c) 196
- (d) 1981

Answer: A

Chemistry

Q93.

For a given reaction, $\text{H}_2\text{O(l)} \rightarrow \text{H}_2\text{O(s)}$ at 0°C and 1 bar, which of the following is true?

- (a) $\Delta G=0$ ■. $\Delta S=0$
- (c) $\Delta H=0$
- (d) $\Delta U=0$

Answer: A

Q94.

Assertion: When an electric current is passed through the copper sulphate electrolyte, the impure copper anode gradually dissolves and pure copper is deposited on the cathode. Reason: Pure metals are deposited in electroplating techniques.

- (a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion
- (b) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion
- (c) Assertion is true but Reason is false
- (d) Assertion is false but Reason is true E. Both Assertion and Reason are false

Answer: B

Physics

Q96.

The water stored in a reservoir possesses:

- (a) Kinetic energy
- (b) Muscular energy
- (c) Potential energy
- (d) Magnetic energy

Answer: C

Q97.

A body covers 26, 28, 30 and 32 meters in 10^{th} , 11^{th} , 12^{th} and 13^{th} seconds respectively. The body starts

- (a) from rest and moves with uniform velocity
- (b) from rest and moves with uniform acceleration
- (c) with an initial velocity and moves with uniform acceleration
- (d) with an initial velocity and moves with uniform velocity

Answer: B

Biology

Q98.

The only plant cells without nucleus among the following is

- (a) Cambium cells
- (b) Root hair
- (c) companion cells
- (d) Xylem vessels

Answer: D

Chemistry

Q99.

Which of the following statements is not true about canal rays?

- (a) They travel in straight line.
- (b) They are made up of positively charged particles.

- (c) Their e/m ratio is same for all gases.
(d) These rays produce fluorescence on ZnS screen.

Answer: C

Biology

Q100.

Which of the following characteristics can be related to cartilaginous joints?

- (a) Permit slight movements
(b) Are found between the bodies of vertebrae
(c) Are found in symphysis
(d) All of the above

Answer: D

Chemistry

Q101.

The two ways of combining atoms are:

- (a) transfer of valence electrons
(b) sharing of valence electrons
(c) both A and B
(d) none of the above

Answer: C

Physics

Q102.

What is the maximum speed at which a railway carriage can move without toppling over along a curve of radius $R=200$ m if the distance from the centre of gravity of the carriage to the level of the rails is $h=1.0$ m, the distance between the rails is $l=2.0$ m and the rails are laid horizontally? (Take $g=10 \text{ m/s}^2$) A · 11.18 m/s^{-1}

- (b) 22.36 m/s^{-1}
(c) 44.72 m/s^{-1}
(d) 74 m/s^{-1}

Answer: C

Chemistry

Q103.

18 g glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is added to 178.2 g of water. The vapour pressure of this aqueous solution at 100°C in torr is:

- (a) 7.60
(b) 76.00
(c) 752.40
(d) 759.00

Answer: C

Physics

Q104.

If mass-energy equivalence is taken into account, when water is cooled to form ice, the mass of water should A . increase

- (b) remain unchanged
- (c) decrease
- (d) first increase then decrease

Answer: C

Biology

Q105.

Carbohydrates are commonly found as starch in plant storage organs. Which of the following five properties of starch makes it useful as a storage material? (A) Easily translocated (B) Chemically non-reactive (C) Easily digested by animals (D) Osmotically inactive (E) Synthesized during photosynthesis The useful properties are A . (A), (C) and (E)

- (b) (A) and (E)
- (c) (B) and (C)
- (d) (B) and (D)

Answer: D

Physics

Q106.

A sinusoidal voltage of peak value 293 V and frequency 50 Hz is applied to a series LCR circuit in which $R=6\ \Omega$, $L=25\text{ mH}$ and $C=750\ \mu\text{F}$. The impedance of the circuit is:

- (a) $7.0\ \Omega$
- (b) $8.9\ \Omega$ c . $9.9\ \Omega$
- (d) $10.0\ \Omega$

Answer: C

Q107.

A power transmission line feeds input power at 2300 V to a step-down transformer with its primary windings having 4000 turns. The output power is delivered at 230 V by the transformer. If the current in the primary of the transformer is 5 A and its efficiency is 90 %, the output current would be :

- (a) 25 A
- (b) 50 A c . 35 A
- (d) 45 A

Answer: D

Q108.

A horizontal magnetic field B is produced across a narrow gap between the two square iron pole pieces. A closed square loop of side a, mass m and resistance R is allowed to fall with the top of the loop in the field. The loop attains a terminal velocity equal to: A . $(R m g)/(B^2 a^2)$

- (b) $(\mu_0)/(4\pi) (R m g)/(B a^2)$
- (c) $(m g B)/(R a^2)$
- (d) $(\mu_0)/(4\pi) (R m B)/(B a^2)$

Answer: A

Biology

Q109.

Coenzymes NAD and NADP contain the vitamin

- (a) Niacin
- (b) Biotin
- (c) Thiamine
- (d) Vitamin B₁₂ E. Vitamin A

Answer: A

Physics

Q110.

The observed wavelength of light coming from a distant galaxy is found to be increased by 0.5 % as compared with that coming from a terrestrial source. The galaxy is.

- (a) Stationary with respect to the earth
- (b) Approaching the earth with velocity of light
- (c) Receding from the earth with velocity of light
- (d) Receding from the earth with a velocity equal to 1.5×10^6 m / s

Answer: D

Biology

Q111.

What is denoted by X and Y in the given graph?

- (a) x -Activation energy without enzyme; Y-Activation energy with enzyme
- (b) X-Activation energy with enzyme; Y-Activation energy without enzyme.
- (c) x -substrate concentration with enzyme; Y - Substrate concentration without enzyme.
- (d) X-Substrate concentration without enzyme; YSubstrate concentration with enzyme.

Answer: B

Chemistry

Q112.

The reagent X used for the following reaction is: $R-C \equiv C R' + H_2 \xrightarrow{X} R-CH=CH-R'$

- (a) Ni / Pd / C
- (c) LiAlH₄
- (d) Ni / Liquid N H₃

Answer: D

Q113.

Generally transition elements form coloured salts due to the presence of unpaired electrons. Which of the following compound will be coloured in solid state?

- (a) Ag₂SO₄ ■ CuF₂
- (c) Cu₂Cl₂
- (d) ZnF₂

Answer: B

Physics

Q114.

Air is not visible because it is nearly a perfectly transparent substance

- (b) neither absorbs nor reflects light
- (c) transmits whole of light
- (d) follows all of the above

Answer: A

Q115.

Two point charges $+8q$ and $-2q$ are located at $x=0$ and $x=L$ respectively. The location of a point on the x axis at which the net electric field due to these two point charge is zero is -

- (a) 2
- (b) $L/4$
- (c) $8L$
- (d) $4L$

Answer: D

Q116.

Assertion: The reaction; ${}_1\text{H}^2 + {}_0\text{n}^1 \rightarrow {}_2\text{He}^4 + {}_0\text{n}^1$ will not happen Reason: The equation given is not balanced.

- (a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion
 - (b) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion
 - (c) Assertion is true but Reason is false
 - (d) Assertion is false but Reason is true
- E. Both Assertion and Reason are false

Answer: A

Chemistry

Q117.

The percentage of lanthanides and iron, respectively, in Misch metal are: A · 50,50

- (b) 75,25
- (c) 90,10
- (d) 95,5

Answer: D

Biology

Q118.

Which is the living component of xylem?

- (a) Tracheids
- (b) Xylem fibre
- (c) xylem parenchyma
- (d) Trachea

Answer: C

Physics

Q119.

Use of thermometer is based on which law of thermodynamics?

- (a) zeroth
- (b) First
- (c) second
- (d) Third

Answer: A

Q120.

g cm s^{-2} is also known as

- (a) newton
- (b) erg
- (c) pascal
- (d) dyne

Answer: D

Chemistry

Q121.

The idea of froth floatation method came from a person X and this method is related to the process Y of ores. X and Y, respectively, are:

- (a) fisher woman and concentration
- (b) washer man and reduction
- (c) washer woman and concentration
- (d) fisher man and reduction

Answer: C

Q122.

Aqueous solution of PBr_3 conducts electricity due to the presence of: A . HOBr ■. ■■■

- (c) H_3PO_4
- (d) H_2O

Answer: B (HBr)

Biology

Q123.

The most nearest prehistoric ancestor of present man may be

- (a) Java ape man
- (b) Cro-Magnon man
- (c) Neanderthal man
- (d) Peking man

Answer: B

Physics

Q124.

The strength of gravity on Mars is only 40 % of that on earth. If a child has a mass of 60 kg on earth, what would the child's weight be on Mars?

- (a) 24 ■. 60 N
- (c) 240 N
- (d) 600 N E. None of the above is correct within 10 %

Answer: C

Q125.

Which of the following statements is not true about Gauss's law?

- (a) Gauss's law is true for any closed surface
- (b) The term q on the right side of Gauss's law includes the sum of all charges enclosed by the surface
- (c) Gauss's law is not much useful in calculating electrostatic field when the system has some symmetry
- (d) Guss's law is based on the inverse square dependence on distance contained in the coulomb's law

Answer: C

Q126.

An open U-tube contains mercury. When 13.6 cm of water is poured into one of the arms of the tube then the mercury rise in the other arm from its initial level is:

- (a) 1 cm
- (b) 0.5 cm
- (c) 10 cm
- (d) 5 cm

Answer: A

Chemistry

Q127.

The reaction that takes place when Cl_2 gas is passed through conc. NaOH solution is :

- (a) oxidation
- (b) reduction
- (c) displacement
- (d) disproportionation

Answer: D

Physics

Q128.

A circuit consists of a resistor, an inductor and a capacitor connected in series. If the resistance of a resistor is $25\ \Omega$ and the value of inductive reactance is equal to capacitive reactance, then the net impedance of the circuit will be :

- (a) $5\ \Omega$
- (b) $25\ \Omega$
- (c) $125\ \Omega$
- (d) $625\ \Omega$

Answer: B

Chemistry

Q129.

i) $\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{O}$ ii) $\text{H}_2\text{O}_2 + 2\text{I}^- \rightarrow \text{I}_2 + 2\text{OH}^-$, H_2O_2 acts as:

- (a) oxidising agent in both (i) and (ii)
- (b) reducing agent in both (i) and (ii)
- (c) oxidising agent in (i) and reducing agent in (ii)
- (d) reducing agent in (i) and oxidising agent in (ii)

Answer: D

Q130.

Some hydrogen-like atom in ground state absorb n photons having the same energy and on de-excitation, it emits exactly n photons. Then the energy of absorbed photon may be: This question has multiple correct options

- (a) 91.8 eV
- (b) 40.8 eV
- (c) 48.4 eV
- (d) 54.4 eV

Answer: B,D

Physics

Q131.

Which statement is incorrect related to induced electric field due to changing magnetic flux?

- (a) It varies with time
- (b) It is non-conservative
- (c) It forms closed loop
- (d) Both (1) & (3)

Answer: D

Chemistry

Q132.

Which property is shown by the alkali sodium hydroxide?

- (a) It has a pH less than pH 7
- (b) It produces a gas when it is warmed with ammonium chloride
- (c) It turns blue litmus red
- (d) It turns Universal Indicator green

Answer: B

Q133.

Protons and neutrons are also called

- (a) nucleons
- (b) isotope
- (c) isobars
- (d) elements

Answer: A

Physics

Q134.

Plum pudding' model of an atom was proposed by:

- (a) C.V.Raman
- (b) N.Bohr
- (c) E.Rutherford
- (d) J.J.Thomson

Answer: D

Biology

Q135.

The synaptonemal complex first appears at

- (a) Leptotene
- (b) Pachytene
- (c) zygotene
- (d) Diplotene

Answer: C

Physics

Q136.

A child is sitting on a swing. Its minimum and maximum heights from the ground are 0.75 m and 2 m respectively, its maximum speed will be:

- (a) 5 m / s ■. 8 m / s
- (c) 10 m / s
- (d) 15 m / s

Answer: A

Q137.

If elements with principal quantum number $n > 4$ were not allowed in nature, the number of possible elements would have been

- (a) 32
- (b) 60
- (c) 18 D

Answer: B

Biology

Q138.

Secretin stimulates secretion of bicarbonate from

- (a) Liver
- (b) Kidneys
- (c) Pancreas
- (d) Stomach

Answer: C

Chemistry

Q139.

The composition of a sample of Wustite is $\text{Fe}_{0.93}\text{O}$. What is the percentage of iron present as Fe^{3+} in total iron?

- (a) 15.05 %
- (b) 25% c .35 %
- (d) 45 %

Answer: A

Q140.

The reaction $\text{C}_2\text{H}_2 + \text{H}_2 \xrightarrow{(250-300^\circ\text{C})} \text{C}_2\text{H}_4$ is called

- (a) Wurtz reaction
- (b) Kolbe's reaction
- (c) Sabatier - Sendern's reaction
- (d) Cabylamine reaction

Answer: C

Biology

Q141.

Succession on sand dune is

- (a) Lithosere

- (b) Psychrosere
- (c) Psammosere
- (d) Halosere

Answer: C

Physics

Q142.

The coefficient of limiting friction between the surfaces of contact is the ratio of the friction to the normal reaction

- (a) Limiting
- (b) sliding
- (c) rolling
- (d) any of these

Answer: A

Chemistry

Q143.

Which of the following hydrocarbon exists mostly in solid or liquid form under the earth surface?

- (a) Ethane
- (b) Methane
- (c) Pentane
- (d) Butane

Answer: C

Physics

Q144.

After the capacitor gets fully charged, S_1 is opened and S_2 is closed so that the inductor is connected in series with the capacitor. Then, A . at $t=0$, energy stored in the circuit is purely in the form of magnetic energy

- (b) at any time $t>0$, current in the circuit is in the same
- (c) at $t>0$, there is no exchange of energy between the inductor and capacitor
- (d) at any time $t>0$, instantaneous current in the circuit may $\propto \sqrt{C/L}$

Answer: D

Q145.

Point where whole weight of body acts vertically is called

- (a) center of mass
- (b) mid point
- (c) center of gravity
- (d) none of above

Answer: C

Q146.

Two waves represented by $y_1 = 10 \sin(2000 \pi t)$ and $y_2 = 10 \sin(2000 \pi t + \pi/2)$ are superimposed at any point at a particular instant. The resultant amplitude is?

- (a) 10 units
- (b) 20 units
- (c) 14.1 units
- (d) zero

Answer: C

Biology

Q147.

The upper chambers of the heart are called

- (a) Auricles
- (b) Ventricles
- (c) spiracles
- (d) Tentacles

Answer: A

Physics

Q148.

Which of the following is not a property of magnetic lines of force?

- (a) They neither have origin nor end.
- (b) These lines do not intersect
- (c) outside the magnet their direction is from north pole to south pole and inside the magnet these are from south to north pole.
- (d) None

Answer: D

Q149.

The ratio (R) of output resistance r_o and the input resistance r_i in measurements of input and output characteristics of a transistor is typically in the range:

- (a) $R \ 10^2$ - 10^3 ■ $R \ 1$ - 10
- (c) $R \ 0.1$ - 1.0
- (d) $R \ 0.1$ - 0.01

Answer: A

Q150.

A man standing unsymmetrically between two parallel cliffs, claps his hand and starts hearing a series of echoes at intervals of 1 s. If the speed of sound in air is 340 ms^{-1} , then the distance between the two parallel cliffs, is

- (a) 170 m
- (b) 340 m c .510 m
- (d) 680 m

Answer: C

Q151.

A dynamo produces an electric current. It is based on the principle:

- (a) magnets have attractive property
- (b) conductors carrying current behave like magnets
- (c) electromagnetic induction
- (d) none of these

Answer: C

Q152.

Suppose that the acceleration of a free fall at the surface of a distant planet was found to be equal to that at the surface of the earth. If the diameter of the planet were twice the diameter of the earth, then the ratio of mean density of the planet to that of the earth would be:

- (a) 4: 1

- (b) 2: 1
(c) 1: 1
(d) 1: 2

Answer: D

Chemistry

Q153.

Which of the following statements is/are correct? This question has multiple correct options

- (a) Liebig's method is used for the quantitative estimation of both C and H
(b) Dumas method is used for the quantitative estimation of N in all nitrogen-containing organic compounds.
(c) In Liebig's combustion method, dry CuO is used.
(d) Silver salt method is a chemical method for the determination of equivalent mass of organic acids.

Answer: A,C,D

Q154.

Heat of combustion ΔH° for C(s), $H_2(g)$ and $CH_4(g)$ are -94, -68 and -213 kcal/mol respectively. Then ΔH° for $C(s) + 2H_2(g) \rightarrow CH_4(g)$ is: A. -17 kcal/mol

- (b) -111 kcal/mol
(c) -170 kcal/mol
(d) -85 kcal/mol

Answer: A

Biology

Q155.

The possible beneficial affect of grazing animals is the

- (a) Removal of wild animals
(b) Eradication of weeds
(c) Removal of wild plants
(d) Addition of their excrete to the soil

Answer: D

Q156.

Compressed Natural Gas (CNG) is

- (a) Propane
(b) Methane
(c) Ethane
(d) Butane

Answer: B

Chemistry

Q157.

A $\text{C}_2\text{H}_4\text{O}$ Compound $+ O_2 \rightarrow X + Y + Z$ Compound (A) in pure form does not give ppt. with AgNO_3 solution. mixture containing 70 % of (A) and 30 % of ether is used as an anaesthetic. Compound (X) and (Y) are oxides while (Z) is a pungent smelling gas. (X is a neutral oxide which turns cobalt chloride paper pink. Compound (Y) turns lime water milky and produces an acidic solution with water. Compounds (A), (X), (Y) and (Z) respectively will be : A. $\text{C}_2\text{H}_4\text{O}$, H_2O , CO_2 , Cl_2 . B. $\text{C}_2\text{H}_4\text{O}$, H_2O , CO_2 , I_2

- (c) $\text{C}_2\text{H}_4\text{O}$, H_2O , CO_2 , N_2
(d) N_2H_4 , CO , N_2O , CO_2

Answer: B

Biology

Q158.

Neanderthal man lived in

- (a) Desert
- (b) Deep forest
- (c) Mountains
- (d) caves

Answer: D

Chemistry

Q159.

Write the correct alternatives: According to Mendeleev's periodic law, properties of elements are periodic function of their.

- (a) Atomic number
- (b) Atomic masses
- (c) Densities
- (d) Boiling points

Answer: B

Physics

Q160.

Who named the negatively charged particles present in cathode rays as electrons?

- (a) J.J. Thomson
- (b) Stoney
- (c) Goldstein
- (d) None of these

Answer: B

Chemistry

Q161.

Which element has the most metallic character?

- (a) Boron
- (b) Aluminium
- (c) Indium
- (d) Thallium

Answer: TI (D)

Q162.

Which of these are particulates?

- (a) Dust and fumes
- (b) Fly ash and smoke
- (c) Mist and fumes
- (d) All of these

Answer: D

Physics

Q163.

Conservation of momentum in a collision between particles can be understood from

- (a) Conservation of energy
- (b) Newton's first law only
- (c) Newton's second law only.
- (d) Both Newton's second and third law.

Answer: D

Chemistry

Q164.

The element which could be extracted by electrolytic reduction of its oxide dissolved in a high temperature melt is:

- (a) sodium
- (b) magnesium
- (c) fluorine
- (d) aluminium

Answer: D

Q165.

Dry ice is:

- (a) ice cube
- (b) sodium chloride
- (c) liquid nitrogen
- (d) solid carbon dioxide

Answer: D

Physics

Q166.

The critical angle of a medium B with respect to another medium A is 45° and the critical angle of a medium C with respect to medium B is 30° . The critical angle of medium C with respect to A is :

- (a) Less than 30°
- (b) Greater than 30°
- (c) 30°
- (d) Cannot be determined

Answer: C

Chemistry

Q167.

What is the electric current required to deposit 0.972 g of chromium in three hours? (E.C.E. of chromium is 0.00018 g/c)

- (a) 0.50 A
- (b) 1A
- (c) zero
- (d) 2.5A

Answer: A

Physics

Q168.

An instrument used to measure high temperature is

- (a) Pyrheliometer
- (b) Pyrometer
- (c) Technometer
- (d) Pyknometer

Answer: B

Q169.

Astigmatism for a human eye can be removed by using an additional:

- (a) concave lens
- (b) convex lens
- (c) cylindrical lens
- (d) prismatic lens

Answer: C

Q170.

Which phenomenon occurs during the following change? Size of naphthalene balls decreases.

- (a) condensation
- (b) sublimation
- (c) evaporation
- (d) melting

Answer: B

Q171.

The velocity of progressive wave which produces the stationary wave, $y = 2 \sin(\pi x/100) \sin(\pi t)$ m:

- (a) 100 m / s
- (b) 1 m / s
- (c) 50 m / s
- (d) 1000 m / s

Answer: A

Q172.

In a physical balance working on the principle of moments, when 5 mg weight is placed on the left pan, the beam becomes horizontal. Both the empty pans of the balance are of equal mass. Which of the following statements is correct?

- (a) Both the arms are of same length.
- (b) Every object that is weighed using this balance appears lighter than its actual weight
- (c) Left arm is longer than the right arm.
- (d) Left arm is shorter than the right arm.

Answer: C

Chemistry

Q173.

With excess of C_2I_2 , ammonia forms:

- (a) ammonium chloride
- (b) nitrogen trichloride
- (c) nitrosyl chloride
- (d) nitrogen

Answer: B

Q174.

Untreated domestic sewage has a low:

- (a) BOD value
- (b) % of dissolved O_2
- (c) amount of dissolved salts
- (d) value of suspended organic matter

Answer: B

Biology

Q175.

Part of the brain acting as relay centre is

- (a) Thalamus
- (b) Hypothalamus
- (d) Pituitary

Answer: A

Q176.

Who proposed cAMP mediated 'Cascade model' of enzyme regulation?

- (a) Fischer
- (b) Sutherland
- (c) Sumner
- (d) Koshland

Answer: B

Physics

Q177.

A particle is thrown vertically upwards with a velocity of 4 m s^{-1} . The ratio of its accelerations after 1 s and 2 s of its motion is

- (a) 2
- (b) 9.8
- (c)
- (d) 4.9

Answer: 1:1 (g/g)

Q178.

In electromagnetic induction, the induced E.M.F is independent of

- (a) Change of flux
- (b) Time
- (c) Tesla
- (d) weber

Answer: C

Q179.

A box is placed on an inclined plane and has to be pushed down. The angle of inclination is :

- (a) Equal to the angle of friction
- (b) More than the angle of friction
- (c) Equal to the angle of repose
- (d) Less than the angle of repose

Answer: D

Q180.

Which one moves with greatest speed:

- (a) α -rays
- (b) β -rays
- (c) γ -rays
- (d) cathode rays

Answer: C

Biology

Q181.

Which of the following fruit developed from bicarpellary syncarpous ovary with false septum?

- (a) siliqua
- (b) Achene
- (c) capsule
- (d) All of the above

Answer: A

Q182.

In nature, the photosynthesis should proceed up to the limit of

- (a) Light
- (b) Temperature $\propto$ $C O_2$
- (d) Moisture and wind

Answer: C

Physics

Q183.

The average half-cycle value of a sine wave with a 40 V peak is

- (a) 25.48 V
- (b) 6.37 V
- (c) 14.14 V
- (d) 50.96 V

Answer: A

Chemistry

Q184.

The element, with atomic number 118 will be

- (a) Alkali
- (b) Noble gas
- (c) Lanthanide
- (d) Transition element

Answer: B

Q185.

Refining of aluminium is done by

- (a) electrolysis
- (b) distillation

- (c) sedimentation
- (d) oxidation

Answer: A

Biology

Q186.

Olecranon process is found in

- (a) Proximal end of ulna
- (b) Distal end of ulna
- (c) Proximal end of tibia
- (d) Proximal end of humerus

Answer: A

Chemistry

Q187.

Acetic acid is obtained when:

- (a) methyl alcohol is oxidised with potassium permanganate
- (b) calcium acetate is distilled in the presence of calcium formate
- (c) acetaldehyde is oxidised with potassium dichromate and sulphuric acid
- (d) glycerol is heated with sulphuric acid

Answer: C

Physics

Q188.

is a property that resists the flow of electrons in a conductor.

- (a) Capacitance
- (b) Inductance
- (c) Resistance
- (d) None of above

Answer: C

Q189.

Name the effect of current responsible for the glow of the bulb in an electric circuit.

- (a) Chemical
- (b) Heating
- (c) Magnetic

Answer: B

Chemistry

Q190.

Assertion A tube light emits white light. Reason Emission of light in a tube takes place at a very high temperature.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: C

Q191.

Which is among the greatest contributors to gaseous pollutions that are responsible for acidic rain?

- (a) Electrical power generation using coal
- (b) Earthquakes
- (c) Air pollution
- (d) All of the above

Answer: A

Physics

Q192.

Assertion (A): Ductile metals are used to prepare thin wires. Reason (R) : In the stress-strain curve of ductile metals, the length between the points representing elastic limit and breaking point is very small.

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (b) Both (A) and (R) are true and (R) is not correct explanation of (A)
- (c) (A) is true but (R) is false.
- (d) (A) is false but (R) is true

Answer: C

Chemistry

Q193.

Which of the following solution in water will have the lowest vapour pressure?

- (a) 0.1 M, NaCl
- (b) 0.1 M, Sucrose ■. 0.1 M, ■■■ I₂
- (d) 0.1 M N a₃ P O₄

Answer: D

Physics

Q194.

To determine the Young modulus of a wire, several measurements are taken. In which row can the measurement not be taken directly with the stated apparatus?

- (a) measurement: area of cross-section of wire apparatus : micrometer screw gauge
- (b) measurement: extension of wire ; apparatus: vernier scale
- (c) measurement: mass of load applied to wire ; apparatus : electronic balance
- (d) measurement: original length of wire ; apparatus metre rule

Answer: B

Q195.

How does the sound produced by a vibrating object in a medium reach your ear?

- (a) Through electron transfer
- (b) Through the vibration of particles
- (c) Sound waves don't need a medium
- (d) Not enough data

Answer: B

Chemistry

Q196.

Complete the reactions with appropriate products. (i) $\text{C}_2\text{H}_5\text{CHO} + \text{NH}_2\text{OH} \rightarrow \text{X}$ (ii) $\text{C}_2\text{H}_5\text{CHO} + \text{PdCl}_2 + \text{H}_2\text{O} \rightarrow \text{Y}$ (iii) $\text{C}_2\text{H}_5\text{CHO} + \text{H}_2\text{SO}_4 \xrightarrow{\lambda} \text{Z}$

- (a) $\text{X} = \text{C}_2\text{H}_5\text{CH=N-OH}$, $\text{Y} = \text{C}_2\text{H}_5\text{CH(OH)OH}$, $\text{Z} = \text{C}_2\text{H}_5\text{COOH}$
(b) $\text{X} = \text{C}_2\text{H}_5\text{CH}_2\text{NH}_2$, $\text{Y} = \text{C}_2\text{H}_5\text{CH}_2\text{CHO}$, $\text{Z} = \text{C}_2\text{H}_5\text{CH}_2\text{COOH}$
(c) $\text{X} = \text{C}_2\text{H}_5\text{CONH}_2$, $\text{Y} = \text{HCHO}$, $\text{Z} = \text{C}_2\text{H}_5\text{COCH}_3$

Answer: A

Q197.

Which type of isomerism is shown by But-2-yne and cyclobutene?

- (a) Ring chain isomerism
(b) Chain isomerism
(c) Position isomerism
(d) Functional isomerism

Answer: D

Physics

Q198.

An infinitely long straight conductor is bent into shape as shown in figure. It carries a current I A. and the radius of circular loop is r metre. Then the magnetic induction at the centre of the circular loop is: A

- (b) $(\mu_0 I)/(\pi r)$
(c) $(\mu_0 I)/(2 \pi r)(\pi+1)$ D · $(\mu_0 I)/(2 \pi r)(\pi-1)$

Answer: D

Chemistry

Q199.

The correct order of decreasing acidity of nitrophenols will be:

- (a) m- nitrophenol > p- nitrophenol > o - nitrophenol
(b) o - nitrophenol > m- nitrophenol > p- nitrophenol
(c) p- nitrophenol > m- nitrophenol > o- nitrophenol
(d) p- nitrophenol > o - nitrophenol > m- nitrophenol

Answer: D

Biology

Q200.

The only ctenophoran to bear cnidoblasts is

- (a) Bolinopsis
(b) Beroe
(c) ctenoplana
(d) Euchlora

Answer: C

Chemistry

Q201.

Which of the following has been used in the manufacture of non-inflammable photographic films?

- (a) Cellulose nitrate
- (b) Cellulose xanthate
- (c) cellulose perchlorate
- (d) Cellulose acetate

Answer: D

Physics

Q202.

Suppose gravity of earth suddenly becomes zero, then in which direction will the moon begin to move if no other celestial body affects it?

- (a) Radial
- (b) Tangential
- (c) cant say
- (d) None

Answer: B

Q203.

A heater of power P W raises the temperature of m k g of a liquid by ΔT K in time t s . Express the specific heat capacity of liquid in terms of above data A · $(P t)/(\Delta T)$

- (b) $(P t(\Delta T + 273))/(m(\Delta T)^2)$
- (c) $(P)/(m \Delta T)$
- (d) $(P t)/(m \Delta T)$

Answer: D

Q204.

Ocean currents are caused by the setting up of in water

- (a) convection current
- (b) heat current
- (c) electric current
- (d) wave current

Answer: A

Q205.

Assertion If objective and eye lenses of a microscope are interchanged then it can work as telescope. Reason The objective lens of telescope has small focal length.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: C

Q206.

For the same mass, which one of the following has the maximum thermal capacity?

- (a) wood
- (b) copper
- (c) water
- (d) Ice

Answer: C

Q207.

Can friction ever be eliminated?

- (a) Yes

- (b) No
- (c) Sometimes
- (d) Depends on the object in regard

Answer: B

Q208.

What is the net charge on an neutral atom?

- (a) positive
- (b) negative
- (c) zero
- (d) none of these

Answer: C

Chemistry

Q209.

At 80°C , the vapour pressure of pure liquid A is 520 mmHg and that of pure liquid B is 1000 mmHg. If a mixture solution of A and B boils at 80°C and 1 atm pressure, the amount of A (mole percent) in the mixture is:

- (a) 50 %
- (b) 54%
- (c) 32 %
- (d) 44%

Answer: D

Physics

Q210.

A uniform rod is resting freely over a smooth horizontal plane. A particle moving horizontally strikes at one end of the rod normally and gets stuck. Then This question has multiple correct options

- (a) the momentum of the particle is shared between the particle and the rod and remains conserved
- (b) the angular momentum about the mid-point of the rod before and after the collision is equal
- (c) the angular momentum about the centre of mass of the combination before and after the collision is equal
- (d) the centre of mass of the rod particle system starts to move translationally with the original momentum of the particle

Answer: A,B,C,D

Biology

Q211.

Cork of commerce obtained from Quercus is part of

- (a) Phellem
- (b) Phellogen
- (c) Phelloderm
- (d) All of the above

Answer: A

Chemistry

Q212.

Isopentane on monochlorination gives isomers and out of them are optically active :

- (a) 3,1
- (b) 4,2 c .3,2
- (d) 4,1

Answer: C

Physics

Q213.

Kilowatt-hour is the unit of

- (a) potential difference
- (b) electric power
- (c) electric energy
- (d) charge

Answer: C

Chemistry

Q214.

If for a reaction $K_c=1$ then $\Delta G^{\text{circ}} =$

- (a) 1.987
- (b) 4.184 c · 0
- (d)

Answer: C

Physics

Q215.

Which of the following factors affect pressure? This question has multiple correct options

- (a) Area
- (b) Acceleration
- (c) Force
- (d) None

Answer: A,C

Biology

Q216.

Which of the following steps of respiration is amphibolic?

- (a) Glycolysis
- (b) Oxidative decarboxylation of pyruvate
- (c) TCA cycle
- (d) Oxidative phosphorylation

Answer: C

Q217.

Assertion Gemmae formation in Funaria occurs in favourable condition Reason The gemmed form on the stem and leaves

- (a) If both the assertion and the reason are true and the reason is a correct explanation of the assertion
- (b) If both the assertion and reason are true but the reason is not a correct explanation of the assertion
- (c) If the assertion is true but the reason is false

(d) If both the assertion and reason are false

Answer: B

Chemistry

Q218.

In equilibrium $\text{C}_2\text{H}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{C}_2\text{H}_3\text{COO}^- + \text{H}^+$ The equilibrium constant may not change when: A. $\text{C}_2\text{H}_3\text{COO}^-$ are added

- (b) $\text{C}_2\text{H}_3\text{COOH}$ is added
- (c) catalyst is added
- (d) mixture is heated

Answer: C

Physics

Q219.

The graphs below show that position versus time for three different cars: 1, 2 and 3. Rank these cars according to the magnitudes of their velocities, greatest first

- (a) 1, 2, 3
- (b) 3, 2, 1
- (c) 2, 1, 3
- (d) 2, 3, 1 E. All tie

Answer: E

Q220.

The current flowing through the resistor in a series LCR a.c. circuit, is $I = \frac{V}{R}$ Now the inductor and capacitor are connected in parallel and joined in series with the resistor as shown in figure. The current in the circuit is now. (Symbols have their usual meaning)

- (a) equal to I
- (b) more than I
- (c) less than I
- (d) zero

Answer: B

Q221.

Atmospheric pressure at sea level is

- (a) 1.013103 Pa
- (b) 1.013 104 Pa
- (c) 1.013 105 Pa
- (d) 1.013 106 Pa

Answer: C

Chemistry

Q222.

Distribution law was given by:

- (a) Henry
- (b) Van't Hoff
- (c) Nernst's
- (d) Ostwald

Answer: C

Q223.

Incorrect statement about given carbohydrate is

- (a) Above compound is a reducing sugar
- (b) Above compound undergo mutarotation
- (c) Above compound is a non-reducing sugar
- (d) Above compound has a glycosidic linkage

Answer: C

Q224.

A solution is prepared by adding 2 g of substance A to 18 g of water. The mass percent of the solute is:

- (a) 10
- (b) 20
- (c) 40
- (d) 25

Answer: A

Physics

Q225.

n capacitors each of capacitance $2 \mu F$ are connected in parallel and a potential difference of 200 V is applied to the combination. The total charge on all the positive plates is 1 Coulomb then n is equal to :

- (a) 3333
- (b) 3000
- (c) 2500
- (d) 25

Answer: C

Q226.

A train is approaching towards a platform with a speed of 10 m s^{-1} while blowing a whistle of frequency 340 Hz. What is the frequency of whistle heard by a stationary observer on the platform? Given speed of sound = 340 m s^{-1}

- (a) 350 Hz
- (b) 340 Hz
- (c) 360 Hz
- (d) 330 Hz

Answer: B

Chemistry

Q227.

Assertion: The specific charge of canal rays is not constant. Reason: The mass of ion varies with speed.

- (a) Assertion is correct but Reason is wrong.
- (b) Assertion is wrong but Reason is correct.
- (c) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion.
- (d) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion.

Answer: A

Q228.

Number of σ and π bonds in acetanilide is respectively.

- (a) 19, 4
- (b) 16, 4
- (c) 20, 3
- (d) 19, 3

Answer: A

Physics

Q229.

A tank, which is open at the top, contains a liquid up to a height H . A small hole is made in the side of a tank at a distance y below the liquid surface. The liquid emerging from the hole lands at a distance x from the tank. This question has multiple correct options

- (a) If y is increased from zero to H , x will first increase and then decrease
- (b) x is maximum for $y = H/2$.
- (c) The maximum value of x is H
- (d) The maximum value of x will depend on the density of the liquid.

Answer: A,B,C

Chemistry

Q230.

Give the type of isomerism exhibited by each of the following pairs. n-Butyl alcohol and diethyl ether

- (a) Functional
- (b) Position
- (c) Tautomerism
- (d) None of These

Answer: A

Physics

Q231.

One curie is equal to 3.7×10^{10} disintegration / sec

- (b) 3.2×10^8 disintegration/sec
- (c) 2.8×10^{10} disintegration / sec
- (d) None of these

Answer: A

Q232.

In the equation of motion, $S = ut + \frac{1}{2}at^2$, S stands for

- (a) displacement in t seconds
- (b) maximum height reached
- (c) displacement in the t^{th} second
- (d) none of these

Answer: A

Chemistry

Q233.

Reagents used for industrial preparation of methanol are:

- (a) $\text{CO} + \text{H}_2$ ■. $\text{HCHO} + \text{H}_2$
- (c) $\text{CO}_2 + \text{H}_2$
- (d) $\text{HCOOH} + \text{H}_2$

Answer: A

Physics

Q234.

When a gold sheet is bombarded by a beam of α -particles, only a few of them get deflected whereas most go straight, undeflected because: This question has multiple correct options

- (a) the force of attraction exerted on the α -particles by the oppositely charged electrons is not sufficient
- (b) a nucleus has a much smaller volume than that of an atom
- (c) the force of repulsion acting on the fast-moving α particles is very small
- (d) the neutrons in the nucleus do not have any effect on the α -particles

Answer: B,C

Chemistry

Q235.

Which of the following statement about Avogadro's hypothesis is correct?

- (a) Under similar conditions of temperature and pressure, gases react with each other in simple ratio.
- (b) Under similar conditions of temperature and pressure, equal volumes of all gases contain same number of molecules
- (c) At NTP all gases contain same number of molecules
- (d) Gases always react with gases only at the given temperature and pressure

Answer: B

Physics

Q236.

If wavelength of photon and electron is same then ratio of total energy of electron to total energy of photon would be:

- (a) (Velocity of electron)/(Light's speed)
- (b) (Light's speed)/(Electron's speed)
- (c) (Light's speed)/(Velocity of electron)
- (d) (Electron's speed)/(Light's speed)

Answer: B

Chemistry

Q237.

Which of the following solutions shows positive deviation from Raoult's law?

- (a) Acetone + Aniline
- (b) Acetone + Ethanol
- (c) water + Nitric acid
- (d) Chloroform + Benzene

Answer: B

Physics

Q238.

It takes 15 minutes for an electrical kettle to heat a certain quantity of water in grams from 0°C to boiling point 100°C . If it takes 81 minutes to boil water into steam, the latent heat of steam is :

- (a) 80 cal / g , ■. 540 cal / g

(c) 100 cal / g

(d) 336 cal / g

Answer: B

Chemistry

Q239.

According to the rules of addition of significant numbers, the addition product of 118.017 and 11.0 will be:

(a) 129.017

(b) 129.0 c · 129

(d) none of these

Answer: B

Physics

Q240.

What will be the input of A and B for Boolean expression $(A+B)(A \cdot B)=1$

(a) 0,0 ■. 0,1

(c) 1,0

(d) 1,

Answer: D

Chemistry

Q241.

Which of the following is one of the postulates of Rutherford's theory?

(a) Electrons continuously lose energy

(b) Electrons move in a spiral path around the nucleus

(c) Electrons revolve in a specific path around the nucleus

(d) Electrons revolve in a stationary orbit

Answer: C

Physics

Q242.

A thin tube of uniform cross-section is sealed at both ends. It lies horizontally. The middle 5 c m contains Hg and two equal ends contain air at the same pressure P_0 . When the tube is held at an angle of 60° with the vertical, the length of the air column above and below the Hg are 46 cm and 44.5 cm . Calculate pressure P_0 in cm of Hg. Assume temperature of the system to be constant.

(a) 55 cm of Hg

(b) 65 cm of Hg

(c) 70.4 cm of Hg

(d) 75.4 cm of Hg

Answer: C

Chemistry

Q243.

Among the following, the species that is both tetrahedral and diamagnetic is: A $[\text{NiCl}_4]^{2+}$

- (b) $\text{Ni}(\text{CO})_4$
 (c) $[\text{Ni}(\text{CN})_4]^{2+}$
 (d) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$

Answer: B

Physics

Q244.

The force of friction is normal reaction between the surfaces.

- (a) Inversely proportional to
 (b) directly proportional to
 (c) proportional to the square of
 (d) independent of

Answer: B

Chemistry

Q245.

No. of 5 membered ring in $[(\text{C}_6\text{H}_5)_3\text{N}]^{2+}$ is.

- (a) 5
 (b) 6
 (c) 7
 (d) 8

Answer: C

Q246.

Butane has a _____ boiling point to that of propanol.

- (a) higher
 (b) similar
 (c) lower
 (d) none of these

Answer: C

Q247.

Chlorobenzene can be prepared by reacting benzene diazonium chloride with:

- (a) HCl
 (b) CuCl_2
 (c) AlCl_3
 (d) HNO_2

Answer: B

Biology

Q248.

Match the following with correct answers. (a) Toucan bird (b) Siberian bird (c) Macaque (d) Macaque IV Migrate to India during winter

- (a) a- II, b- I, c- II, d- IV
 (b) a- III, b- IV, c- I, d- I
 (c) a- I, b- II, c- II, d- IV
 (d) a- V, b- II, c- I, d- I

Answer: C

Physics

Q249.

The total kinetic energy of 1 mole of N_2 at 27 C will be approximately

- (a) 3739.662
- (b) 1500 calorie
- (c) 1500 kilo calorie
- (d) 1500 erg.

Answer: A

Chemistry

Q250.

Which derivative of coal is more viscous in nature?

- (a) coke
- (b) Coal tar
- (c) coal gas
- (d) Graphite

Answer: B

Biology

Q251.

In Ginkgoales, the male gametes are

- (a) Motile
- (b) Non-motile
- (c) Amoeboid
- (d) Absent

Answer: A

Q252.

Holotype is

- (a) Specimen used by author as nomenclature type.
- (b) Specimen referred along with original description.
- (c) Duplicate of nomenclature type.
- (d) Specimen selected from original, when nomenclature type is missing.

Answer: A

Physics

Q253.

During the melting of a slab of ice at 273 K at atmospheric pressure: This question has multiple correct options

- (a) Positive work is done by the ice-water on the atmosphere
- (b) Positive work is done on the ice-water by the atmosphere
- (c) The internal energy of the ice-water increases
- (d) The internal energy of the ice-water system decreases

Answer: A,C

Chemistry

Q254.

On reduction of glycolic acid with H I the product formed is:

- (a) acetic acid
- (b) iodo aetic acid
- (c) formic acid
- (d) none of these

Answer: A

Q255.

Which among the following plots are liner (a-x) is the concentration of the reactant remaining after time, t? (1):- (a-x) vs t, for a first order reaction (2) :- (a-x) vs t, for a zero order reaction (3):- (a-x) vs t, for a second order reaction (4):- $1/(a-x)$ vs t, for a second order reaction

- (a) 1 and 2
- (b) 1 and 3 c · 2 and 3
- (d) 2 and 4

Answer: D

Biology

Q256.

Ozone layer of upper atmosphere is being destroyed by

- (a) Sulphurdioxide
- (b) Carbondioxide
- (c) chlorofluorocarbon
- (d) Smogg

Answer: C

Q257.

Dialysis fluid contains all the constituents as in plasma except

- (a) Electrolytes
- (b) Proteins
- (c) Nitrogenous wastes
- (d) All the above

Answer: B

Chemistry

Q258.

The graph below shows the atomic radii of atoms in picometers (pm) as a function of atomic number. Which letter on this graph most closely shows the location of the transition metals of period 5? Atomic Radius vs. Atomic Number

- (a) Region E
- (b) Region D
- (c) Region c
- (d) Region B

Answer: C

Q259.

A sigma bond forms:

- (a) when two orbitals directly overlap but there is only one bonding interaction.
- (b) when two orbitals directly overlap and there are two bonding interaction.
- (c) when two valence electrons combine into one orbital.

(d) when two atoms overlap and the valence electrons bond E. the creation of an S shaped orbital.

Answer: A

Biology

Q260.

The smallest unit of classification is known as

- (a) species
- (b) De condole
- (c) Linnaeus
- (d) Mayr

Answer: A

Chemistry

Q261.

For the following equilibrium, $N_2 O_4 \rightleftharpoons 2 N O_2$ in gaseous phase, $N O_2$ is 50 % of the total volume when equilibrium is set up. Hence, percent of dissociation of $N_2 O_4$ is:

- (a) 50%
- (b) 25%
- (c) 66.66 %
- (d) 33.33%

Answer: D

Q262.

Pick out the pair in which the energy change of one is reverse of the energy change in the other : (1) radio (2) fluorescent lamp (3) toaster (4) photoelectric cell

- (a) 1 and 2
- (b) 2 and 3 c · 2 and 4
- (d) 3 and 4

Answer: C

Physics

Q263.

Find out the missing particle in the following nuclear reaction? ${}_1^2H + {}_{29}^{63}Cu \rightarrow {}_{30}^{64}Zn + (?)$

- (a) Proton
- (b) Neutron
- (c) Electron
- (d) Positron E . Deuteron

Answer: B

Biology

Q264.

Who proposed the term ecosystem?

- (a) Odum
- (b) Gardner
- (c) Warming
- (d) Tansley

Answer: D

Physics

Q265.

A solid sphere rolls without slipping on a rough horizontal floor, moving with a speed v . It makes an elastic collision with a smooth vertical wall. After impact, This question has multiple correct options

- (a) it will move with a speed v initially.
- (b) its motion will be rolling without slipping
- (c) its motion will be rolling with slipping initially and its rotational motion will stop momentarily at some instant.
- (d) its motion will be rolling without slipping only after some time.

Answer: C,D

Q266.

In which of the following doped semiconductors majority of charge carriers are holes?

- (a) Antimony doped with germanium
- (b) Boron doped with silicon
- (c) Arsenic doped with silicon
- (d) Phosphorus doped with germanium

Answer: B

Q267.

Name the high energetic invisible electromagnetic wave which helps in the study of structure of crystals.

- (a) X -rays
- (b) Cathod rays
- (c) Gamma rays
- (d) All of the above

Answer: A

Q268.

A constant force acting on a body of mass 3.0 kg changes its speed from 2.0 ms^{-1} to 3.5 ms^{-1} in 25 s. The direction of the motion of the body remains unchanged. What is the magnitude of the force (in newton)?

- (a) 0.18 ■. 0.36
- (c) 0.72
- (d) 0.24

Answer: A

Q269.

Bats detect the obstacle in their path by receiving the detected

- (a) Infrasonic waves
- (b) Ultrasonic waves
- (c) Both
- (d) None

Answer: B

Q270.

The displacement current was first populated by

- (a) Maxwell
- (b) Marconi
- (c) Ampere
- (d) Hertz

Answer: A

Q271.

An overhead tank having some water possesses energy.

- (a) Kinetic
- (b) Potential
- (c) Thermal
- (d) Electrical

Answer: B

Chemistry

Q272.

Which of the following is used for making optical instruments:

- (a) SiO_2
- (c) SiH_4
- (d) SiC

Answer: A

Q273.

Which of the following have identical bond order? (I) C N^- (II) O_2^- (III) N O^+ (V) C N^+

- (a) I, II
- (b) II, IV
- (c) I, II, III
- (d) I, IV

Answer: C

Q274.

The function of 'buffer solution' is:

- (a) to increase the pH value in chemical reaction
- (b) to decrease the pH value on chemical reaction
- (c) to increase or decrease the pH value of according the need
- (d) to keep the pH value constant in chemical reaction

Answer: D

Biology

Q275.

Assertion Forest influences local air circulation and may thus retain solid suspensions and gaseous elements.
Reason Forests control the wind velocity and air flows.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: A

Physics

Q276.

In case of refraction of light

- (a) Frequency changes
- (b) Speed changes
- (c) Wavelength changes

- (a) a is correct
(b) b and c are correct c . a, b, c are correct
(d) a and b are correct

Answer: B

Biology

Q277.

One of them is absent in closed vascular bundles.

- (a) Cambium
(b) Pith
(c) Ground tissue
(d) All of the above

Answer: A

Chemistry

Q278.

After how many seconds will the concentration of the reactant in a first order reaction be halved, if the rate constant is $1.155 \times 10^{-3} \text{ sec}^{-1}$?

- (a) 600
(b) $100 \text{ c} \cdot 60$
(d) 10

Answer: A

Physics

Q279.

Which of the following type of radiations are radiated by on oscillating electric charge?

- (a) Electric
(b) Magnetic
(c) Thermoelectric
(d) Electromagnetic

Answer: D

Q280.

The missing particle in the reaction: ${}_{92}^{253}\text{Es} + {}_2^4\text{He} \rightarrow {}_{101}^{256}\text{Md} +$

- (a) deuteron
(b) proton
(c) neutron
(d) β - particle

Answer: B

Q281.

Microwaves are detected by

- (a) barometer
(b) point contact diodes
(c) thermopiles
(d) the eye

Answer: B

Chemistry

Q282.

Q Type your question to X). Assume that Y and Z are on the same horizontal line, and z and X are on the same vertical line. The area inside the shape X-Y-Z-X is 1 0 0 0 J If the heat absorbed by the gas during process X-Y is 2000 J and the heat released by the gas during process Y Z is 3000 J , how much heat is lost or gained during process Z-X ? A .2000 J is gained

- (b) 2000 J is lost
- (c) 1000 J is gained
- (d) 1000 J is lost E. 5000 J is gained

Answer: B

Q283.

Among BF_3 , $\text{N}(\text{SiH}_3)_3$, $\text{N}(\text{CH}_3)_3$, $\text{O}(\text{SiH}_3)_2$ the number of molecules or molecular ions which have back bonding is:

- (a) 1
- (b) 3 c . 4
- (d) all I

Answer: B

Q284.

PVC is prepared by the polymerisation of:

- (a) ethylene
- (b) 1-chloropropene
- (c) propene
- (d) 1-chloroethene

Answer: D

Physics

Q285.

A conductor is moved in a varying magnetic field. Name the law which determines the direction of current induced in the conductor:

- (a) Fleming's right hand rule
- (b) Mohr's right hand rule
- (c) Fleming's left hand rule
- (d) Mohr's left hand rule

Answer: A

Q286.

Three simple harmonic motions in the same direction having the same amplitude a and same period are superposed. If each differs in phase from the next by 45° , then : This question has multiple correct options

- (a) The resultant amplitude is $(1+\sqrt{2}) a$
- (b) The phase of the resultant motion relative to the first is 90°
- (c) The energy associated with the resulting motion is $(3+2\sqrt{2})$ times the energy associated with any single motion
- (d) The resulting motion is not simple harmonic

Answer: A,C

Q287.

A magnifying glass is used to read the newspaper. As it is moved far away from the newspaper

- (a) Text becomes blurred and magnification reduces
- (b) Text becomes more focussed and magnification reduces
- (c) Text becomes more focused and magnification increases

(d) Text becomes blurred and magnification increases

Answer: D

Chemistry

Q288.

When a bomb calorimeter is used to determine the heat of reaction, which property of the system under investigation is most likely to remain constant?

- (a) Number of molecules
- (b) Pressure
- (c) Temperature
- (d) volume

Answer: D

Physics

Q289.

Assertion: Virtual object can't be seen by human eye. Reason: Virtual object is formed by converging rays.

- (a) Both assertion and reason are correct and reason is the correct explanation of the assertion
- (b) Both assertion and reason are correct, but reason is not the correct explanation of the assertion
- (c) Assertion is correct but reason is incorrect
- (d) Assertion is incorrect but Reason is correct E. Both assertion and reason are incorrect

Answer: D

Chemistry

Q290.

When heat is given to gas in an isothermal change, the result will be

- (a) external work done
- (b) rise in temperature
- (c) increase in internal energy
- (d) external work done and also rise in temperature

Answer: A

Biology

Q291.

Which among the following is the longest phase of cell cycle?

- (a) Prophase
- (b) Telophase c · G₁
- (d) G₂

Answer: C

Chemistry

Q292.

One mole of an organic compound requires 0.5 moles of oxygen to produce a carboxylic acid. The compound may be:

- (a) alcohol

- (b) ether
- (c) ketone
- (d) aldehyde

Answer: D

Physics

Q293.

A train of length 100 m is moving in a hilly region. At what speed must it approach a tunnel of length 80 m so that a person at rest with respect to the tunnel will see that the entire train is in the tunnel at one time?

- (a) 1.25 c
- (b) 0.8 c
- (c) 0.64 c
- (d) 0.6 c E .0 .36 c

Answer: B

Q294.

Optical fibre works on the principle of

- (a) Scattering
- (b) Refraction
- (c) Total internal reflection
- (d) Interference

Answer: C

Chemistry

Q295.

Which of the following statement(s) is/are correct about the modern periodic table? This question has multiple correct options

- (a) It consists of s, p, d and f blocks
- (b) The energy levels filling order in 6 th period is 6 s, 4 f, 5 d and 6 p
- (c) III B group does not contain maximum number of elements
- (d) It consists only metals

Answer: A,B

Biology

Q296.

Mangifera is a..

- (a) Variety
- (b) Genus
- (c) Species
- (d) Class

Answer: B

Physics

Q297.

Electric switches and appliances should be tested only with

- (a) Right hand

- (b) Left hand
- (c) Both hands
- (d) Electric tester

Answer: D

Chemistry

Q298.

Why is catalyst added while preparing oxygen from decomposition of KClO_3 ?

- (a) to increase the rate of reaction
- (b) KClO_3 decomposes at lower temperature
- (c) catalyst undergoes no change chemically or in mass
- (d) all of the above

Answer: D

Biology

Q299.

Flattened sacs of Golgi are separated from each other by a space which is A . 13 A

- (b) 130 A
- (c) 125 ,
- (d) 300 A

Answer: B

Physics

Q300.

What characteristic property of light is responsible for the blue colour of the sky?

- (a) Interference
- (b) Diffraction
- (c) scattering
- (d) Dispersion

Answer: C

Biology

Q301.

In *Brassica oleracea* var. capitata (cabbage), the edible part is

- (a) Inflorescence
- (b) Flower bud
- (c) vegetative bud
- (d) Stem

Answer: C

Physics

Q302.

The solid angle subtended by the total surface area of a sphere at the centre is

- (a) 4π ■. 2π c . π

(d) 3π

Answer: A

Chemistry

Q303.

The correct IUPAC name of the following hydrocarbon is :

- (a) 4,5 -diisopropyl-3-methylheptane
- (b) 3-methyl-4,5-diisopropylheptane
- (c) 5 -ethyl-4-isopropyl-3,6-dimethylheptane
- (d) 3-Ethyl-4-isopropyl-2,5-dimethylheptane

Answer: C

Q304.

Guanine is an example of:

- (a) Nucleotide
- (b) Nucleoside
- (c) Purine base
- (d) Pyrimidine base

Answer: C

Physics

Q305.

Assertion At a point in space, the electric field points towards north. In the region, surrounding this point the rate of change of potential will be zero along the east and west Reason Electric field due to a charge is in the space around the charge

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Assertion is incorrect but Reason is correct

Answer: B

Biology

Q306.

Organisms called methanogens are most abundant in a

- (a) Hot spring
- (b) Sulphur rock
- (c) cattle yard
- (d) Polluted stream

Answer: C

Chemistry

Q307.

Assertion Assertion: A solution of NH_4Cl in water is acidic in nature. Reason Reason : Ammonium ions undergo hydrolysis to form NH_4OH

- (a) If both the assertion and reason are true and reason is the correct explanation of assertion.
- (b) If both the assertion and reason are true but reason is not the correct explanation of assertion

- (c) If assertion is true but reason is false
(d) If both assertion and reason are false

Answer: C

Q308.

If we plot volume of a certain mass of a gas against temperature at constant pressure, we get a straight line intersecting on the negative side at -273°C which explains about absolute zero. This graph is known as:

- (a) isochor
(b) isotherm c . isotone
(d) isobar

Answer: D

Q309.

During adsorption, enthalpy and entropy of the system are but ΔG must be

- (a) negative, positive
(b) negative, negative
(c) positive, negative
(d) positive, positive

Answer: B

Q310.

Match List-I & List-II (A) $\text{O}_3 + \text{C}_2\text{N}_2$ & (1) Black colour (B) $\text{H}_2\text{O}_2 + \text{O}_3$ & (2) Rocket fuel (C) $\text{Ag} + \text{O}_3$ & (3) Tanning (D) $\text{O}_3 + \text{H}_2\text{g}$ & (4) $\text{O}_2 + \text{H}_2$ Oarray The correct match is:

- (a) A-1, B-3, C-2, D-4
(b) A-2, B-4, C-1, D-3
(c) A-3, B-2, C-3, D-1
(d) A-4, B-1, C-4, D-2

Answer: B

Physics

Q311.

A positive charge and a negative charge are held in place and initially form a dipole. Then an uncharged conducting bar is placed in between them. How is the force between the charges affected?

- (a) The force increases.
(b) The force is unaffected.
(c) The force decreases.
(d) The answer cannot be determined.

Answer: C

Chemistry

Q312.

In answering the above questions, which law has been used? [C=12, N=14 O=16]

- (a) Avogadro's law
(b) Mass balance
(c) Equilibrium
(d) All of the above

Answer: B

Physics

Q313.

Sound is produced by vibration. Define vibration.

- (a) Vibration is a rapid to and fro motion of the object.
- (b) Vibration is a circular motion of the object.
- (c) Vibration is a translational motion of the object.
- (d) None of the above

Answer: A

Q314.

Assertion (A): According to Newton's third law sum of action and reaction is not equal to zero Reason (R) : The forces action and reaction acts on different bodies.

- (a) A and R are correct and R is the correct explanation of
- (b) A and R are correct and R is not the correct explanation A
- (c) A is true and R is false
- (d) A is false and R is true

Answer: D

Q315.

A solid iron ball A of radius r collides head on with another stationary solid iron ball B of radius $2r$. The ratio of their speeds just after the collision ($e=0.5$) is:

- (a) 3
- (b) $4c \cdot 2$
- (d)

Answer: C

Biology

Q316.

Who proposed the cycle of events leading to the fixation of $C O_2$ in mesophyll and its reduction in bundle sheath?

- (a) Emerson
- (b) Melvin Calvin
- (c) Hatch and Slack
- (d) Hill and Bendall

Answer: C

Q317.

A novel species with double stranded genetic material consists of 5 bases namely P.Q. R. S. T. with percentages given below. Percentage 22 28 22 Based on the above information, which ONE of the following inferences is NOT supported by the observation?

- (a) S- base pairs with T, and Q base pairs with R
- (b) S base pairs with Q ,and T base pairs with Q
- (c) P base pairs with R , and S base pairs with Q
- (d) P base pairs with R , and T base pairs with Q

Answer: B

Physics

Q318.

A rubber ball of mass 4 kg has the same diameter as a plastic ball of mass 0.5 kg. Both the balls are dropped simultaneously from the roof of a building. When they are 8 m above the ground, they have the same

- (a) Kinetic energy

- (b) Potential energy
- (c) Momentum
- (d) Acceleration

Answer: D

Chemistry

Q319.

Which of the following statement (s) differentiate between electrochemical cell and electrolytic cell? This question has multiple correct options

- (a) Spontaneous or non-spontaneous nature of the chemical process.
- (b) Chemical reactions occurring at the electrodes
- (c) Positive and negative nature of anode
- (d) Dependence on Faraday's law

Answer: A,C

Q320.

Which of the following forms an ideal solution? This question has multiple correct options

- (a) Acetone and phenol
- (b) Bromoethane and chloroethane
- (c) CCl_4 and SiCl_4
- (d) Bromobenzene and Chlorobenzene

Answer: B,D

Physics

Q321.

Choose the proper food from following, to keep our eyes healthy:

- (a) Green leafy vegetables
- (b) Fish
- (c) orange
- (d) All of these

Answer: D

Chemistry

Q322.

Reactive dyes utilize a attached to a substituent that is capable of directly reacting with the fibre substrate.

- (a) chromophore
- (b) auxochrome
- (c) chlorophyll
- (d) none of the above

Answer: A

Q323.

Which of the following will show a negative deviation from Raoult's law?

- (a) Acetone-benzene
- (b) Acetone-ethanol
- (c) Benzene-methanol
- (d) Acetone-chloroform

Answer: D

Q324.

The presence of nitrogen in the organic compound is tested as:

- (a) N_2 ■. NH_3
- (c) NO
- (d) CN

Answer: D

Physics

Q325.

Two charged balls of the same radius and weight suspended on threads of equal length are immersed into a liquid having density of d_1 and a dielectric constant K . The density d of the material of the balls for the angles of divergence of the threads in the air and in the dielectric to be the same is: $A \cdot (Kd)/(K-1)$

- (b) $(K-1)/(Kd)$
- (c) $d/(K-1)$
- (d) $(K-1)/d$

Answer: A

Chemistry

Q326.

Which one of the following is a strong electrolyte?

- (a) CH_3 ■■■■ ■. $HCOOH$
- (c) H_2CO_3
- (d) $NaOH$

Answer: D

Q327.

What is the IUPAC name of $K_2[PtCl_6]$?

- (a) Potassium hexachloro platinum
- (b) Potassium hexachloro platinum (IV)
- (c) Potassium hexachloro platinate (IV)
- (d) Dipotassium hexachloro platinum

Answer: C

Q328.

Treatment of calcium carbide with water gives:

- (a) ethene
- (b) ethyne
- (c) ethane
- (d) benzene

Answer: B

Physics

Q329.

A body in a laboratory takes 4 minutes to cool from $61^{\circ}C$ and $59^{\circ}C$. If the laboratory temperature is $30^{\circ}C$ then the time taken by it to cool from $51^{\circ}C$ to $49^{\circ}C$ will be: $A \cdot 8$

- (b) 6 min c .5 min

(d) 4 min

Answer: A

Q330.

The excess temperature of a body falls from 12°C to 6°C in 5 minutes, then the time to fall the excess temperature from 6°C to 3°C is (assume the Newton's cooling is valid)

- (a) 10 minutes
- (b) 7.5 minutes
- (c) 5 minutes
- (d) 2.5 minutes

Answer: A

Chemistry

Q331.

Which of the following does not result in an increase in entropy?

- (a) crystallisation of sucrose from solution
- (b) rusting of iron
- (c) conversion of ice to water
- (d) vapourisation of camphor

Answer: A

Q332.

The element placed with cerium, in the table formed by Newlands' was:

- (a) actinium
- (b) lanthanum
- (c) lawrencium
- (d) zirconium

Answer: B

Biology

Q333.

Which of the following processes takes place in the presence of ATP? I. Diffusion II. Active transport III. Facilitated diffusion

- (a) I only
- (b) II only
- (c) III only
- (d) I and III E. II and III

Answer: B

Q334.

An adult has - teeth.

- (a) 20
- (b) 24 c · 28
- (d) 32

Answer: D

Physics

Q335.

Eye lens is a:

- (a) transparent double-convex lens
- (b) transparent double-concave lens
- (c) transparent concavo-convex lens
- (d) None of these

Answer: A

Chemistry

Q336.

Which one of the above oxidation curve is wrongly presented in the given Gibbs free energy vs temperature graph?

- (a) $C \rightarrow CO_2$
- (b) $Ti \rightarrow TiO_2$, $Pb \rightarrow PbO$
- (d) $Mg \rightarrow MgO$

Answer: C

Q337.

Which of the following molecules has two sigma (σ) and two pi (π) bonds?

- (a) C_2H_4 , N_2F_2
- (c) $C_2H_2Cl_2$
- (d) HCN

Answer: B

Physics

Q338.

Displacement of a body is given by $x = 5 + 3t + 2t^2$. The value instantaneous acceleration is:

- (a) 4 units
- (b) 6 units
- (c) 5 units
- (d) 2 units

Answer: A

Q339.

The distance between the sun and the earth is

- (a) 150 million kilometers
- (b) 100 million kilometers
- (c) 2 billion kilometers
- (d) 1.2 light years

Answer: A

Chemistry

Q340.

Which of the following statements is not true? CO_2 is the best fire extinguisher

- (b) Global warming can cause acid rain
- (c) Burning of coal and diesel releases sulphur dioxide gas
- (d) LPG has higher calorific value than biogas

Answer: B

Q341.

When $\text{Ca}(\text{OH})_2$ attains solubility equilibrium, the:

- (a) solution is saturated
- (b) pH will be less than 7
- (c) K_{sp} is less than the K_{sp}
- (d) concentrations of the ions are equal

Answer: A

Q342.

Calculate the % composition of Carbon in CO_2 . Molar mass is 44.01

- (a) 40 %
- (b) 67%
- (c) 30%
- (d) 27%

Answer: D

Biology

Q343.

The spray of PMA causes

- (a) Decrease in transpiration
- (b) Increase in transpiration
- (c) Increase in absorption
- (d) Increase in guttation

Answer: A

Q344.

Salivary glands of cockroach open on

- (a) Maxilla
- (b) Hypopharynx
- (c) Labium
- (d) Labrum

Answer: B

Physics

Q345.

Magnetic field lines:

- (a) starts from north pole
- (b) starts from south pole
- (c) starts from current-carrying wires
- (d) are closed loops

Answer: D

Biology

Q346.

When a cell is placed in pure water then?

- (a) Wall pressure is increased
- (b) Turgor pressure is increased

- (c) water potential of cell is increased
(d) All the above

Answer: D

Chemistry

Q347.

Antimicrobial drugs that are derived from the natural processes of some microorganisms and that can inhibit or destroy other microorganisms, are specifically known as:

- (a) synthetic drugs
(b) chemotherapeutic drugs
(c) antibiotics
(d) prophylactic drugs

Answer: C

Q348.

An aqueous solution of salt is alkaline. This shows that salt is made from:

- (a) a strong acid and a strong base
(b) a strong acid and a weak base
(c) a weak acid and a weak base
(d) a weak acid and a strong base

Answer: D

Q349.

Which of the following cyanohydrins on hydrolysis give an optically active acid giving iodoform reaction?

- (a) Acetone cyanohydrin
(b) Propionaldehyde cyanohydrin
(c) Acetaldehyde cyanohydrin
(d) Formaldehyde cyanohydrin

Answer: C

Biology

Q350.

Which of the following may show photosynthesis in moonlight

- (a) some thermal algae
(b) some marine algae
(c) some fresh water algae
(d) None of the above

Answer: B

Chemistry

Q351.

The colloidal system in which the disperse phase and dispersion medium are both liquids is known as:

- (a) an emulsion
(b) an aerosol
(c) a gel
(d) a foam

Answer: A

Q352.

What is the name for red azo dye?

- (a) p-hydroxy azo benzene
- (b) β - naphthyl azo benzene
- (c) p-amino azo benzene
- (d) p- N, N dimethyl amino azo benzene

Answer: D

Q353.

List I

- (a) Salt bridge Reduction
- (b) Standard hydrogen electrode 2) Oxidation
- (c) Cathode 3) Reference electrode
- (d) Anode 4) KCl, KNO_3 (or) $\text{NH}_4 \text{NO}_3$ The correct match is :
 - (a) 3421
 - (b) 4 312
 - (c) 4321
 - (d) 1234

Answer: C

Physics

Q354.

Assertion (A): To protect any instrument from external magnetic field, it is put inside an iron box Reason (R) : Iron is a ferromagnetic substance

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true and R is not correct explanation of
- (c) A is true, but R is false
- (d) A is false, but R is true

Answer: A

Biology

Q355.

For which of the following, 'Wharton's jelly' term is used?

- (a) Elastic tissue
- (b) Muroid tissue
- (c) Reticular tissue
- (d) All of the above

Answer: B

Q356.

Inspiration occurs when

- (a) Pressure inside lungs is lower than atmospheric pressure
- (b) Pressure inside lungs is higher than atmospheric pressure
- (c) Pressure inside lungs is same as atmospheric pressure
- (d) Pressure has no effect

Answer: A

Physics

Q357.

Which of the following is unipolar transistor?

- (a) p-n-p transistor
- (b) n-p-n transistor
- (c) Field effect transistor
- (d) Point contact transistor

Answer: C

Biology

Q358.

Which one of the following diseases is caused due to ozone depletion on human?

- (a) Huntington's chorea
- (b) Small pox
- (c) Plague
- (d) Erythema

Answer: D

Chemistry

Q359.

Among the following, the strongest acid is :

- (a) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$ ■. C_6H_5 ■■■■
- (c) $\text{m-C}_6\text{H}_3\text{OC}_6\text{H}_4\text{COOH}$
- (d) $\text{p-C}_6\text{H}_3\text{OC}_6\text{H}_4\text{COOH}$

Answer: C

Q360.

From the α -particle scattering experiment, Rutherford concluded that: This question has multiple correct options

- (a) α -particles can approach within a distance of the order of 10^{-14} m of the nucleus
- (b) the radius of the nucleus is less than 10^{-14} m
- (c) scattering follows Coulomb's law
- (d) the positively charged parts of the atom move with extremely high velocities

Answer: A,B,C

Physics

Q361.

The characteristic spectrum of an atom is observed as

- (a) Pure line spectrum
- (b) Emission band spectrum
- (c) Absorption line spectrummmmmmm
- (d) Absorption band spectrum

Answer: A

Biology

Q362.

What is the function of copper-T?

- (a) Stops zygote formation
- (b) Checks mutation
- (c) stops obliteration of blastocoels
- (d) stops fertilization

Answer: D

Physics

Q363.

A body executing SHM has its velocity 10 cm/s and 7 cm/s when its displacements from mean position are 3 cm and 4 cm respectively. The length of path is $A \cdot 10 \text{ cm}$

- (b) 9.5 cm
- (c) 4 cm
- (d) 11.36 cm

Answer: A

Q364.

The moment of inertia of a flywheel is 0.2 kg m^2 which is initially stationary. constant external torque 5 N m acts on the wheel. The work done by this torque during 10 sec is:

- (a) 1250 J
- (b) 2500 J
- (c) 5000 J
- (d) 6250 J

Answer: D

Chemistry

Q365.

The set representing the correct order of first ionisation potential is :

- (a) $K > N > Li > B > Mg > Ca$
- (c) $B > C > N$
- (d) $G > S > i > C$

Answer: B

Q366.

During electrolysis of a salt solution 0.05 F of electricity deposits 1.6 g of divalent metal. The atomic wt. of the metal is :

- (a) 32
- (b) 16
- (c) 64
- (d) 1.6

Answer: C

Physics

Q367.

The property to restore the natural shape or to oppose the deformation is called:

- (a) elasticity
- (b) plasticity
- (c) ductility
- (d) none of the above

Answer: A

Biology

Q368.

The blind sac body plan is applicable to members of of kingdom Animalia

- (a) Porifera
- (b) Annelida and Arthropoda
- (c) Cnidaria and Platyhelminthes
- (d) None of the above

Answer: C

Physics

Q369.

A particle moves along the X-axis according to the equation $x = 10^3(\pi t)$. the amplitudes and frequencies of component SHMs are

- (a) amplitude $30/4, 10/4$; frequencies $3/2, 1/2$ ■. amplitude $30/4, 10/4$; frequencies $1/2, 3/2$
- (c) amplitude $10, 10$; frequencies $1/2, 1/2$
- (d) amplitude $30/4, 10$; frequencies $3/2,$

Answer: B

Q370.

If the velocity of a particle is $v = A t + B t^2$, where A and B are constants, then the distance travelled by it between 1 s and 2 s is $A \cdot 3/2 + 4 B$

- (b) $3 A + 7 B$
- (c) $3/2 A + 7/3 B$
- (d) $A/2 + B/3$

Answer: A

Chemistry

Q371.

In the Arrhenius equation $k = A e^{E_a / R T}$ rate will be constant at:

- (a) infinite T or zero E_a
- (b) infinite E_a or zero T
- (c) infinite E_a and zero T
- (d) none of these

Answer: A

Q372.

One mole of ethanol is treated with one mole of ethanoic acid at 25 . One-fourth of the acid changes into ester at equilibrium. The equilibrium constant for the reaction will be:

- (a) $1/9$
- (b) $4/9$
- (c) 9
- (d) $9/4$

Answer: D

Physics

Q373.

Water is filled upto same height in two identical closed containers A and B . Container A has vacuum over the water while container B has air over the water. At the same depth of both the containers there is an opening on which identical balloons A and B are attached as shown in the figure given below. Then

- (a) Balloon - A will bulge more than balloon - B
- (b) Balloon - B will bulge more than balloon - A
- (c) Both the balloons will bulge equally
- (d) None of the balloons will bulge

Answer: B

Chemistry

Q374.

Give the hybridization of carbon in correct showing A . C O_3^{2-} S P^3 d

- (b) Diamond S P^3
- (c) Graphite S P^3
- (d) Fullerene S P^3

Answer: B

Physics

Q375.

Where should the object be placed to get the virtual image in case of convex lens?

- (a) object should be placed between F and 2 F
- (b) object should be placed between F and 0
- (c) object should be placed beyond 2F
- (d) object should be placed at 0

Answer: B

Chemistry

Q376.

Which is the condition, unrelated to quantities, that indicates that the rate going into solution is equal to the rate coming out of solution?

- (a) Dilute
- (b) concentrated
- (c) Unsaturated
- (d) Saturated E. Supersaturated

Answer: D

Q377.

In N_2 molecule, the atoms are bonded by:

- (a) one σ , two π bonds ■. one σ , one π bonds
- (c) t w o σ , one π bonds
- (d) three σ , one π bonds

Answer: A

Physics

Q378.

If a proton and anti-proton come close to each other and annihilate, how much energy will be released?

- (a) $1.5 \times 10^{-10} \text{ J}$
- (b) $3 \times 10^{-10} \text{ J}$
- (c) $4.5 \times 10^{-10} \text{ J}$
- (d) $2 \times 10^{-10} \text{ J}$

Answer: B

Chemistry

Q379.

Qualitative analysis of nitrogen is done by which test/ method?

- (a) Duma's method
- (b) Kjeldahl's method
- (c) Lassaigne's test
- (d) silver mirror test

Answer: C

Q380.

toppr Q Type your question 4+3 0+1 ■

- (b) $(\text{C}_2\text{H}_5)_2\text{C}=\text{CH}_2 + \text{HCl} \rightarrow (\text{C}_2\text{H}_5)_2\text{CH}-\text{CH}_2\text{Cl}$
- (c) $(\text{C}_2\text{H}_5)_3\text{C}-\text{CH}_2\text{OH} + \text{HBr} \rightarrow (\text{C}_2\text{H}_5)_3\text{C}(\text{Br})-\text{CH}_2\text{CH}_3$
- (d) $\text{C}_2\text{H}_5\text{CH}_2\text{Br} + \text{H}_2\text{O}^{\text{lo minus}} \rightarrow \text{C}_2\text{H}_5\text{CH}=\text{CH}_2$
- (a) a. Nucleophilic substitution and 1, 2-Methyl shift
- (b) Nucleophilic substitution reaction
- (c) Elimination reaction
- (d) Addition reaction
- (b) a. Addition reaction
- (b) Nucleophilic substitution reaction
- (c) Elimination reaction
- (d) Nucleophilic substitution and 1, 2-Methyl shift
- (c) a. Nucleophilic substitution reaction
- (b) Addition reaction
- (c) Nucleophilic substitution and 1, 2-Methyl shift
- (d) Elimination reaction
- (d) a. Addition reaction
- (b) Nucleophilic substitution reaction
- (c) Nucleophilic substitution and 1, 2-Methyl shift
- (d) Elimination reaction

Answer: B

Q381.

ASSERTION: In the extraction of Ag, complex $\text{Na}[\text{Ag}(\text{CN})_2]$ is reacted with Zn. REASON: Zn is a d-block transition metal.

- (a) If both assertion and reason are CORRECT, and reason is the CORRECT explanation of the assertion
- (b) If both assertion and reason are CORRECT, but reason is NOT the CORRECT explanation of the assertion
- (c) If assertion is CORRECT but reason is INCORRECT
- (d) If assertion is INCORRECT but reason is CORRECT

Answer: B

Physics

Q382.

Voltmeter is an instrument used to measure

- (a) Electric current
- (b) Electric charge
- (c) Resistance
- (d) Potential difference

Answer: D

Chemistry

Q383.

For the equilibrium, $\text{H}_2\text{O(s)} \rightleftharpoons \text{H}_2\text{O(l)}$ which of the following is true?

- (a) The pressure changes do not affect the equilibrium.
- (b) The amount of ice that melts is more, if the pressure on the system is increased
- (c) The amount of liquid that freezes is more, if the pressure on the system is increased.
- (d) At low pressure, the nature of equilibrium changes to forward direction

Answer: B

Q384.

The stabilities of coordination compounds containing the ions Cu^{2+} , Ni^{2+} , Co^{2+} , Fe^{2+} and Mn^{2+} follow the order: $\text{Mn}^{2+} > \text{Fe}^{2+} > \text{Co}^{2+} > \text{Ni}^{2+} > \text{Cu}^{2+}$

- (b) $\text{Ni}^{2+} > \text{Cu}^{2+} > \text{Fe}^{2+} > \text{Co}^{2+} > \text{Mn}^{2+}$
- (c) $\text{Cu}^{2+} > \text{Ni}^{2+} > \text{Co}^{2+} > \text{Fe}^{2+} > \text{Mn}^{2+}$
- (d) $\text{Fe}^{2+} > \text{Mn}^{2+} > \text{Ni}^{2+} > \text{Co}^{2+} > \text{Cu}^{2+}$

Answer: C

Physics

Q385.

The typical operating current for a lightemitting diode is approximately:

- (a) 1 to 5 m A
- (b) 5 to 20 m A
- (c) 20 to 500 m A
- (d) 500 to 2000 m A

Answer: B

Chemistry

Q386.

In the structure of $\text{H}_2\text{C(SF)}_4$, to decide the plane in which C=S is present the following bond angle values are given: Axial F-S-F angle (idealised = 180) $\Rightarrow$ 170 Equatorial F-S-F angle (idealised = 120) $\Rightarrow$ 97 After deciding the plane of double bond, which of the following statement is/are correct?

- (a) Two C-H bonds are in the same plane of axial S-F bonds
- (b) Two C-H bonds are in the same plane of equatorial S-F bonds
- (c) Total five atoms are in the same plane
- (d) Equatorial S-F bonds are perpendicular to plane of π - bond

Answer: A,C,D

Q387.

The incorrect statement(s) among the following is (are): i) Primary valency of a transition metal ion in complex represent its oxidation number. ii) Zn^{2+} is colourless. iii) MnCl_2 exhibits green color in aqueous solution. iv) F

e, C o, N i complexes are diamagnetic in nature.

- (a) i and ii
- (b) ii only
- (c) iii and iv
- (d) iv only

Answer: C

Biology

Q388.

Annual rings and growth rings are formed due to the fluctuations in the activity of

- (a) Xylem
- (b) Phloem
- (c) xylem and phloem
- (d) Cambium

Answer: D

Physics

Q389.

The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of $10\ \Omega$ is $A \cdot 0.2\ \Omega$

- (b) $0.5\ \Omega$
- (c) $0.8\ \Omega$
- (d) $1.0\ \Omega$

Answer: B

Biology

Q390.

The molecular weight of chlorophyll α is

- (a) 907
- (b) 893
- (c) 91
- (d) 712

Answer: B

Chemistry

Q391.

Water plays an important role in :

- (a) hydrological cycle
- (b) water cycle
- (c) both A and B
- (d) none of above

Answer: C

Physics

Q392.

The radius of first Bohr orbit in hydrogen atom is r_0 then the radius of first orbit in helium atom will be $A \cdot 2 r_0$

- (b) $4 r_0$
- (c) $(r_0)/(2)$
- (d) r

Answer: C

Q393.

Assertion A load when lifted vertically up requires greater effort than to roll the same load up the ramp. Reason The work done in lifting the load vertically up is less.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: C

Chemistry

Q394.

Which of the following is a polysaccharide?

- (a) Cellulose
- (b) Sucrose
- (c) Galactose
- (d) Maltose

Answer: A

Physics

Q395.

A cylindrical bar magnet is lying along the axis of a circular coil. If the magnet is rotated about the axis of the coil then

- (a) emf will be induced in the coil
- (b) only induced current will be generate in the coil
- (c) no current will be induced in the coil
- (d) both emf and current will be induced in the coil

Answer: C

Q396.

A wire 50 cm long and 1 mm^2 in cross section carries a current of 4 A when connected to a 2 V battery. The resistivity of the wire is

- (a) $2 \times 10^{-7} \Omega \text{ m}$ ■ $5 \times 10^{-7} \Omega \text{ m}$
- (c) $4 \times 10^{-6} \Omega \text{ m}$
- (d) $1 \times 10^{-6} \Omega \text{ m}$

Answer: A

Chemistry

Q397.

Which of the following compound cannot be oxidised by O_3 ?

- (a) K I ■ K M n O_4

- (c) K_2O_4
(d) $FeSO_4$

Answer: B

Q398.

Method used to separate Petrol from crude oil is:

- (a) Distillation
(b) chromatography
(c) suballisation
(d) fractional distillation

Answer: D

Q399.

In iodine charge transfer test the color of ether compound formed is:

- (a) blue
(b) pale yello
(c) brown
(d) dark red

Answer: D

Physics

Q400.

If the air density were uniform, then the height of the atmosphere above the sea level to produce a normal atmospheric pressure of 1.0×10^5 Pa is (density of air is 1.3 kg / m^3 , $g=10 \text{ m / s}^2$):

- (a) 0.77 km
(b) 7.7 km
(c) 77 km
(d) 0.077 km

Answer: B

Chemistry

Q401.

Which of the following is used as a food preservative?

- (a) Aldehydes
(b) Ketones
(c) carboxylic acid
(d) Alcohol

Answer: C

Biology

Q402.

Anaerobic process that takes place after glycolysis is called as

- (a) TCA
(b) Calvin cycle
(c) Krebs cycle
(d) Fermentation

Answer: D

Physics

Q403.

Buntly mixed 440 gm of ice at 0°C with 540 gm of water at 80°C in a bowl. Then what would remain after sometime in the bowl?

- (a) only ice
- (b) only water
- (c) ice and water in same amount
- (d) ice and water will vapourise

Answer: B

Q404.

The force between the plates of a parallel plate capacitor of capacitance C and distance of separation of the plates d with a potential difference V between the plates, is: $\frac{1}{2} C V^2 / d$

- (b) $(C^2 V^2) / (2 d^2)$
- (c) $(C^2 V^2) / (d^2)$
- (d) $(V^2 d) / (C)$

Answer: A

Q405.

Fuse wire should be placed in the path of wire

- (a) Neutral
- (b) Phase
- (c) Earth
- (d) None of these

Answer: B

Chemistry

Q406.

S O_2 and C l_2 both are used as bleaching agent. Factors that cause bleaching are respectively:

- (a) H (that causes reduction), O (that causes oxidation)
- (b) O (that causes reduction), H (that causes oxidation)
- (c) H (that causes oxidation), O (that causes reduction)
- (d) O (that causes oxidation), H (that causes reduction)

Answer: D

Biology

Q407.

Hormone responsible for ageing is

- (a) GA
- (b) IAA
- (c) Ethylene
- (d) Cytokinin

Answer: C

Q408.

The number of time a person breathes in a minute is known as This question has multiple correct options

- (a) Respiration rate
- (b) Breathing rate
- (c) combustion rate

(d) None of the above

Answer: A,B

Q409.

Blood passes from left ventricle to right atrium. It is

- (a) Pulmonary circulation
- (b) Systemic circulation
- (c) Coronary circulation
- (d) Arterovenous circulation

Answer: B

Q410.

All the reactions in glycolysis are reversible except

- (a) 3 -Phosphoglycerate to 1,3 -Biphosphoglycerate.
- (b) Glucose 6-phosphate to fructose - 6-phosphate.
- (c) 1,3 -Biphosphoglycerate to glyceraldehyde 3 -P.
- (d) Fructose - 6 -phosphate to fructose- 1,6 -diphosphate.

Answer: D

Chemistry

Q411.

Bleaching powder is obtained by the interaction of Cl_2 with:

- (a) dilute solution of Ca(OH)_2
- (b) concentrated solution of Ca(OH)_2
- (c) dry CaO
- (d) dry slaked lime

Answer: D

Physics

Q412.

The lighting circuit is used for running:

- (a) devices which draw small current
- (b) devices which draw large current
- (c) devices which do not draw current
- (d) devices which are weightless

Answer: A

Chemistry

Q413.

What is the physical state of PCl_5 under normal condition?

- (a) Solid
- (b) Liquid
- (c) Gaseous
- (d) None of these

Answer: B

Physics

Q414.

A block is thrown with a velocity of 2 m / s (relative to ground) on a belt, which is moving with velocity 4 m / s in opposite direction of the initial velocity of block. If the block stops slipping on the belt after 4 s of the throwing then choose the correct statement(s) This question has multiple correct options

- (a) Displacement with respect to ground is zero after 2.66 and displacement with respect to ground is 12 m after 4 s
- (b) Displacement with respect to ground in 4 s is 4 m .
- (c) Displacement with respect to belt in 4 s is -12 m .
- (d) Displacement with respect to ground is zero in 8/3 s

Answer: A,C,D

Q415.

In a metallic conductor, electric current is thought to be due to the movement of

- (a) ions
- (b) amperes
- (c) electrons
- (d) protons

Answer: C

Biology

Q416.

Which one of the following statements pertaining to plant structure is correct?

- (a) cork lacks stomata but lenticels carry out transpiration.
- (b) Passage cells help in transfer of food from cortex to phloem
- (c) Sieve tube elements possess cytoplasm but no nuclei
- (d) The shoot apical meristem has a quiescent centre.

Answer: C

Chemistry

Q417.

Which of the following rules regarding the significant figures and calculations involving them is not correct?

- (a) The result of an addition or subtraction is reported to the same number of decimal places as present in number with least decimal places.
- (b) Result of multiplication or division should have same number of significant figures as present in most precise figure
- (c) The result of multiplication or division should be rounded off to same number of significant figures as present in least precise figure.
- (d) The non-significant figures in the measurements are rounded off.

Answer: B

Biology

Q418.

Taxonomically, what is species?

- (a) A group of evolutionary related populations
- (b) A group of organism which can interbreed
- (c) Category to which most taxonomic information is attached
- (d) Fundamental unit of phylogenetic history of organisms

Answer: B

Q419.

Q Type your question sheets: I. Plants should be collected in the flowering stage. II. Every detail regarding the plant such as locality, ecological conditions, plants vegetative and floral characters, etc should be noted. III. Plants are evenly pressed by unfolding a plant part between blotting papers (or newspapers) with the help of plant presser. IV. Blotting paper need not be changed until plants get dried. V. After drying, the plant's specimen is carefully mounted/pasted on the herbarium sheets. VI. The herbarium sheet is labeled on the lower right hand corner representing the number of plant specimen, date of collection, etc. Which of the above statement(s) is/are incorrect?

- (a) Only I
- (b) Only IV
- (c) I and IV
- (d) III and IV

Answer: B

Physics

Q420.

Two spheres of the same metal have radii 1 m and 4 m and temperature 4000 K and 2000 K respectively. The ratio of the energy radiated per second by the first to the second is :

- (a) 1:
- (b) 1:
- (c) 2:
- (d) 1:

Answer: A

Q421.

Which of the following will not increase voltage in a simple generator?

- (a) Increasing the number of turns in armature coil
- (b) Winding the coil on a soft armature
- (c) Increasing the size of the gap in which the armature turns
- (d) Increasing the speed of rotation.

Answer: C

Q422.

Which of the following is an isotope of hydrogen?

- (a) Deuterium
- (b) Trinium
- (c) Promium
- (d) All of the above

Answer: A

Q423.

The capacity of a parallel plate condenser is $10 \mu F$ without the dielectric. Material with a dielectric constant of 2 is used to fill half- thickness between the plates. The new capacitance is μF

- (a) 10
- (b) 20
- (c) 15
- (d) 13.33

Answer: D

Chemistry

Q424.

Which of the following statement(s) about the Modern periodic table is/are incorrect? (i) The elements in the Modern periodic table are arranged on the basis of their decreasing atomic number. (ii) The elements in the Modern periodic table are arranged on the basis of their increasing atomic number (iii) Isotopes are placed in an adjoining group(s) in the periodic table.

- (a) (i) only
- (b) (i) and (ii)
- (c) (ii) and (iii)
- (d) (i) and (iii)

Answer: D

Physics

Q425.

For a radioactive element, the correct representation of number of nuclei (N) decayed after time (t) is represented by

- (a)
- (b) c
- (d)

Answer: B

Biology

Q426.

The diatomaceous earth is used to insulate boilers and steam pipes because

- (a) The wall of diatoms is deposited with calcium.
- (b) The diatomaceous earth is cheap.
- (c) It is a good conductor of heat.
- (d) The wall of diatoms is made of silica

Answer: D

Physics

Q427.

The dimensions of strain is:

- (a) $L B \cdot L^2$
- (c) It is dimensionless
- (d) $M L^2 T^{-2}$

Answer: C

Chemistry

Q428.

A l has lower ionization potential than that of M g because:

- (a) A l atom is bigger than M g atom
- (b) M g atom is bigger than A l atom
- (c) all electrons in M g are paired, but those of A l are not
- (d) all belongs to a higher group

Answer: C

Q429.

An aqueous solution of methanol in water has vapour pressure:

- (a) equal to that of water
- (b) equal to that of methanol
- (c) more than that of water
- (d) less than that of water

Answer: C

Physics

Q430.

If a charge is moved against the coulomb force of an electric field, then :

- (a) work is done by the electric field
- (b) energy is used from some outside source
- (c) the strength of the field is decreased
- (d) the energy of the system is decreased

Answer: B

Chemistry

Q431.

The following reaction occurs in the Blast Furnace where iron ore is reduced to iron metal: $\text{Fe}_2\text{O}_3(\text{s}) + 3\text{CO}(\text{g}) \rightleftharpoons 2\text{Fe}(\text{l}) + 3\text{CO}_2(\text{g})$ Using Le Chatelier's principle, predict which of the following will not disturb the equilibrium?

- (a) Addition of Fe_2O_3
- (b) Removal of CO_2
- (c) Removal of CO
- (d) Addition of CO_2

Answer: A

Biology

Q432.

In Colon, constrictions of the wall form a series of small pockets called

- (a) Haustra
- (b) Crypts of lieberkuhn
- (c) zymogen cells
- (d) Taenial

Answer: A

Physics

Q433.

A man running along a straight road with uniform velocity $\mathbf{u} = u\mathbf{i}$ feels that the rain is falling vertically down along $-\mathbf{j}$. If he doubles his speed, he finds that the rain is coming at an angle θ with the vertical. The velocity of the rain with respect to the ground is :

- (a) $u\mathbf{i} - u\theta\mathbf{j}$
- (b) $u\mathbf{i} - u/\theta\mathbf{j}$
- (c) $u\theta - u\mathbf{j}$
- (d) $u/\theta\mathbf{i} - u\mathbf{j}$

Answer: D

Chemistry

Q434.

The structures of $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}((\text{P}(\text{H}_3)_3)_2\text{C}_2\text{I}_2)]$ are:

- (a) square planar in both cases
- (b) tetrahedral and square planar respectively
- (c) tetrahedral in both cases
- (d) square planar and tetrahedral respectively

Answer: B

Physics

Q435.

Threshold wavelength of tungsten is 2300 angstrom. If ultraviolet light of wavelength 1800 angstrom is incident on it, then the maximum kinetic energy of photoelectrons would be?

- (a) 1.5 eV
- (b) 2.5 eV
- (c) 3.0 eV
- (d) 5.0 eV

Answer: A

Biology

Q436.

Spongy parenchyma is arranged in

- (a) One layer
- (b) Loosely arranged
- (c) Compactly arranged
- (d) Regularly arranged around large cavities

Answer: B

Chemistry

Q437.

A sample of acetaldehyde contained some ethyl alcohol as impurity. The reagent useful for the purification of CH_3CHO is :

- (a) NaHCO_3
- (b) $\text{Na}_2\text{CO}_3 \cdot \text{NaHSO}_3$
- (d) PCl_5

Answer: C

Physics

Q438.

Assertion A body with constant acceleration always moves along a straight line. Reason A body with constant magnitude of acceleration may not speed up.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion

- (c) Assertion is correct but Reason is incorrect
(d) Assertion is incorrect but Reason is correct

Answer: D

Q439.

The de-Broglie wavelength of a proton and alpha particle is same, the ratio of their velocities is :

- (a) 1: 2
(b) 2: c · 1: 4
(d) 4:

Answer: A

Q440.

A parallel plane capacitor C with plates of unit area and separation d is filled with a liquid of dielectric constant $k=2$. The level of liquid is $u/3$ initially. Suppose the liquid level decreases at a constant speed v , the time constant as a function of time t is $A (6 \epsilon_0 R)/(5 d + 3 v t)$

- (b) $((15 d + 9 v t) \epsilon_0 R)/(2 d^2 - 3 d u t - 9 v^2 t^2)$
(c) $(6 \pi \epsilon_0 R)/(5 d - 3 v t)$
(d) $((15 d - 9 v t) \epsilon_0 R)/(2 d^2 + 3 d v t - 9 v^2 t^2)$

Answer: D

Q441.

A reflecting telescope utilises:

- (a) a convex mirror
(b) a concave mirror & a prism
(d) a plano-concave lens

Answer: B

Chemistry

Q442.

K_a for acetic acid in water is 1.7×10^{-5} at 25°C . The pH of a mixture of 25 ml of 0.02 N acetic acid and 2.5 ml of 0.1N NaOH (neglecting volume change) will be ($1.7=0.23$)

- (a) 2.
(b) 4.8 & 7.5
(d) 1.0.

Answer: B

Physics

Q443.

Assertion If the length of the conductor is doubled, the drift velocity will become half of the original value (keeping potential difference unchanged). Reason At constant potential difference, drift velocity is inversely proportional to the length of the conductor.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
(b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
(c) Assertion is correct but Reason is incorrect
(d) Both Assertion and Reason are incorrect

Answer: A

Biology

Q444.

Red blood cells

- (a) Divide by mitotic division
- (b) Divide by meiotic division
- (c) Divide amitotically
- (d) Do not divide

Answer: D

Q445.

An example of Liliaceae family is

- (a) Lupin
- (b) Soyabean
- (c) Petunia
- (d) Tulip

Answer: D

Chemistry

Q446.

Which of the following is a pentose sugar?

- (a) Glucose
- (b) Fructose
- (c) Ribose
- (d) Cellulose

Answer: C

Physics

Q447.

A particle projected from the origin ($x=y=0$) moves in x y plane so that its velocity is $v=(2i+4xj)$ m / s, when it is at point (x, y) m. (i and j are the unit vectors along x and y axis). What is the value of y , when $x=3$

- (a) 15
- (b) $10\sqrt{3}$
- (d) 8

Answer: C

Q448.

$H e^+$ is in n^{th} state. It emits two successive photons of wavelength 103.7 nm and 30.7 nm, to come to ground state the value of n is:

- (a) 4
- (b) $5\sqrt{3}$
- (d)

Answer: B

Q449.

The energy required to transfer an electron from second Bohr orbit to third Bohr orbit in a hydrogen like atom with nuclear charge Z_e is 6.8 eV. The wavelength of radiation required to shift the electron from first orbit to infinite orbit will be : $A \cdot 2.527 A^\circ$

- (b) $25.27 A^\circ$
- (c) $252.7 A^\circ$
- (d) $2527 A^\circ$

Answer: B

Chemistry

Q450.

The kinetic energy of electrons ejected by using light having frequency equal to threshold frequency (ν_0) is: A .

$h \nu_0$

- (b) almost zero
- (c) very large
- (d) h / ν_0

Answer: B

Biology

Q451.

The system of giving one name to the individual is called

- (a) Quasibinomial system
- (b) Semibinomial system
- (c) Mononomial system
- (d) Uninomial system

Answer: C

Physics

Q452.

When white light is incident on a prism, which of the following ray will experience lesser deviation?

- (a) Black
- (b) Green
- (c) violet
- (d) Red

Answer: D

Chemistry

Q453.

An aldehyde on oxidation gives

- (a) an acid
- (b) an alcohol
- (c) an ether
- (d) a ketone

Answer: A

Q454.

Reaction, $\text{C}_2\text{H}_2 + 2\text{NaOH} \rightarrow \text{P}$ What are the products?

- (a) $\text{NaC}\equiv\text{CN} + \text{NaOCClH}_2$ ■. $\text{NaC}\equiv\text{CN} + \text{NaOCClH}_2\text{O}_2$
- (c) $\text{NaC}\equiv\text{CN} + \text{NaOCClH}_2\text{O}$
- (d) $\text{NaC}\equiv\text{CN} + \text{NaOCClO}_2$

Answer: C

Q455.

What type of bond does Mg form with Cl ?

- (a) Covalent
- (b) Ionic
- (c) co-ordinate
- (d) None of these

Answer: B

Q456.

The chloro-bis (ethylenediamine) nitrocobalt(III) ion is - A . $[\text{Co}(\text{N O}_2)_2(\text{e n})_2 \text{C l}_2]^+$

- (b) $[\text{CoCl}(\text{N O}_2)_2(\text{e n})_2]^+$
- (c) $[\text{Co}(\text{N O}_2) \text{C l}(\text{e n})_2]$
- (d) $[\text{Co}(\text{en}) \text{C l}_2(\text{N O}_2)_2]$

Answer: C

Q457.

The percentage of formaldehyde in formalin is:

- (a) 40 %
- (b) 30%
- (c) 20 %
- (d) 10%

Answer: A

Physics

Q458.

Ray optics is valid when characteristic dimension are

- (a) much smaller than wavelength of light.
- (b) of same order as wavelength of light
- (c) much larger than wavelength of light
- (d) none of the above

Answer: C

Q459.

Name the rays or waves of wavelength nearly 0.1 n m

- (a) IR rays
- (b) UV rays
- (c) Gamma rays
- (d) x-rays

Answer: D

Biology

Q460.

The tissue is defined as a group of

- (a) Cells which are dissimilar in origin, but are similar in form and function
- (b) Cells which are similar in origin, but dissimilar in form and function
- (c) Cells which are dissimilar in origin, form and function
- (d) Cells which are similar in origin, form and function

Answer: D

Physics

Q461.

If the minimum deviation produced by a prism of refracting angle 60° is 30° , then calculate the refractive index of the material of the prism.

- (a) 2
- (b) 1.414
- (c) $\sqrt{13}$
- (d) 4

Answer: C

Chemistry

Q462.

Roasting is carried out to This question has multiple correct options

- (a) convert sulphide ore to oxide
- (b) remove water of hydration
- (c) melt the ore
- (d) remove carbon and sulphur impurities

Answer: A,D

Biology

Q463.

Prenatal growth in higher vertebrate is

- (a) Auxetic
- (b) Accretionary
- (c) Multiplicative
- (d) None of the above

Answer: C

Chemistry

Q464.

Out of the following redox reactions I. $\text{N}_2\text{H}_4 + \text{N}_2\text{O}_3 \xrightarrow{\Delta} \text{N}_2 + \text{O}_2 + 2\text{H}_2\text{O}$ II. $\text{N}_2\text{H}_4 + \text{N}_2\text{O}_2 \xrightarrow{\Delta} \text{N}_2 + 2\text{H}_2\text{O}$ III. $\text{PCl}_5 \xrightarrow{\Delta} \text{P} + \text{Cl}_2$ disproportionation is not shown in:

- (a) I and II
- (b) II and III
- (c) I and III
- (d) I, II and III

Answer: B

Physics

Q465.

Two identical copper spheres carrying charges $+Q$ and $-9Q$ separated by a certain distance has attractive force F . If the spheres are allowed to touch each other and moved to distance of separation 'x'. So that the force between them becomes $(4F)/9$. Then x is equal to

- (a) d
- (b) $2d$
- (c) $d/2$
- (d) $4d$

Answer: B

Q466.

Two identical coils carry equal currents have a common centre and their planes are at right angles to each other. The ratio of the magnitude of the resultant magnetic field at the centre and the field due to one coil is :

- (a) 2: 1
- (b) $1: 2\sqrt{2}$
- (c) $1: \sqrt{2}$
- (d) $1: 2$

Answer: C

Chemistry

Q467.

Which compound is used as an antacid?

- (a) Soda ash
- (b) Baking soda
- (c) Nitrous oxide
- (d) Magnesium carbonate

Answer: D

Physics

Q468.

A H e^+ ion is in its first excited state. Its ionization energy is?

- (a) 6.04 eV
- (b) 13.60 eV
- (c) 54.40 eV
- (d) 48.36 eV

Answer: B

Q469.

A nucleus with atomic number Z and neutron number N undergoes two decay processes. The result is a nucleus with atomic number $Z-3$ and neutron number $N-1$. Which decay processes took place?

- (a) Two β^- decays
- (b) Two β^+ decays
- (c) An α -decay and a β^- decay
- (d) An α -decay and a β^+ decay

Answer: D

Biology

Q470.

Dryopithecus is considered to be

- (a) Ancestral form of ape
- (b) Ancestral form of monkeys
- (c) Connecting links between ape and man
- (d) Common ancestor of ape and man

Answer: D

Chemistry

Q471.

A dispersion of liquid in water is an example of a colloidal

- (a) Aerosol
- (b) Emulsion
- (c) Foam
- (d) Gel

Answer: B

Physics

Q472.

A body having kinetic energy k moving on a rough horizontal surface is stopped at a distance x by constant frictional force. The force of friction exerted on the body is $A \cdot k/x$

- (b) $(\sqrt{k})/(x)$
- (c) $(k)/(\sqrt{x})$
- (d) $k \cdot x$

Answer: A

Chemistry

Q473.

1.2 mol of $\text{C}_6\text{H}_5\text{NH}_2$ ($pK_b = 3.3$) is added to 0.8 moles of HCl and the solution is diluted to one litre, resulting pH of solution is:

- (a) 10.7
- (b) 3.6
- (c) 10.4
- (d) 11.3

Answer: A

Physics

Q474.

The resultant of two forces $3P$ and $2P$ is R . If the first force is doubled then the resultant is also doubled. The angle between the two forces is

- (a) 60°
- (b) 120°
- (c) 30°
- (d) 135°

Answer: B

Chemistry

Q475.

Benzenediazonium chloride on treatment with water gives:

- (a) Benzene
- (b) o-chlorophenol
- (c) Anisole
- (d) Phenol

Answer: D

Physics

Q476.

In Young's experiment with white light, the central fringe is white. If now a transparent film is introduced in the upper beam coming from the top slit, then the white fringe:

- (a) moves downward
- (b) moves upward
- (c) remains at the same place
- (d) totally disappears

Answer: A

Q477.

A point charge is brought in an electric field. The electric field at a nearby point: This question has multiple correct options

- (a) Will increase if the charge is positive
- (b) Will decrease if the charge is negative
- (c) May increase if the charge is positive
- (d) May decrease if the charge is negative

Answer: C,D

Chemistry

Q478.

In the following redox reaction: $Zn(s) + NO_3^-(aq) + H^+(aq) \rightarrow Zn^{2+}(aq) + NH_4^+(aq)$ $Zn(s)$ and $NO_3^-(aq)$ respectively are respectively:

- (a) oxidant and reductant
- (b) reductant and oxidant
- (c) both oxidant
- (d) both reductant

Answer: B

Physics

Q479.

Two concentric loops are insulated from each other. The entire loop lies in the plane (of the paper). A uniform magnetic field B points into the plane of the paper. At $t=0$, the loop starts rotating about the common diameter as axis with a constant angular velocity ω in the magnetic field. Which of the following options is/are correct? This question has multiple correct options

- (a) The amplitude of the maximum net e.m.f. induced due to both the loops is equal to the amplitude of maximum e.m.f. induced in the smaller loop alone
- (b) The rate of change of the flux is maximum when the plane of the loops is perpendicular to plane of the paper
- (c) The e.m.f. induced in the loop is proportional to the sum of the areas of the two loops
- (d) The net e.m.f. induced due to both the loops is proportional to $\cos \omega t$

Answer: A,B

Q480.

An object is placed 30 cm away from a convex lens of focal length 10 cm and a sharp image is formed on a screen. Now a concave lens is placed in contact with the convex lens. The screen now has to be moved by 45 cm to get a sharp image again. The magnitude of focal length of the concave lens is (in cm)

- (a) 72
- (b) 60
- (c) 36

(d) 20

Answer: B

Chemistry

Q481.

A metal oxide is acidic when : $\chi < 2.1$

(b) $\chi = 2.1$ to 3.2

(c) $\chi > 3.2$

(d) None of these

Answer: C

Physics

Q482.

In common base circuit of a transistor, current amplification factor is 0.95 Calculate the emitter current if base current is 0.2 mA

(a) 2 mA

(b) 4 mA

(d) 8 mA

Answer: B

Chemistry

Q483.

In the extraction of aluminium by electrolytic process, cryolite is added to

(a) lower the melting point of alumina

(b) increase the electrical conductivity

(c) minimise the anode effect

(d) remove impurities from alumina

Answer: A,B

Biology

Q484.

Heart beat is accelerated by

(a) Sympathetic nerves and noradrenaline

(b) Cranial nerves and adrenaline

(c) Cranial nerves and acetylcholine

(d) Sympathetic nerves and acetylcholine

Answer: A

Chemistry

Q485.

Freshly prepared precipitate sometimes gets converted to colloidal solution by:

(a) coagulation

(b) electrolysis

(c) diffusion

(d) peptisation

Answer: D

Q486.

Among the following, the couple with most positive reduction potential is :

- (a) $\text{Mn}^{+3} / \text{Mn}^{+2}$
- (b) $\text{Cr}^{+3} / \text{Cr}^{+2}$
- (c) $\text{Fe}^{+3} / \text{Fe}^{+2}$
- (d) $\text{Co}^{+3} / \text{Co}^{+2}$

Answer: D

Physics

Q487.

A bar magnet is dropped along the axis of copper ring held horizontally. The acceleration of fall is

- (a) equal to g at the place.
- (b) less than g
- (c) more than g
- (d) depends upon diameter of the ring and length of the magnet

Answer: B

Chemistry

Q488.

4 protons, 5 neutrons, and 4 electrons are there in an atom of beryllium. Its mass number is:

- (a) 10
- (b) 5
- (c) 8
- (d) 9

Answer: D

Biology

Q489.

Which is not an example of cytoplasmic inheritance?

- (a) Female sterility in maize
- (b) Sigma particle inheritance
- (c) Kappa particle inheritance
- (d) Plastid inheritance

Answer: B

Chemistry

Q490.

For a general reaction $\text{X} \rightarrow \text{Y}$, the plot of conc. of X vs time is given in the figure. What is the order of the reaction and what are the units of rate constant?

- (a) zero, $\text{mol L}^{-1} \text{s}^{-1}$
- (b) First, $\text{mol L}^{-1} \text{s}^{-1} \text{c}$. First, s^{-1}
- (d) zero, $\text{L mol}^{-1} \text{s}^{-1}$

Answer: C

Q491.

Haber's process is a/an:

- (a) endothermic process
- (b) exothermic process
- (c) spontaneous with no heat liberation
- (d) None

Answer: B

Q492.

On igniting Fe_2O_3 at 1400°C the product obtained is

- (a) Fe_2O_3 melt
- (b) FeO & Fe_3O_4
- (d) metallic iron

Answer: C

Physics

Q493.

A step up transformer is used to:

- (a) increase the current and increase the voltage
- (b) decrease the current and increase the voltage
- (c) increase the current and decrease the voltage
- (d) decrease the current and decrease the voltage

Answer: B

Biology

Q494.

Drones in a colony of honey bees originate by

- (a) Thelytoky
- (b) Arrhenotoky
- (c) Amphitoky
- (d) Deuterotoky

Answer: B

Q495.

The advantage/s associated with water harvesting at the community level is/are

- (a) The ground water level increases due to recharging of wells
- (b) Ground water keeps the layers of soil above it moist and prevents loss of water by evaporation.
- (c) Ground water keeps the layers of soil above it dry and prevents loss of water by evaporation
- (d) Both A and B

Answer: D

Physics

Q496.

A double convex lens of focal length 6 cm is made of glass of refractive index 1.5. The radius of curvature of one surface which is double that of the other surface will be A .3 c m

- (b) 9 cm & 12 cm

(d) 24 cm

Answer: C

Biology

Q497.

Mists are produced by:

- (a) smoke formed during combustion of organic matter
- (b) particles of spray liquids and by condensation of vapours in air
- (c) fine solid particles produced during crushing and grinding
- (d) condensation of vapours during chemical reactions

Answer: B

Q498.

Malleus is formed by modification of articular bone of lower vertebrates, while incus is modification of

- (a) Quadrate
- (b) Jugal
- (c) Hyomandibularis
- (d) Both A and B

Answer: A

Q499.

Humans and animals are dependent on plants.

- (a) Directly
- (b) Indirectly
- (c) Both A and B
- (d) None of the above

Answer: C

Q500.

External ear is found in

- (a) Reptiles
- (b) Mammals
- (c) Amphibians
- (d) Fishes

Answer: B

Physics

Q501.

A particle of mass m moving with a velocity $(3\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-1}$ collides with another body of mass M and finally moves with velocity $(-2\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$. Then during the collision: This question has multiple correct options

- (a) Impulse received by m is $m(5\mathbf{i} + \mathbf{j})$
- (b) Impulse received by m is $m(-5\mathbf{i} - \mathbf{j})$
- (c) Impulse received M is $m(-5\mathbf{i} - \mathbf{j})$
- (d) Impulse received by M is $m(5\mathbf{i} + \mathbf{j})$

Answer: B,C

Biology

Q502.

Root pressure is a

- (a) Pulling force
- (b) Pushing force
- (c) Both A and B
- (d) None of these

Answer: B

Q503.

Flower is a modified shoot as

- (a) Thalamus may elongate to show internodes.
- (b) There is aggregation into inflorescence.
- (c) It bears essential organs.
- (d) It may have epicalyx.

Answer: A

Physics

Q504.

A plane changes its pitch so that it is flying 100 m / s in an area of the Earth's magnetic field that is perpendicular to the plane and equal to 5.010^5 T. What is the emf induced between the wing tips? Assume that the plane was previously flying parallel to the earth's magnetic field.

- (a) 0.12 v
- (b) 0.20
- (c) 1.2 v
- (d) 1.2 v D

Answer: A

Biology

Q505.

In humans, the concentration of carbon dioxide in the plasma

- (a) Causes increased production of hydrochloric acid
- (b) Regulates gastric acid production by forming carbonic acid
- (c) Regulates breathing rate by its effect on the medulla
- (d) Causes inflammation of the tissues of the bronchial tubes

Answer: C

Physics

Q506.

The gravitational potential difference between the surface of a planet and a point 20 m above it is 16 J / kg. Then the work done in moving a 2 kg mass by 8 m on a slope 60 degree from the horizontal, is:

- (a) 11.1 J
- (b) 5.5 J
- (c) 16 J
- (d) 27.7 J

Answer: A

Q507.

The photoelectric effect represents that

- (a) Light has a particle nature
- (b) Electron has a wave nature

- (c) Proton has a wave nature
(d) None of the above

Answer: A

Chemistry

Q508.

Hair shampoos belong to which class of synthetic detergent?

- (a) Anionic
(b) Cationic
(c) Neutral
(d) None of these

Answer: A

Physics

Q509.

The equation of an equipotential line in an electric field is $y=2x$, then the electric field strength vector at (1,2) may be: A .4 i+3 j

- (b) 4 i+8 j
(c) 8 i+4 j
(d) -8 i+4 j

Answer: D

Q510.

Which of the following is different from the rest?

- (a) bar
(b) torr
(c) kg-wt
(d) Pa

Answer: C

Biology

Q511.

Deuterostomial coelomates are

- (a) Chordates
(b) Mollusk
(c) Arthropods
(d) Annelids

Answer: A

Chemistry

Q512.

The IUPAC name of $[\text{Ni}(\text{NH}_3)_4][\text{NiCl}_4]$ is :

- (a) tetrachloronickel (II)-tetraaminenickel (II)
(b) tetramminenickel (II)-tetrachloronickel (II)
(c) tetraamminenickel (II)-tetrachloronickelate (II)
(d) tetrachloronickel (II)-tetraaminenickelate (0)

Answer: C

Biology

Q513.

Which of the following is not true about photosynthesis?

- (a) In non-cyclic photophosphorylation, water is split and oxygen is released
- (b) Photosystem-II makes noncyclic photophosphorylation more efficient than does cyclic photophosphorylation
- (c) Cyclic system is used when there is insufficient N A D P⁺ present to absorb electrons in plant cells
- (d) More carbohydrate is produced during cyclic photophosphorylation than during non-cyclic photophosphorylation

Answer: D

Physics

Q514.

Sensitivity of a moving coil galvanometer can be increased by Fill in the blank.

- (a) decreasing number of turns in rectangular coil
- (b) decreasing magnetic induction of magnetic field
- (c) increasing area of rectangular coil
- (d) increasing twist constant of phosphor bronze fiber

Answer: C

Q515.

According to Maxwell's hypothesis, changing of electric filed give rise to

- (a) magnetic field
- (b) pressure gradient
- (c) charge
- (d) voltage

Answer: A

Chemistry

Q516.

10 ml of CO is mixed with 25 ml air having 20 per cent O₂ by volume. What would be the final volume if none of CO and O₂ is left after the reaction?

- (a) 30 m l
- (b) 40 m
- (c) 43 ml
- (d) 49 m

Answer: A

Q517.

The IUPAC name of:

- (a) 2 -methyl-3-bromo hexanal
- (b) 3-bromo-2-methyl butanal
- (c) 2 -bromo-3-bromo butanal
- (d) 3-bromo-2-methyl pentanal

Answer: D

Physics

Q518.

Refraction of light in the eye occurs at

- (a) The pupil only
- (b) The cornea only
- (c) The lens only
- (d) Both the cornea and the lens

Answer: D

Chemistry

Q519.

Copper metal is used in making:

- (a) electric wires
- (b) fuse wires
- (c) filling cavity in teeth
- (d) preparation of hydrogen

Answer: A

Q520.

$B e^{2+}$ is isoelectronic with which of the following ions?

- (a) H^+
- (b) Li^+
- (c) Na^+
- (d) Mg^{2+}

Answer: B

Physics

Q521.

The radii and Young's moduli of two uniform wires A and B are in the ratio 2: 1 and 1: 2 respectively. Both wires are subjected to the same longitudinal force. If the increase in length of the wire A is one percent, the percentage increase in length of the wire B is:

- (a) 1.0
- (b) 1.5
- (c) 2.0
- (d) 3.0

Answer: D

Q522.

A swimmer while jumping into water from a height easily forms a loop in air, if

- (a) He pulls his arms and leg in
- (b) He spreads legs and his arms
- (c) He keeps himself straight
- (d) His body is so formed

Answer: A

Chemistry

Q523.

To prevent food from spoilage by microorganism, which substance is used?

- (a) Aspartame
- (b) Arneto
- (c) Salt of sorbic acid

(d) Tetrazine

Answer: C

Physics

Q524.

Sun radiates energy at the rate of 3.6×10^{26} Joule/s. The rate of decrease in mass of the sun is (in kgs^{-1}) A .12 $\times 10^{10}$

- (b) 1.3×10^{20}
- (c) 4×10^9
- (d) 3.6×10^{36}

Answer: C

Biology

Q525.

Gluconeogenesis is controlled by

- (a) Cortisol
- (b) Corticosterone
- (c) Thyroxine
- (d) All of the above

Answer: D

Q526.

A zygote which has an X chromosome inherited from the father will develop into a A . B O y

- (b) Girl
- (c) x chromosome does not determine the sex of a child
- (d) Either boy or girl

Answer: B

Chemistry

Q527.

A proton and an α -particle are accelerated through the same potential difference. The ratio of their De Broglie wavelength is:

- (a) $\sqrt{2}$
- (b) $(1)/(\sqrt{2})$
- (c) $2 \sqrt{2}$
- (d) 2

Answer: A

Q528.

Choose the correct answer from the alternative given. Complete the following reactions. (i) $\text{MnO}_4^- + 2\text{H}_2\text{O} + 3\text{e}^- \rightarrow +40\text{H}^-$ (i i) $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow +4\text{H}_2\text{O}$ (i i i) $\text{MnO}_4^- + \text{e}^- \rightarrow \text{A} \cdot \text{MnO}_2, \text{Mn}^{2+}, \text{MnO}_4^{2-}$

- (b) $\text{Mn}^{2+}, \text{MnO}_2, \text{MnO}_4^{2-}$
- (c) $\text{MnO}_4^{2-}, \text{Mn}^{2+}, \text{MnO}_4^{2-}$
- (d) $\text{MnO}_2, \text{MnO}_4^{2-}, \text{Mn}^{2+}$

Answer: A

Q529.

When molten sulphur is suddenly cooled by pouring into water, it takes the form of :

- (a) Milk of sulphur.

- (b) Colloidal sulphur
- (c) Flower of sulphur.
- (d) Plastic sulphur.

Answer: D

Q530.

The following general structure is representative of sulphonamides. Which of the following statements is true for active sulphonamides? A · R¹ can be H or an alkyl group

- (b) R² must be hydrogen
- (c) The aromatic ring is essential
- (d) The sulphonamide functional group can be replaced with an ester

Answer: C

Q531.

Which of the following factor generally favours electrovalency? This question has multiple correct options

- (a) Low charge on ions
- (b) High charge on ions
- (c) Large cation and small anion
- (d) Small cation and large anion

Answer: A,D

Physics

Q532.

A: Coloured spectrum is seen when we look through a cloth R: Diffraction of light takes place when light is travelling through the pores of cloth

- (a) Both A and R are true, and R is not correct explanation of A
- (b) Both A and R are true, and R is correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

Answer: B

Chemistry

Q533.

The drug cimetidine was designed to: This question has multiple correct options

- (a) prevent the interaction of histamine with receptors
- (b) stimulates the secretion of pepsin
- (c) reduce gastro intestinal juices
- (d) all of the above

Answer: A,C

Physics

Q534.

The acceleration of a particle which moves along the positive x-axis varies with its position as shown. If the velocity of the particle is 0.8 m / s at x=0, the velocity of the particle at x= 1.4 is (in m / s)

- (a) 1.6
- (b) 1.2
- (c) 1 .
- (d) None of these

Answer: B

Biology

Q535.

Taste buds for bitter taste are found on tongue at

- (a) Tip
- (b) On basal surface
- (c) Posterior part
- (d) Lateral sides

Answer: C

Q536.

If following data belongs to an adult male Cardiac index = cardiac output / Total body surface area in adult = 1.7 m^2 Cardiac index $4 \text{ litre / mt. / m}^2$ Heart rate 68 / minute End systolic volume Total body surface area -1.7 m^2 Then choose the correct set of option.

- (a) cardiac output = 5 litre/minute. ; stroke volume = 70 ml . ; EDV=120 ml
- (b) c.O. = 5 litre/minute. ; SV = 100 ml.; EDV = 50 ml

Answer: A

Chemistry

Q537.

Shape of CO_2 is:

- (a) pyramidal
- (b) triangular
- (c) linear
- (d) spherical

Answer: C

Biology

Q538.

In catabolic series of reaction, complex molecules are broken down into smaller molecules releasing

- (a) water
- (b) Oxygen
- (c) Energy
- (d) Nutrients

Answer: C

Physics

Q539.

Due to air a falling body faces a resistive force proportional to square of velocity v , consequently its effective downward acceleration is reduced and is given by $a = g - kv^2$ where $k = 0.002 \text{ m}^{-1}$. The terminal velocity of the falling body is (in m / s)

- (a) 60
- (b) 70
- (c) 80
- (d) 90

Answer: B

Chemistry

Q540.

NaHCO_3 can be used on its own to make cakes or bread. Baking powder is more commonly used, and contains NaHCO_3 , $\text{Ca}(\text{H}_2\text{PO}_4)_2$ and starch. An improved combination baking powder contains about 40 % starch, 30 % NaHCO_3 , 20 % $\text{NaAl}(\text{SO}_4)_2$ and 10 % $\text{Ca}(\text{H}_2\text{PO}_4)_2$. Answer the following questions based on above study. In improved baking powder, $\text{NaAl}(\text{SO}_4)_2$ is used for :

- (a) makes the decomposition of NaHCO_3 faster
- (b) slows the decomposition of NaHCO_3
- (c) acts as filler
- (d) makes the bread rise

Answer: B

Physics

Q541.

Generators used in power stations to generate electricity are

- (a) DC generators
- (b) AC generators
- (c) Both
- (d) None

Answer: B

Chemistry

Q542.

In a certain Bohr orbit, the total energy is -4.9 eV . For this orbit, the kinetic energy, and potential energy are respectively:

- (a) 9.8 eV , -4.9 eV
- (b) 4.9 eV , -4.9 eV
- (c) 4.9 eV , -9.8 eV
- (d) 9.8 eV , -9.8 eV

Answer: B

Physics

Q543.

Gravitational constant is experimentally measured using a

- (a) physical balance
- (b) torsional balance
- (c) beam balance
- (d) gyrometer

Answer: B

Chemistry

Q544.

Mg is present in

- (a) Chlorophyll

- (b) Haemoglobin
- (c) Vitamin-12
- (d) Vitamin-B

Answer: A

Physics

Q545.

A Gaussian surface in the figure is shown by dotted line. The electric field on the surface will be:

- (a) due to q_1 and q_2 only
- (b) due to q_2 only
- (c) zero
- (d) due to all

Answer: D

Q546.

A microscope is used with sodium light and its resolving power is not sufficiently large. Higher resolution will be obtained by using wavelength of

- (a) 20 micron
- (b) 2 micron
- (c) 1 micron
- (d) $400 \text{ \AA}$

Answer: D

Biology

Q547.

Pepsin acts in

- (a) Basic medium
- (b) Acidic medium
- (c) Neutral medium
- (d) All types of media

Answer: B

Physics

Q548.

An image formed by a plane mirror is:

- (a) erect and smaller than the object
- (b) erect and of same size as the object
- (c) inverted and larger than the object
- (d) inverted and smaller than the object

Answer: B

Chemistry

Q549.

Which of the following statements is/are correct? This question has multiple correct options

- (a) Molecular formula or molecular mass of a gaseous hydrocarbon can be determined even without knowing their percentage composition by audiometry

- (b) In Lassaigne's test, N and S both present in the organic compounds are converted into CNS^- ion. $\text{K}_2\text{S O}_4$ and Cu S O_4 are added in Kjeldahl's method. $\text{K}_2\text{S O}_4$ acts as a catalyst while Cu S O_4 raises the boiling point of $\text{H}_2\text{S O}_4$.
- (d) Layer test is used to distinguish Cl^- and Br^- ions.

Answer: A,B,D

Q550.

Chlorofluorocarbons (CFCs) are widely used in air conditioners, refrigerators etc, because of being:

- (a) highly reactive
- (b) flammable
- (c) non-reactive
- (d) all of these

Answer: C

Q551.

In long form of periodic table, the total number of periods is:

- (a) 5
- (b) 7
- (c) 8
- (d)

Answer: B

Q552.

Predict if there will be any precipitate by mixing 50 mL of 0.01 M NaCl and 50 mL of 0.01 M AgNO_3 solution. The solubility product of AgCl is 1.5×10^{-10} .

- (a) since ionic product is greater than solubility product no precipitate will be formed
- (b) since ionic product is lesser than solubility product. precipitation will occur
- (c) since ionic product is greater than solubility product. precipitation will occur.
- (d) since ionic product and solubility product are same, precipitation will not occur

Answer: C

Physics

Q553.

Under same depth, atmospheric pressure is water pressure.

- (a) less than
- (b) greater than
- (c) equal to
- (d) none of these

Answer: A

Q554.

A wire carrying a current of 100 A is bent into the form of a circle of radius 5cm. The flux density at the centre of the coil is Wb/m^2 tabular 12.57×10^{-5} tabular

- (b) 125.7×10^{-5}
- (c) 1.257×10^{-5}
- (d) 1257×10^{-5}

Answer: B

Q555.

A body is accelerated by applying a force of 30 N. The change in the momentum of the body after 2 sec is?

- (a) 7.5 kg-m / s
- (b) 30 kg-m / s
- (c) 120 kg-m / s

(d) 60 kg-m / s

Answer: D

Biology

Q556.

Obturator foramen is found in

- (a) Pectoral girdle of rabbit
- (b) Pectoral girdle of frog
- (c) Pelvic girdle of rabbit
- (d) Pelvic girdle of frog

Answer: C

Physics

Q557.

In the two dimensional motions:

- (a) x-t graph gives actual path of the particle
- (b) y-t graph gives actual path of the particle
- (c) $\sqrt{x^2+y^2}$ versus t graph gives the actual path of the particle
- (d) y-x graph gives actual path of particle

Answer: D

Q558.

The resistance of an iron wire is $10\ \Omega$ and its temperature coefficient of resistance is $5 \times 10^{-3} / ^\circ\text{C}$. A current of 30 mA is flowing in it at 20^{circ}C . Keeping potential difference across its ends constant, if its temperature is increased to $120^{\text{circ}}\text{C}$ then the current flowing in the wire will be (in mA)

- (a) 20
- (b) 35
- (c) 10
- (d) 40

Answer: A

Biology

Q559.

During respiration there is

- (a) gain in dry weight
- (b) loss in dry weight
- (c) no change in dry weight
- (d) increase in the overall weight

Answer: B

Chemistry

Q560.

If three elements X, Y, and Z form a Dobereiner's triad and atomic weights of X and Z are 9 and 40 respectively then the atomic weight of the element Y is approximately:

- (a) 24.
- (b) 49

(c) 34.5

(d) 29

Answer: A

Q561.

Which of the following substituents will decrease the acidic strength of phenol?

(a) $-\text{NO}_2$

(b) $-\text{CN}$ $-\text{CH}_3$

(d) $-\text{CHO}$

Answer: C

Physics

Q562.

Who proposed the concept of Inertia?

(a) Galileo

(b) Newton

(c) Pascal

(d) Einstein

Answer: A

Chemistry

Q563.

Which of the following is a nonbiodegradable pollutant?

(a) DDT

(b) Sewage

(c) Plant leaves

(d) Domestic wastes

Answer: A

Q564.

In which of the following reactions, the product obtained is chiral?

(a) $\text{CH}_3\text{COCH}_3 + \text{NaBH}_4$

(b) CH_3COCl Rosenmund's reduction

(c) $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3 + \text{HCl}$

(d) $\text{CH}_3\text{CH}_2\text{COCH}_3 + \text{LiAlH}_4$

Answer: D

Physics

Q565.

A passenger in moving train tosses a coin which falls behind him. It means that the motion of the train is

(a) Accelerated

(b) Uniform

(c) Retarded

(d) Circular motion

Answer: A

Q566.

Assertion A capacitor can be given only a limited quantity of charge Reason A capacitor is an arrangement which can store sufficient quantity of charge

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Assertion is incorrect but Reason is correct

Answer: C

Biology

Q567.

How many organisms in the list given below are autotrophs? Lactobacillus, Nostoc, Chara, Nitrosomonas Nitrobacter, Streptomyces, Sacharomyces Trypanosoma, Porphyra, Wolfia

- (a) Five
- (b) Six
- (c) Three
- (d) Four

Answer: B

Physics

Q568.

Center of gravity is usually located where

- (a) less mass is concentrated
- (b) less weight is concentrated
- (c) more mass is concentrated
- (d) more weight is concentrated

Answer: D

Q569.

Mercury boils at 356°C . However mercury thermometers are made such that they can measure temperatures up to 500°C . This is done by :

- (a) maintaining vacuum above the mercury column in the stem of the thermometer
- (b) filling nitrogen gas at high pressure above the mercury column
- (c) filling nitrogen gas at low pressure above the mercury column
- (d) filling oxygen gas at high pressure above the mercury column

Answer: B

Q570.

The static frictional force is

- (a) self adjustable.
- (b) not self adjustable.
- (c) scalar quantity.
- (d) equal to the limiting force.

Answer: A

Biology

Q571.

In autogenic succession

- (a) Community itself modifies its own environment, thus causing its own replacement by new community
- (b) Early dominance of heterotrophs
- (c) Replacement of existing communities due to external conditions
- (d) Early and continued dominance of autotrophic organisms

Answer: A

Chemistry

Q572.

Which of the following acts as a biological catalyst?

- (a) Lipids
- (b) Proteins
- (c) carbohydrates
- (d) Nucleic acids

Answer: B

Q573.

The atomic number of potassium is 19. Hence the number of electrons in N shell will be

- (a) 9
- (b)
- (c) 8
- (d) 2

Answer: C

Biology

Q574.

A typical gymnospermous plant has 8 chromosomes in leaf cells. What will be the number of chromosomes in the cells of its gametophyte?

- (a) 16
- (b) $8 \div 4$
- (d) 24

Answer: C

Chemistry

Q575.

Due to the nature of catenation, carbon can form:

- (a) Open chain compounds as well as branched chain compounds.
- (b) Open chain compounds as well as closed chain compounds.
- (c) Both A and B are correct
- (d) None of these.

Answer: C

Physics

Q576.

The area of cross section of the wider tube shown in figure is 900 cm^2 . If the boy standing on the piston weighs 45 kg, the difference in the levels of water in the two tubes.

- (a) 50 cm

- (b) 25cm
- (c) 75 cm
- (d) 100 cm

Answer: B

Q577.

An electromagnetic wave has energy in the form of

- (a) variable electric field
- (b) variable magnetic field
- (c) both A and B
- (d) none of the above

Answer: C

Chemistry

Q578.

Which of the following statements is/are true? This question has multiple correct options

- (a) It is impossible to satisfy the octet rule for all atoms in XeF_2
- (b) MgSO_4 is soluble in water because hydration energy of MgSO_4 is higher in comparison to its lattice energy.
- (c) The bond in NO^+ should be stronger than the bond in NO^-
- (d) For ozone molecule, one oxygen-oxygen bond is stronger than the other oxygen-oxygen bond.

Answer: B,C

Q579.

What is geothermal energy?

- (a) Ocean thermal energy
- (b) Interior heat of the earth.
- (c) Energy obtained from the heat of hot dry rocks
- (d) None of these

Answer: B

Physics

Q580.

Figures (a),(b),(c) and (d) respectively correspond to

- (a) The short-sighted eye, the correction of long-sight, the long-sighted eye and the correction of short-sight
- (b) The short-sighted eye, the correction of short-sight, the long-sighted eye and the correction of long-sight
- (c) The long-sighted eye, correction of short-sight, the short-sighted eye and the correction of long-sight
- (d) The long-sighted eye, the correction of long-sight, the long-sighted eye and the correction of short-sight

Answer: B

Q581.

A soap bubble of radius 3 cm is charged with surface charge density σ such that $(\sigma^2)/(\epsilon_0) = 0.4 \text{ SI units}$. The excess pressure inside the bubble in SI unit (surface tension of soap solution $= 3 \times 10^{-3} \text{ N/m}$) is:

- (a) 0
- (b) 0.2
- (c) 0.5
- (d) 0.25

Answer: B

Chemistry

Q582.

toppr OG Q Type your question- carried out slowly and reversibly. Starting at state a, where its volume and pressure are (V_0, p_0) , the gas undergoes an isothermal expansion until it reaches state b where its volume and pressure are $(8V_0, p_0/8)$. In process $b \rightarrow c$ the gas is cooled at constant volume to $(8V_0, p_0/32)$. Process $c \rightarrow a$ is an adiabatic compression back to state a. Select incorrect statement: This question has multiple correct options

- (a) $a \rightarrow b$, there is no change in internal energy of gas
- (b) $c \rightarrow a$, there is no change in entropy of gas
- (c) $b \rightarrow c$, there is no work done by gas
- (d) $a \rightarrow b \rightarrow c \rightarrow a$, there is work done by gas in cyclic process

Answer: D

Physics

Q583.

A 20 kg load is suspended from the lower end of a wire 10 cm long and 1 mm^2 in cross sectional area. The upper half of the wire is made of iron and the lower half with aluminium. The total elongation in the wire is $(Y_{\text{iron}} = 20 \times 10^{10} \text{ N/m}^2, Y_{\text{Al}} = 7 \times 10^{10} \text{ N/m}^2)$

- (a) $1.92 \times 10^{-4} \text{ m}$
- (b) $17.8 \times 10^{-3} \text{ m}$
- (c) $1.78 \times 10^{-3} \text{ m}$
- (d) $1.92 \times 10^{-3} \text{ m}$

Answer: A

Chemistry

Q584.

When ethyl bromide is heated with silver nitrate the product obtained is:

- (a) ethyl nitrite
- (b) ethyl nitrate
- (c) nitropropane
- (d) none of the above

Answer: B

Q585.

Which has highest bond energy and stronger bond? A. F_2

- (b) Cl_2
- (c) Br_2
- (d) I_2

Answer: B

Q586.

The freezing point of a solution containing 0.2 g of acetic acid in 20.0 g benzene is lowered by 0.45°C . The degree of association of acetic acid in benzene is: [Assume acetic acid dimerizes in benzene and K_f for benzene $= 5.12 \text{ K} \cdot \text{kg mol}^{-1}$]

- (a) 94.5 %
- (b) 54.9 %
- (c) 78.2 %
- (d) 100%

Answer: A

Q587.

Trisilylamine $[\text{N}(\text{SiH}_3)_3]$ has a

- (a) planar geometry

- (b) tetrahedral geometry
- (c) pyramidal geometry
- (d) none of these

Answer: A

Physics

Q588.

The B-H curve (i) and (ii) shown in fig., associated with

- (a) (i) diamagnetic and (ii) paramagnetic substance
- (b) (i) paramagnetic and (ii) ferromagnetic substance
- (c) (i) soft iron and (ii) steel respectively
- (d) (i) steel and (ii) soft iron respectively

Answer: D

Q589.

The P E of a 2 k g particle, free to move along x -axis is given by $V(x) = ((x^3)/(3) - (x^2)/(2))$ J . The total mechanical energy of the particle is 4 J . Maximum speed (in m s^{-1}) is $A \cdot (1)/(\sqrt{2})$

- (b) $\sqrt{2}$
- (c) $(3)/(\sqrt{2})$
- (d) $(5)/(\sqrt{6})$

Answer: C

Chemistry

Q590.

$[\text{Cr}(\text{NH}_3)_6][\text{Cr}(\text{SCN})_6]$ and $[\text{Cr}(\text{NH}_3)_2(\text{SCN})_4][\text{Cr}(\text{NH}_3)_4(\text{SCN})_2]$ are the examples of what type of isomerism?

- (a) Ionisation isomerism
- (b) Linkage isomerism
- (c) coordination isomerism
- (d) Solvate isomerism

Answer: C

Q591.

2- pentanone and 3- pentanone are:

- (a) Chain isomers
- (b) Tautomers
- (c) Metamers
- (d) Positional isomers

Answer: D

Q592.

The element with highest boiling point in the following is :

- (a) oxygen
- (b) Sulphur
- (c) Tellurium
- (d) Polonium

Answer: D

Q593.

Calculate the volume occupied by 40 g. of a CH_4 at S.T.P if its V . D . is 8

- (a) 24 litres
- (b) 46 litres
- (c) 34 litres
- (d) 56 litres

Answer: D

Physics

Q594.

A 10 W electric heater is used to heat a container filled with 0.5 kg of water. It is found that the temperature of water and the container rises by 3 K in 15 min. The container is then emptied, dried and filled with 2 kg of oil. The same heater now raises the temperature of container-oil system by 2 K in 20 min. Assuming that there is no heat loss in the process and the specific heat of water is $4200 \text{ J kg}^{-1} \text{ K}^{-1}$, the specific heat of oil in the same unit is equal to:

- (a) 1.50×10^3 ■ 2.55×10^3
- (c) 3.00×10^3
- (d) 5.10×10^3

Answer: B

Biology

Q595.

Incorrect statement in the following

- (a) Glomerulus is a group of fine blood capillaries
- (b) Loop of Henle is a part of malpighian body
- (c) Jatropa is the source of bio-diesel
- (d) Symptoms of uremia is swelling of feet

Answer: B

Q596.

Stomochord is found in

- (a) Urochordata
- (b) Hemichordata
- (c) cephalochordata
- (d) Both A and B

Answer: B

Q597.

Choose the correct answers from the alternatives given : Meiosis involves:

- (a) two nuclear divisions with chromosomes dividing only once
- (b) two nuclear divisions with chromosomes dividing twice
- (c) two nuclear divisions with chromosomes dividing four times
- (d) one nuclear division with chromosomes dividing only once

Answer: A

Physics

Q598.

The SI unit of gravitational potential is :

- (a) Joule/kg
- (b) Joule/kg

- (c) kg/Joule
- (d) Joule/kg²

Answer: A

Q599.

Alternating current is converted to direct current by

- (a) rectifier
- (b) dynamo
- (c) transformer
- (d) motor

Answer: A

Chemistry

Q600.

At constant temperature, the osmotic pressure of a solution is: This question has multiple correct options

- (a) directly proportional to the concentration
- (b) inversely proportional to the molar mass of solute
- (c) directly proportional to the square of the concentration
- (d) directly proportional to the square root of the concentration

Answer: A,B

Q601.

One mole of HNO_3 absorbs 5 moles of electrons in a redox reaction. The reduced product of this reaction is :

- (a) 1 mole N_2
- (b) 1 mole N_2O
- (c) 1 mole NO_2
- (d) 1 mole NH_3

Answer: D

Q602.

Silica gel is a very useful:

- (a) raw material for soap
- (b) drying agent
- (c) wetting agent
- (d) solvent

Answer: B

Biology

Q603.

Pea plants were more suitable than cattle for Mendel's experiment because

- (a) There were no breeding records of cattles
- (b) Pea plants can be self-fertilised
- (c) cattle are not easy to maintain
- (d) All pea plants have 2n chromosomes and fewer genetic traits.

Answer: B

Chemistry

Q604.

Surfactants, having both cationic and anionic centres attached to the same molecule, are called:

- (a) cationic surfactant
- (b) anionic surfactant
- (c) zwitter-ionic surfactants
- (d) neutral surfactants

Answer: C

Physics

Q605.

A spherical mirror and a thin spherical lens have each a focal length of -15 cm. The mirror and lens are likely to be:

- (a) both concave
- (b) both Convex
- (c) the mirror is concave and lens is convex
- (d) the mirror is convex but lens is concave

Answer: A

Q606.

Assertion: Electromotive force is a force which helps the electrons to flow and produce current. Reason: Electromotive force is independent of the voltage across the cell.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: D

Q607.

A steady current is passing through a linear conductor of non-uniform cross-section. The net quantity of charge crossing any cross-section per second is

- (a) Independent of area of cross-section
- (b) Directly proportional to the length of the conductor
- (c) Directly proportional to the area of cross-section.
- (d) Inversely proportional to the length of the conductor

Answer: A

Q608.

Three point charges of $2q$, $-q$ and $-q$ are placed at the corners of an equilateral triangle of side a . Then: This question has multiple correct options.

- (a) the potential at the centroid of the triangle is zero
- (b) the electric field at the centroid of the triangle is zero
- (c) the dipole moment of the system is $\sqrt{2} q a \hat{j}$
- (d) the dipole moment of the system is $\sqrt{3} q a \hat{j}$

Answer: A,D

Q609.

Assertion: nm is not the same as mN . Reason: $\text{m} = 10^{-9} \text{ m}$ and $1 \text{ mN} = 10^{-3} \text{ N}$.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Assertion is incorrect but Reason is correct

Answer: A

Q610.

Two convex lenses have focal lengths 10 cm and 40 cm . If a telescope has to be made by using these two lenses 1. The distance between the lenses should be 50 cm 2. The distance between the lenses should be 30 cm 3. The magnifying power of this telescope will be 5 4. An inverted image will be produced in this telescope Which two of the give are correct?

- (a) 1 and 4
- (b) 2 and 4
- (c) 1 and 3
- (d) 3 and 4

Answer: A

Chemistry

Q611.

The decomposition of an equal mole of two reactants P and Q obey 1 st order kinetics with half-life 54 and 18 minutes respectively. If both are allowed to run for 54 minutes. The ratio of mole of Q and P left after decomposition will be:

- (a) 3
- (b) $\frac{1}{4}$
- (c) 4
- (d) 16

Answer: B

Biology

Q612.

Which one of the following animals has a nervous system but no brain?

- (a) Earthworm
- (b) Hydra
- (c) Amoeba
- (d) cockroach

Answer: B

Chemistry

Q613.

The number of faradays of electricity required to decompose 100 ml water (density =0.99 gm / ml) is:

- (a) 2
- (b) 11
- (c) 100
- (d) 5.5

Answer: D

Biology

Q614.

Tidal volume of air in a normal healthy man during inspiration is about

- (a) 300-400ml
 - (b) 500-700ml
 - (c) 900-1000 ml
 - (d) 100-250ml
-

Answer: B

Physics

Q615.

When a divergent beam of light is incident on a plane mirror, the image formed is

- (a) Upright and real
- (b) Upright and virtual
- (c) Inverted and virtual
- (d) Inverted and real

Answer: B

Chemistry

Q616.

Which of the following has the highest boiling point?

- (a) Ne ■. ■■
- (c) C H₄
- (d) xe

Answer: D

Physics

Q617.

are used in making electric generators and electric motors.

- (a) electric current
- (b) magnetes
- (c) cables
- (d) all

Answer: B

Chemistry

Q618.

2 moles of P C I₅ when heated in a closed vessels of 2 litre capacity at equilibrium 40 % of P C I₅ dissociated in P C I₃ and C I₂ . What is the value of the equilibrium constant?

- (a) 0.267
- (b) 0.786
- (c) 0.345
- (d) 1.879

Answer: A

Physics

Q619.

is the distance between consecutive corresponding points of the same phase, such as crests, troughs, compressions, rarefactions or zero crossings and is a characteristic of both travelling waves and standing waves, as well as other spatial wave patterns.

- (a) frequency

- (b) amplitude
- (c) wavelength
- (d) none of these

Answer: C

Q620.

An electron accelerated by 200 V enters a magnetic field .If its velocity is 100 m / sec . then (e / m) for it will be:
(C / Kg)

- (a) 1.75×10^{10}
- (b) 1.75×10^{11}
- (c) 1.75×10^9
- (d) 1.75×10^6

Answer: B

Q621.

A steady current of magnitude I and an A C current of peak value I are allowed to pass through identical resistor for the same time. The ratio of heat produced in the two resistors will be :

- (a) 2: 1
- (b) 1: 2
- (c) 1: 1
- (d) None of these

Answer: A

Q622.

The wavelength of a certain electron transition in the hydrogen spectrum is 4864 Å. Identify the transition.

- (a) Third line Balmer
- (b) First line Lyman
- (c) First line Paschen
- (d) second line Balmer

Answer: D

Chemistry

Q623.

An electron in the hydrogen atom is excited from 1st orbit to 5th orbit. Find the change in energy of the electron in e V:

- (a) $(13.6 \times 24) / 25$
- (b) $(13.6 \times 4) / 5$ c $-(0.529 \times 4) / 5$
- (d) $(13.6 \times 4) / 9$

Answer: A

Q624.

All alkane forming isomers if the number of least carbon atom is :

- (a) 1
- (b) 2 c .3 D

Answer: C

Biology

Q625.

Which of these is an example for zygomorphic flower with imbricate aestivation?

- (a) Calotropis

- (b) Mustard
- (c) canna
- (d) cassia E. Cucumber

Answer: D

Physics

Q626.

Large astronomical telescopes always use as objective

- (a) lens
- (b) mirror
- (c) combinations of lenses
- (d) none of the above

Answer: B

Biology

Q627.

What is a rhytidome?

- (a) It is a bark
- (b) Dead tissue separated by periderm on outside
- (c) Dead tissue separated by periderm on inner side
- (d) Secondary cortex

Answer: B

Chemistry

Q628.

Cassiterite ore consists of magnetic impurity named as:

- (a) chromite
- (b) wolframite
- (c) magnetite
- (d) Limonite

Answer: B

Physics

Q630.

On the earth surface, 'g' is a vector and its direction is oriented towards the centre of the A · body

- (b) sun
- (c) earth
- (d) none of these

Answer: C

Q631.

Assertion The average mass of one Mg atom is 24.305 amu, which is not equal to the actual mass of one Mg atom. Reason Three isotopes, ^{24}Mg , ^{25}Mg and ^{26}Mg of Mg can be found in nature.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect

(d) Assertion is incorrect but Reason is correct

Answer: A

Q632.

Assertion Path of a particle in SHM is always a straight line. Reason All straight line motions are not simple harmonic.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Assertion is incorrect but Reason is correct

Answer: B

Chemistry

Q633.

Out of $\text{Ni}(\text{CO})_4$, $\text{Ni}(\text{CN})_4^{2-}$ and NiCl_4^{2-} select paramagnetic and diamagnetic. A. $\text{Ni}(\text{CO})_4$ and $\text{Ni}(\text{CN})_4^{2-}$ are diamagnetic

- (b) NiCl_4^{2-} is paramagnetic
- (c) Both (A) and (B) Correct
- (d) None of these

Answer: C

Physics

Q634.

Assertion The density of water remains constant as it is cooled from room temperature until its temperature reaches 4°C Reason Below 4°C , the density increases.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: D

Q635.

A cell sends a current through a resistance R for time t; next the same cell sends current through another resistance r for the heat is developed in both the resistance, then the internal resistance of the cell is:

- (a) $(R+r)/2$
- (b) $(R-r)/2$
- (c) $\sqrt{Rr}$
- (d) $\sqrt{(R+r)/2}$

Answer: C

Chemistry

Q636.

Some gaseous equilibrium are following: $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$ K_1 $2\text{CO} + \text{O}_2 \rightleftharpoons 2\text{CO}_2$ K_2 $2\text{H}_2 + \text{O}_2 \rightleftharpoons 2\text{H}_2\text{O}$ then find out the relation between equilibrium constants :- A. $K = K_1 K_2$

- (b) $K = (K_1 K_2)^2$
- (c) $K = (K_1 K_2)^{1/2}$
- (d) $K = (K_1 / K_2)^{1/2}$

Answer: D

Physics

Q637.

A uniform sphere of radius R is placed on a smooth horizontal surface and a horizontal force F is applied on it at a distance h above the centre. The acceleration of the centre of mass of the sphere is

- (a) maximum when $h=0$
- (b) maximum when $h=R$
- (c) maximum when $h=R/2$
- (d) independent of h

Answer: D

Chemistry

Q638.

In isomeric alkyl halides:

- (a) the branch chain isomer have relatively high boiling point as compared to its straight chain isomer
- (b) the branch chain isomer have relatively low boiling point as compared to its straight chain isomer
- (c) the branch chain isomer boiling point equal to its straight chain isomer
- (d) none of these

Answer: B

Q639.

Examples of lyophilic colloids are: This question has multiple correct options

- (a) albumin ■. R■■■
- (c) smoke
- (d) metalsulphide

Answer: A,B

Physics

Q640.

How should people wearing spectacles work with a microscope?

- (a) They should keep on wearing their spectacles
- (b) They should never use the microscope
- (c) They should take off their spectacles
- (d) They may either put on their spectacles or they may take off their spectacles

Answer: D

Chemistry

Q641.

Boron can not form more than 4 bonds because: This question has multiple correct options

- (a) of unavailability of d-orbitals
- (b) of unavailability of empty p-orbitals
- (c) Boron is so smaller
- (d) it does not have more valence electrons

Answer: A,C,D

Q642.

Two molecules indicated below are capable to intramolecular H-bonding. Which is likely to form more stable hydrogen bonds? A

- (b) II
- (c) Both are equally capable
- (d) Cannot be predicted

Answer: B

Q643.

Assertion Element can form different compound. Reason Any substance that contains only one kind of an atom is known as an element.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: B

Physics

Q644.

Which of the following applications is not based on use of ICs?

- (a) Microprocessor chip.
- (b) Laptops and personal computers.
- (c) Mobile phones.
- (d) Transformers

Answer: D

Chemistry

Q645.

Assertion Adsorption is exothermic and spontaneous inspite of the fact that adsorption is accompanied with decrease in entropy. Reason The factor $T \Delta S$ is lesser than ΔH

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Assertion is incorrect but Reason is correct

Answer: A

Physics

Q646.

Imagine an atom made up of proton and a hypothetical particle of double the mass of the electron but having the same charge as the electron. Apply the Bohr atom model and consider all possible transitions of this hypothetical particle to the first excited level. The longest wavelength photon that will be emitted has wavelength λ (give in term of the Rydberg constant R for the hydrogen atom) equal to $A \cdot 9/5 R$

- (b) $36/5 R$
- (c) $18/5 R$
- (d) $4/R$

Answer: B

Chemistry

Q647.

$\text{C}_6\text{H}_5-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{NH}_2$ The IUPAC name of the compound is :

- (a) 1- Amino -1 - phenyl - 2 - methyl propane
- (b) 2- Methyl-1- phenyl propan-1 - amine
- (c) 2- Methyl-1- amino -1 - phenyl propane
- (d) 1- -Isopropyl -1 - phenyl methyl amine

Answer: B

Physics

Q648.

The electromagnetic waves do not transport

- (a) energy
- (b) charge
- (c) momentum
- (d) information

Answer: B

Chemistry

Q649.

What is the molecular mass of glucose $\text{C}_6\text{H}_{12}\text{O}_6$ molecule up to 6 significant figures?

- (a) 0180.16
- (b) 180.16
- (c) 180.1620
- (d) 180.162

Answer: D

Biology

Q650.

Darwin was most influenced by :

- (a) Lamarck's theory of acquired characters
- (b) Weisman's theory of origin of species
- (c) Wallace's theory of origin of species
- (d) Essay on Population by Malthus

Answer: D

Chemistry

Q651.

Helium is used in balloons in place of hydrogen because it is:

- (a) Incombustible
- (b) Lighter than hydrogen
- (c) Radioactive
- (d) More abundant than hydrogen

Answer: A

Q652.

Urea on reacting with water gives:

- (a) N H_3 ■. $\text{N}_2 \text{O}_3$
(c) H N O_2
(d) H N O_3

Answer: A

Physics

Q653.

How is the unit bar related to the S.I. unit pascal?

- (a) $1 \text{ bar} = 10^5 \text{ pascal}$
(b) $1 \text{ pascal} = 10^5 \text{ bar}$
(c) $1 \text{ bar} = 10 \text{ pascal}$
(d) $1 \text{ bar} = 1 \text{ pascal}$

Answer: A

Q654.

The electromagnetic waves travel in free space with the velocity of

- (a) Sound
(b) Light
(c) Greater than that of light
(d) Greater than that of sound

Answer: B

Chemistry

Q655.

Identify the incorrect statement(s) with respect to CO : This question has multiple correct options

- (a) it combines with water to give carbonic acid.
(b) it combines with haemoglobin of the blood.
(c) it is a powerful oxidising agent.
(d) CO is weak field ligand.

Answer: A,C,D

Physics

Q656.

Work done in moving an object through an equipotential surface is

- (a) Positive
(b) Negative
(c) zero
(d) Depends on the field direction

Answer: C

Chemistry

Q657.

A student takes 2 mL acetic acid in a dry test tube and adds a pinch of sodium hydrogen carbonate to it. He makes the following observations: I. A colourless and odourless gas evolves with a brisk effervescence. II. The gas turns lime water milky when passed through it. III. The gas burns with an explosion when a burning splinter is brought near it. IV. The gas extinguishes the burning splinter that is brought near it. The correct observations

are:

- (a) I, II, III and IV
- (b) II, III and IV
- (c) I, II, III and IV
- (d) IV, I and II

Answer: D

Biology

Q658.

Cross bridges between actin and myosin is broken up by

- (a) Hydrolysis of AP
- (b) Binding of ATP to the myosin head
- (c) Binding of calcium to the subunit of troponin
- (d) Exposure of tropomyosin

Answer: B

Physics

Q659.

Assertion A body on an inclined plane just tends to slide down if the angle of inclination is equal to angle of friction. Reason When the angle of inclination is equal to the angle of friction, the component of weight parallel to the plane is just sufficient to overcome the frictional force.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: A

Biology

Q660.

Who is called as 'father of Indian plant physiology'?

- (a) J. C. Bose
- (b) Calvin
- (c) R. Mishra
- (d) K. K. Nanda

Answer: A

Chemistry

Q661.

A system is said to be in thermodynamic equilibrium with Surrounding if

- (a) it is only in thermal equilibrium with surrounding.
- (b) it is in both thermal and chemical equilibrium with surrounding.
- (c) it is in thermal, chemical as well as mechanical equilibrium with the surrounding.
- (d) it is in thermal and mechanical equilibrium, but not in chemical equilibrium with surrounding.

Answer: C

Q662.

Maximum change in p H is observed when 50 of 1 M K O H solution is added to

- (a) 25 ml. of 2.5 N HCl
- (b) 50 ml of 1 N HCl
- (c) 30 ml of 2 N HCl
- (d) 50 ml distilled water

Answer: D

Physics

Q663.

An object with a mass M is suspended from an elastic spring with a spring constant k. The object oscillates with period T. If the mass of oscillations is quadrupled, how it will change the period of oscillations?

- (a) The period is increased by factor two
- (b) The period is increased by factor four
- (c) The period is decreased by factor two
- (d) The period remains the same

Answer: A

Chemistry

Q664.

Which of the following is a property of nitrogen?

- (a) Supports combustion
- (b) Allows combustion
- (c) Supports but does not allow combustion
- (d) Neither supports nor allows combustion

Answer: D

Q665.

Which of these should be a characteristic of the semi permeable membrane used in ultra filtration method?

- (a) The membrane should be a polymer material
- (b) The pore size of the membrane should be very small
- (c) Both A and B
- (d) None of the above

Answer: C

Physics

Q666.

A block of metal weighing 2 k g is resting on a frictionless plane. It is struck by a jet releasing water at a rate of 1 k g / s and at a speed of 5 m / s .The initial acceleration of the block will be A .2 .5 m / s²

- (b) 0.5 m / s²
- (c) 10 m / s²
- (d) 20 m / s²

Answer: A

Q667.

Motion of a bus moving on a crowded road is an example of

- (a) uniform motion
- (b) non-uniform motion

- (c) either uniform or non-uniform motion
- (d) neither uniform nor non-uniform motion

Answer: B

Q668.

In Rutherford's α -rays scattering experiment, gold foils are used because of

- (a) high malleability
- (b) ductility
- (c) high melting point
- (d) high ionisation energy

Answer: A

Chemistry

Q669.

$A+B \rightleftharpoons C+D$, K_c for this reaction is 10. If 1,2,3,4 mole/litre of A, B, C and D respectively are present in a container at 25^oC, the direction of reaction will be:

- (a) from left to right
- (b) from right to left
- (c) reaction is at equilibrium
- (d) unpredictable

Answer: B

Q670.

Which of these is true for most of the organic compounds?

- (a) Low melting and boiling points
- (b) High melting and boiling points
- (c) Low melting point but high boiling point
- (d) High melting point but low boiling point

Answer: A

Q671.

Discharge of industrial waste into the soil causes

- (a) soil pollution
- (b) air pollution
- (c) water pollution
- (d) all of above

Answer: A

Physics

Q672.

The correct way to represent an unit amount of force is :

- (a) 1 newton
- (b) 1 Newton
- (c) 1 NEWTON
- (d) None of these

Answer: A

Biology

Q673.

Chief merit of Bentham and Hooker's classification is that

- (a) The system mostly based on evolutionary concepts.
- (b) It is a natural systems of classification of all groups of plants.
- (c) It also considered the phylogenetic aspects.
- (d) The description of taxa are based on actual examination of specimen.

Answer: D

Physics

Q674.

If the electric flux entering and leaving a closed surface are 6×10^6 and 9×10^6 S I units respectively, then the charge inside the surface of permittivity of free space ϵ_0 is

- (a) $\epsilon_0 \times 10^6$
- (b) $-2 \epsilon_0 \times 10^6$
- (c) $3 \epsilon_0 \times 10^6$
- (d) $2 \epsilon_0 \times 10^6$

Answer: D

Q675.

Weightlessness experienced in a spaceship is due to

- (a) absence of inertia.
- (b) absence of gravity
- (c) absence of accelerating force.
- (d) free fall of the spaceship

Answer: D

Chemistry

Q676.

Assertion Micelles are formed by surfactant molecules above the critical micellar concentration (CMC). Reason The conductivity of solution having surfactant molecules decreases sharply at the CMC.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Assertion is incorrect but Reason is correct

Answer: C

Biology

Q677.

The gas that usually causes explosion in coal mines is

- (a) methane
- (b) carbon monoxide
- (c) carbon dioxide
- (d) hydrogen

Answer: A

Chemistry

Q678.

1,4-dimethylbenzene on heating with anhydrous AlCl_3 and HCl produces :

- (a) 1,2 -dimethylbenzene
- (b) 1, 3-dimethylbenzene
- (c) 1,2,3 -trimethylbenzene
- (d) ethylbenzene

Answer: B

Q679.

Which is responsible for ozone depletion?

- (a) CFCs
- (b) Hydrocarbons $\cdot \text{CO}_2$
- (d) CO

Answer: A

Q680.

Which group of hydrocarbon is most toxic?

- (a) Saturated hydrocarbon
- (b) Long chain hydrocarbon
- (c) Polycyclic aromatic hydrocarbon
- (d) Unsaturated hydrocarbon

Answer: C

Physics

Q681.

An alternating voltage given as $V = 100\sqrt{2} \sin 100\pi t$ is applied to a capacitor of $1 \mu\text{F}$. The current reading of the ammeter will be equal to mA

- (a) 10
- (b) 20
- (c) 40
- (d) 80

Answer: B

Chemistry

Q682.

The weakest base among the following is A . H^+

- (b) CH_3^-
- (c) CH_3O^-
- (d) Cl^-

Answer: D

Physics

Q683.

The moving air from the hair dryer removes the molecules of liquids as soon as they escape from the surface of your hair. Which factor of evaporation is responsible for the above example?

- (a) Temperature
- (b) Motion of air
- (c) Nature of liquid

(d) Humidity of surrounding

Answer: B

Chemistry

Q684.

Which is not isoelectronic with the other three?

- (a) C O
- (b) N O⁺ c · C N⁻
- (d) O₂

Answer: D

Physics

Q685.

A field of strength $5 \times 10^4 / \pi$ ampere turns / meter acts at right angles to the coil of 50 turns of area 10^{-2} m^2 . The coil is removed from the field in 0.1 second. Then the induced e.m.f. in the coil is :

- (a) 0.1 V
- (b) 0.2 V
- (c) 1.96 V D · 0.98 V

Answer: D

Chemistry

Q686.

When an amino acid solution which is slightly alkaline is subjected to electrolysis, the resulting component migrates towards

- (a) Cathode due to zwitter ion formation.
- (b) Anode due to the conversion of zwitter ion to negative ion
- (c) Migrate to cathode due to the conversion to negative ion
- (d) One component towards cathode and the other towards anode due to the zwitter ion formation

Answer: B

Physics

Q687.

What is minimum frequency of the photon required to carry out transition A · $1.21 \times 10^{15} \text{ H z}$

- (b) $1.61 \times 10^{15} \text{ Hz}$
- (c) $1.21 \times 10^{14} \text{ H z}$
- (d) $1.61 \times 10^{14} \text{ Hz}$

Answer: A

Chemistry

Q688.

Peroxydisulphuric acid and its salts are powerful oxidizing agents. Thus it can be estimated reducing agents and also study the kinetics of the reaction. Peroxydisulphuric acid:

- (a) is also called Marshall's acid
- (b) has two peroxy linkages

- (c) liberates I_2 with KI in a fast reaction
(d) has oxidation number of sulphur as +7

Answer: A

Q689.

This is the branch of chemistry dealing with the separation, identification and quantitative determination of the composition of different substances. Name is :

- (a) Physical chemistry
(b) Inorganic chemistry
(c) organic chemistry
(d) Analytical chemistry

Answer: D

Biology

Q690.

Which is the short day plant from the following?

- (a) cotton
(b) Rice
(c) sugarcane
(d) All of the above

Answer: B

Chemistry

Q691.

In the reaction, $3Cu + 2NH_3 + 3H_2O \rightarrow 3Cu + N_2$ the change of NH_3 to N_2 involves:

- (a) Loss of 6 electrons per mole of N_2
(b) Loss of 3 electrons per mole of N_2
(c) Gain of 6 electrons per mole of N_2
(d) Gain of 3 electrons per mole of N_2

Answer: A

Q692.

fibres are made from cellulose.

- (a) Vegetable
(b) Animal
(c) Both vegetable and animal
(d) None of them

Answer: A

Physics

Q693.

An electric dipole made up of a positive and negative charge, each of $1 \mu C$ separated by a distance of 2 cm, is placed in an electric field of $10^5 N/C$. The work done in rotating the dipole from the position of stable equilibrium through an angle of 180° is : $A \cdot 2 \times 10^{-3}$ Joule

- (b) 2×10^{-8} Joule
(c) 4×10^{-3} Joule
(d) Zero

Answer: C

Chemistry

Q694.

Number of electrons in 1.8 mL of H_2O are: A . 6.02×10^{23}

- (b) 6.02×10^{24}
- (c) 6.02×10^{22}
- (d) 6.02×10^{25}

Answer: B

Physics

Q695.

A tuning fork of frequency 500 Hz is sounded on a resonance tube. The first and second resonances are obtained at 17 cm and 52 cm . The velocity of sound is

- (a) 170 m / s
- (b) 350 m / s
- (c) 520 m / s
- (d) 850 m / s

Answer: B

Biology

Q696.

All chemical characteristics are common to living beings, except

- (a) Type of protein present in the body.
- (b) Similar triple code for amino acids.
- (c) Energy is stored by high phosphate bonds.
- (d) Ribosomes are the sites of protein synthesis.

Answer: A

Physics

Q697.

What is the relation between restoring force (F) and displacement (x) of a particle to perform SHM. A . $F \propto x^2$

- (b) $F \propto x^{-2}$
- (c) $F \propto x^{1/2}$
- (d) $F \propto x$

Answer: D

Chemistry

Q698.

Which of the following are examples of aerosols? This question has multiple correct options

- (a) Whipped cream
- (b) Clouds
- (c) Fog
- (d) Soap lather

Answer: B,C

Physics

Q699.

A thick strip of copper is mounted as a compound pendulum about O. If it is made to swing through a uniform magnetic field B normal to the plane of the strip then (neglecting air resistance) it is found that

- (a) strip swings almost freely.
- (b) motion of the strip is heavily damped.
- (c) strip does not oscillate at all but immediately comes to rest in the vertical position
- (d) strip swings almost freely but its temperature decreases

Answer: B

Biology

Q700.

Which of the following is not a function of lymph?

- (a) Supplies oxygen and prevents spreading of infections
- (b) Returns proteins to the blood
- (c) Absorbs vitamins and minerals
- (d) None of the above

Answer: A

Q701.

At the time of fertilization, sperm head enters in the egg from

- (a) Any where
- (b) Animal pole
- (c) vegetal pole
- (d) Lateral side of egg

Answer: B

Chemistry

Q702.

In phenol carbon atom attached to $-OH$ group undergoes: sp^3 hybridisation

- (b) sp hybridisation
- (c) sp^2 hybridisation
- (d) No hybridisation

Answer: C

Q703.

What is the correct formula for calcium chloride?

- (a) $CaCl_2$
- (b) $CaCl$
- (c) $CaCl_2$
- (d) $CaCl_3$

Answer: C

Biology

Q704.

Transporting pull arise mainly from

- (a) Roots
- (b) Leaves
- (c) Stems
- (d) All

Answer: B

Physics

Q705.

Read the following statements and pick the correct choice Statement A: With increase in temperature, viscosity of a gas increases and that of a liquid decreases. Statement B: If the density of a small sphere is equal to the density of the liquid in which it is dropped, then the terminal velocity of the sphere will be zero.

- (a) Both A and B are true
- (b) Both A and B are false
- (c) A is true and B is false
- (d) B is true but A is false.

Answer: A

Biology

Q706.

Atul is traveling from Mumbai to Nagpur by train, in between, he felt uneasy because

- (a) He used an improperly sanitized public toilet
- (c) He was already sick.
- (d) None of the above

Answer: A

Q707.

Two snails in the same class can also belong to which group?

- (a) Order
- (b) species
- (c) Genus
- (d) Family E. Phylum

Answer: E

Physics

Q708.

A body is dropped from a certain height from the ground. When it is halfway down, it possess,

- (a) Only K.E.
- (b) Both K.E. and P.E.
- (c) only P.E.
- (d) zero energy

Answer: B

Q709.

The following three objects (1) a metal tray, (2) a block of wood, and (3) a wooden cap are left in a closed room overnight. Next day the temperature of each is recorded as T_1 , T_2 and T_3 respectively. The likely situation is

- (a) $T_1 = T_2 = T_3$ ■. $T_3 > T_2 > T_1$
- (c) $T_3 = T_2 > T_1$
- (d) $T_3 > T_2 = T_1$

Answer: A

Biology

Q710.

Select the correct option: During electron transport system(ETS), electron transport proceeds from carriers that have potential to those having redox potential. This electron transport down the energy gradient leads to the formation of ATP from ADP and P_i , which is referred to as

- (a) Low, high, oxidative phosphorylation
- (b) Low, high, oxidative decarboxylation
- (c) High, low, oxidative phosphorylation
- (d) High, low, oxidative decarboxylation

Answer: C

Chemistry

Q711.

A weak mono acidic base is 5 % ionized in 0.01 M solution. The hydroxide ion concentration in the solution is:

- (a) 5×10^{-2} ■. 5×10^{-4}
- (c) 5×10^{-10}
- (d) 2×10^{-11}

Answer: B

Q712.

Which of the following statements is not correct about the structure of PCl_5 ?

- (a) PCl_5 has a trigonal bipyramidal structure
- (b) Three equatorial P-Cl bonds are equivalent
- (c) The two axial bonds are different and longer than equatorial bonds
- (d) Equatorial bond pairs suffer more repulsion than that of the axial bond pairs

Answer: D

Physics

Q713.

What is the electric potential at the centre of square of side 1 m? The charges $1 \times 10^{-8} \text{ C}$, $-2 \times 10^{-8} \text{ C}$, $3 \times 10^{-8} \text{ C}$ and $2 \times 10^{-8} \text{ C}$ are placed at the corners of the square. A · 205 V

- (b) 308 V c .509 V
- (d) 110 V

Answer: A

Chemistry

Q714.

Which tool allows liquids to be swirled with a low risk of spilling?

- (a) Crucible
- (b) Evaporating dish
- (c) Tongs
- (d) Erlenmeyer flask E. Mortar and pestle

Answer: D

Biology

Q715.

Identify the wrong statement about meiosis.

- (a) Pairing of homologous chromosomes
- (b) Four haploid cells are formed
- (c) At the end of meiosis number of chromosomes are reduced to half
- (d) Two cycle of DNA replication occur

Answer: D

Chemistry

Q716.

The stability of ferric ion is due to

- (a) half filled f-orbitals
- (b) half filled d-orbitals
- (c) completely filled f-orbitals
- (d) completely filled d-orbitals

Answer: B

Biology

Q717.

Normal cardiac output is

- (a) 15 litres/min
- (b) 520 litres/min
- (c) 5 litres/min
- (d) None of the above

Answer: C

Chemistry

Q718.

The resistance of 1 N solution of acetic acid is $250\ \Omega$ when measured in a cell having a cell constant of $1.15\ \text{cm}^{-1}$. The equivalent conductance (in $\text{ohm}^{-1}\ \text{cm}^2\ \text{equiv}^{-1}$) of 1 N acetic acid is

- (a) 2.3
- (b) 4.6
- (c) 9.2
- (d) 18.4

Answer: B

Q719.

Some salts, although containing two different metallic elements, give test for one of them in solution. Such salts are:

- (a) complex salt
- (b) double salt
- (c) normal salt
- (d) none of these

Answer: A

Physics

Q720.

Two isolated metallic solid sphere of radius R and $2R$ are charged such that both of these have same charge density

- (a) the spheres are located far away from each other and are connected by a thin conducting wire. the new charge density on the bigger sphere is $A \cdot (5\sigma)/(6)$ ■. $(56\sigma)/(5)$
(c) $(3\sigma)/(5)$
(d) $(52\sigma)/(6)$

Answer: C

Biology

Q721.

Compensation point is the value of the factor where there is

- (a) Neither photosynthesis nor respiration
(b) Little photosynthesis
(c) Photosynthesis is equal to the rate of respiration
(d) Beginning of photosynthesis

Answer: C

Chemistry

Q722.

The velocity of the electron in 3rd orbit of hydrogen atom in cm s^{-1} A. 4.88×10^4

- (b) 2.44×10^8
(c) 7.29×10^7
(d) 2.44×10^{12}

Answer: B

Q723.

All dilute acids react with bleaching powder, liberating:

- (a) chlorine gas
(b) hydrogengas
(c) nitrogen gas
(d) HCl gas

Answer: A

Physics

Q724.

E.O. Lawrence was awarded the 1939 Physics Nobel Prize for invention and development of

- (a) diode
(b) microwave transmitter
(c) cyclotron
(d) none of these

Answer: C

Q725.

kg m s^{-1} is the SI unit of

- (a) Impulse

- (b) Force
- (c) Angular velocity
- (d) None of these

Answer: A

Q726.

Band gap in a semiconductor is of the order

- (a) 6 eV
- (b) 0.60 eV
- (c) 60 eV
- (d) 0 eV

Answer: B

Q727.

Sublimation is a technique that helps chemists to:

- (a) break
- (b) purify
- (c) melt
- (d) freeze

Answer: B

Q728.

When light propagates in vacuum there is an electric field and a magnetic field. Which of the following is not true about these fields?

- (a) They are constant in time
- (b) They have zero average value
- (c) They are perpendicular to the direction of propagation of light
- (d) They are mutually perpendicular

Answer: A

Q729.

When the switch of an electric bell is pushed

- (a) Flow of the current stops through the electromagnet in the bell
- (b) Current starts to flow through the electromagnet
- (c) voltage decreases in the current flowing through the electromagnet
- (d) None of the above

Answer: B

Q730.

A ball is thrown from rear end of the compartment of train to the front end which is moving at a constant horizontal velocity. An observer A sitting in compartment and another observer B standing on the ground draw the trajectory. They will have

- (a) equal horizontal and equal vertical ranges.
- (b) equal vertical ranges but different horizontal ranges
- (c) different vertical ranges but equal horizontal ranges
- (d) different vertical ranges and different horizontal ranges

Answer: B

Chemistry

Q731.

At temperature 1000°C , ΔG of the reaction $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$ is Temperature

- (a) -ve

- (b) +ve c . zero
(d) nothing can be said

Answer: A

Q732.

Select the species having zero oxidation state at the underlined elements. This question has multiple correct options A · (C H₃)₂ S O

- (b) C₁₂ H₂₂ O₁₁
(c) H₂ S O₃
(d) N₂ H₄

Answer: B,D

Q733.

Which of the following statement is correct? A · E_{text {cell}}} and ■_r G of cell reaction both are extensive properties

- (b) E_{text {cell}}} and ■_r G of cell reaction both are intensive properties
(c) E_{text {cell}}} in the intensive property while ■_r G of cell reaction is an extensive property.
(d) E_{text {cell}}} in the intensive property while ■_r G of cell reaction is an intensive property.

Answer: C

Physics

Q734.

A string vibrates in 4 segments to a frequency of 400 Hz. What frequency will cause it to vibrate into 7 segments?

- (a) 700 H z ■. 500 H z
(c) 400 Hz
(d) 100 H z

Answer: A

Q735.

Assertion The sensitivity of a moving coil galvanometer is increased by placing a suitable magnetic material as a core inside the coil. Reason Soft iron has a high magnetic permeability and cannot be easily magnetized or demagnetized.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
(b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
(c) Assertion is correct but Reason is incorrect
(d) Assertion is incorrect but Reason is correct

Answer: C

Biology

Q736.

Match the following and choose the correct option. A

- (b) A- 4, B-1, C-2, D-3
(c) A-4, B-1, C-3, D-4
(d) A-4, B-2, C-3, D-4

Answer: A

Physics

Q737.

Assertion Assertion: When a light wave travels from a rarer to a denser medium, its speed decreases. The decrease in speed imply a reduction in energy carried by the light wave. Reason Reason: The energy of a wave is inversely proportional to velocity of wave

- (a) If both assertion and reason are true and reason is the correct explanation of assertion
- (b) If both assertion and reason are true and reason is not the correct explanation of assertion.
- (c) If assertion is true but reason is false
- (d) If both assertion and reason are false

Answer: C

Biology

Q738.

Cell division can not be stopped in phase of the cell cycle.

- (a) G₁ phase
- (b) G₂ phase
- (c) S phase
- (d) Prophase

Answer: D

Chemistry

Q739.

The frequency of the spectral line obtained when the electron in $n=3$ of Hydrogen atom drops to the ground state is: A · 2.925×10^{15} Hertz

- (b) 2.925×10^{13} Hertz
- (c) 2.925×10^{14} Hertz
- (d) 36559×10^{10} Hertz

Answer: A

Q742.

Which of the correct statement for the given acids?

- (a) Phosphinic acid is a diprotic acid while phosphonic acid is a monoprotic acid.
- (b) Phosphinic acid is a monoprotic acid while phosphonic acid is a diprotic acid
- (c) Both are triprotic acids.
- (d) Both are diprotic acids

Answer: B

Biology

Q743.

Biogas plant is more useful than burning of dung cakes because

- (a) It is non-pollutant
- (b) It yields manure as well as fuel gas.
- (c) It yields important biochemicals.
- (d) Both A and B

Answer: D

Q744.

Which of the following animal respire but there is no respiratory organ in them? A · Frog

- (b) Cockroach
- (c) Fish
- (d) Earthworm

Answer: D

Physics

Q745.

A circular road of radius r is banked for a speed $v=40 \text{ km/hr}$. A car of mass m attempts to go on the circular road. The friction coefficient between the tyre and the road is negligible.

- (a) The car cannot make a turn without skidding
- (b) if the car turns at a speed less than 40 km/hr , it will slip down.
- (c) If the car turns at the correct speed of 40 km/hr , the force by the road on the car is equal to $m v^2 / r$
- (d) If the car turns at the correct speed of 40 km/hr , the force by the road on the car is greater than mg as well as greater than $m v^2 / r$

Answer: D

Q746.

A microscope is focused on a mark, then a glass slab of refractive index 1.5 and thickness of 3 cm is placed on the mark to get the mark again in focus, the microscope should be moved

- (a) 4 cm
- (b) 2 cm
- (c) 6 cm
- (d) 8 cm

Answer: B

Chemistry

Q747.

Identify the group 14 A element that best fits each of the following descriptions: (a) prefers the +2 oxidation state. (b) forms the strongest π bonds. (c) is the second most abundant element in the earth's crust. (d) forms the most acidic oxide. A · (a) P b ; (b) C ; (c) S i ; (d) C

- (b) (a) P b ; (b) C ; (c) P b ; (d) C c . (a) P b ; (b) C ; (c) S i ; (d) S i
- (d) (a) S n ; (b) C ; (c) S i ; (d) C

Answer: C

Physics

Q748.

The equation of a plane progressive wave is given by $y=A(\omega t-kx)$ The dimension of (ω/k) is that of

- (a) frequency
- (b) velocity
- (c) wavelength
- (d) inverse of velocity

Answer: B

Q749.

If the dimension of a physical quantity is given by $[M^a L^b T^c]$ then the physical quantity will be

- (a) acceleration, if $a=1, b=-1, c=-2$
- (b) velocity, if $a=1, b=0, c=-1$
- (c) pressure, if $a=1, b=-1, c=-2$
- (d) force, if $a=0, b=-1, c=-2$

Answer: C

Q750.

If the mass of the body on the surface of the earth is 50 k g, its mass at the centre of the earth is:

- (a) zero
- (b) More than 50 kg
- (c) Less than 50 kg
- (d) Equal to 50 kg

Answer: D

Chemistry

Q751.

As the concentration of reactants increases:

- (a) rate of the reaction decreases
- (b) rate of the reaction increases
- (c) rate of the reaction remains the same
- (d) reaction stops

Answer: B

Q752.

When alpha particles are sent through a thin metal foil, most of them go straight through the foil because:

- (a) alpha particles are much heavier than electrons
- (b) alpha particles are positively charged
- (c) most part of the atom is empty space
- (d) alpha particles move with very high velocity

Answer: C

Q753.

Evaporation of the solution of copper sulfate solution helps in:

- (a) making the solution concentrated
- (b) crystallization of copper sulfate
- (c) both A and B
- (d) none of these

Answer: C

Biology

Q754.

Which organ is part of circulatory system but does not take part in pumping?

- (a) Heart
- (b) Liver
- (c) spleen
- (d) Kidney

Answer: C

Chemistry

Q755.

The name of the element having atomic number 100, was adopted in the honour of :

- (a) Alfred Nobel

- (b) Enrico Fermi
- (c) Dmitri Mendeleev
- (d) Albert Einstein

Answer: B

Q757.

What is the common name of propanone?

- (a) Oxobutane
- (b) Acetone
- (c) Ethyl acetone
- (d) Ethyl methyl ketone

Answer: B

Q758.

Which of the following solution of NaCl has the higher specific conductance?

- (a) 1 N
- (b) 0.01 N & 0.1 N
- (d) 0.001 N

Answer: A

Biology

Q759.

Carbonic anhydrase is mostly active in

- (a) RBC
- (b) WBC
- (c) Blood Plasma
- (d) Blood Platelets

Answer: A

Q760.

Primary roots and its branches constitutes to

- (b) Tap root system
- (c) Fibrous roots
- (d) Seminal roots

Answer: B

Chemistry

Q761.

The heat of atomization of methane and ethane are 360 kJ / mol and 620 KJ / mol respectively. The longest wavelength of light capable of breaking the C-C bond is: (Avogadro number = 6.02×10^{23} , $h = 6.62 \times 10^{-34}$ Js A · 2.48×10^4 nm

- (b) 1.49×10^3 nm
- (c) 2.48×10^3 nm
- (d) 1.49×10^4 nm

Answer: C

Q762.

Assertion BaCO_3 is more soluble in HNO_3 than in plain water. Reason Carbonate is a weak base and reacts with the H^+ from the strong acid causing the barium salt to dissociate.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion

- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
(c) Assertion is correct but Reason is incorrect
(d) Both Assertion and Reason are incorrect

Answer: A

Q763.

The modern periodic table is ordered on the basis of:

- (a) atomic mass
(b) atomic radius
(c) atomic charge
(d) atomic number

Answer: D

Physics

Q764.

In a healthy human eye, the focussing is done by the

- (a) To and fro movement of the eye-lens
(b) Changing the curvature of the retina
(c) Change in the convexity of the lens through ciliary tubes
(d) Change in the refractive index of the eye fluid

Answer: C

Biology

Q765.

During photorespiration, when two O_2 molecules react with 2 moles of RUBP then how many 3-P G A molecules are formed from 2 molecules of 2- phosphoglycolate

- (a) 2
(b) 1 c .3
(d) 4

Answer: A

Physics

Q766.

The concave mirrors are used in

- (a) reflecting telescope
(b) magic-laterns
(c) cinema projectors
(d) all the above

Answer: D

Chemistry

Q767.

Which of the following pentahalides does not exist?

- (a) PF_5
(b) PCl_5
(c) PBr_5

(d) PI

Answer: D

Q768.

Assertion Zinc is used in the galvanization of iron. Reason Its coating on iron articles increases their life by protecting them from rusting.

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion
- (b) If both Assertion and Reason are correct, but Reason is not the correct explanation of Assertion
- (c) If Assertion is correct but Reason is incorrect
- (d) If Assertion is incorrect but Reason is correct

Answer: A

Biology

Q769.

Identify the incorrect statement.

- (a) Forest prevents soil erosion.
- (b) Forest influences climate and water cycle.
- (c) Soil helps forest to regenerate.
- (d) Plants and animals in the forest are not dependent on one another.

Answer: D

Physics

Q770.

Assertion Pascal's law is the working principle of hydraulic lift. Reason $\text{Pressure} = (\text{thrust})/(\text{area})$

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: B

Chemistry

Q771.

Which of the following is true for Thomson's model of the atom?

- (a) The radius of an electron can be calculated using Thomson's model
- (b) In an undisturbed atom, the electrons will be at their equilibrium positions, where the attraction between the cloud of positive charge and the electrons balances their mutual repulsion
- (c) When the electrons are disturbed by collision, they will vibrate around their equilibrium positions and emit electromagnetic radiation whose frequency is of the order of magnitude of the frequency of electromagnetic radiation of a vibrating electron
- (d) Both (B) and (C)

Answer: D

Physics

Q772.

A magnet makes 40 oscillations per minute at a place having magnetic field intensity of $0.1 \times 10^{-5} \text{ T}$. At another place, it takes 2.5 seconds to complete one vibration. The value of earth's horizontal field at that place is

- (a) $0.25 \times 10^{-6} \text{ T}$ ■ $0.36 \times 10^{-6} \text{ T}$ ■
(c) $0.66 \times 10^{-8} \text{ T}$
(d) $1.2 \times 10^{-6} \text{ T}$ ■

Answer: A

Q773.

A reversible engine takes heat from a reservoir at 527°C and gives out heat to a k at 127°C . How many Joules/s must it take from the reservoir to perform useful mechanical work at the rate of 750 W ?

- (a) 257.14
(b) 1500
(c) 457.14
(d) 557.14

Answer: B

Chemistry

Q774.

Silver is extracted from Ag_2S by

- (a) fusing it with KCl and electrolyzing the melt
(b) reducing it with Zinc
(c) treating with sodium cyanide followed by zinc
(d) roasting it and reducing the resultant product by smelting

Answer: C

Biology

Q775.

At the poles, the air is colder than that at latitudes about?

- (a) 30 degrees
(b) 60 degrees
(c) 90 degrees
(d) 80 degrees

Answer: B

Chemistry

Q776.

Name the compounds used in preparation of nylon-66.

- (a) E-caprolactum
(b) Hexamethylenediamine and adipic acid
(c) Dimethyl tetraphthalate
(d) Hexamethylene diamine

Answer: B

Q777.

Which of the following are the correct statements regarding the catalyzed and uncatalyzed reactions at constant temperature? This question has multiple correct options

- (a) Catalyzed reaction does not alter ΔH of the reaction
(b) At constant temperature, ΔG^0 of the catalysed and uncatalyzed reaction is same
(c) Catalyzed reaction alters equilibrium constant even at constant temperature
(d) Catalysed reaction alters r_f and r_b values equally at constant temperature

Answer: A,B,D

Q778.

A dilute ferrous sulphate solution was gradually added to the beaker containing acidified potassium permanganate solution. The light purple colour of the solution fades and finally disappears. Which of the following is the correct explanation for the observation? A · K Mn O_4 is an oxidising agent, and it oxidises Fe S O_4

(b) Fe S O_4 acts as an oxidising agent and oxidises K Mn O_4

(c) The colour disappear due to dilution; no reaction is involved

(d) K Mn O_4 is an unstable compound and decomposes in presence of Fe S O_4 to a colourless compound

Answer: A

Biology

Q779.

Which of the following substances, if introduced into the bloodstream, would cause coagulation of blood at the site of its introduction?

(a) Fibrinogen

(b) Prothrombin

(c) Heparin

(d) Thromboplastin

Answer: D

Chemistry

Q780.

Which of the following analgesics is not habit forming?

(a) Morphine

(b) Aspirin

(c) codeine

(d) Heroin

Answer: B

Biology

Q781.

Amoeba sticks to substratum by means of

(a) Rough surface

(b) Plasmalemma

(c) Protein

(d) None of the above

Answer: B

Q782.

Which one of the following statements is correct?

(a) At present it is not possible to grow maize without chemical fertilizers.

(b) Extensive use of chemical fertilizer may lead to eutrophication of nearby water bodies.

(c) Both Azotobacter and Rhizobium fix atmospheric nitrogen in root nodules of plants.

(d) Cyanobacteria such as Anabaena and Nostoc are important mobilizers of phosphates and potassium for plant nutrition in the soil.

Answer: B

Physics

Q783.

What bring the rolling ball to rest?

- (a) Force of friction
- (b) Gravitational force
- (c) Air drag
- (d) Buoyant force

Answer: A

Q784.

The mean value of velocity vector projection average over $3/8$ th of a period after the start is $A \cdot (2\sqrt{2} A \omega)/(\pi)$

- (b) $(2 A \omega)/(3\sqrt{2} \pi)$
- (c) $(A \omega)/(3 \pi)$
- (d) $(2\sqrt{2} A \omega)/(3 \pi)$

Answer: D

Chemistry

Q785.

A transition metal 'X' has a configuration $[\text{Ar}] 3d^4$ in its +3 oxidation state, its atomic number is:

- (a) 25
- (b) $26 - 3 = 23$
- (d) 19

Answer: A

Physics

Q786.

Looming' in polar region A · Is an optical illusion

- (b) Arise due to the variation of refractive index of atmosphere
- (c) Is an example of total internal reflection
- (d) All of the above

Answer: D

Biology

Q787.

After the secondary growth, the oldest layer of secondary phloem in a dicot stem is located

- (a) Just outside the vascular cambium
- (b) Just inside the vascular cambium
- (c) Just inside the vascular primary phloem
- (d) Just outside the secondary xylem

Answer: C

Q788.

Green leaf like one internode long stem branches/phyllodes are called

- (a) Phyllodes
- (b) Phylloclades

- (c) Bulbils
- (d) Cladodes

Answer: D

Q789.

The repository of plant taxonomy and systematics is

- (a) Red data book
- (b) Herbarium
- (c) Botanical garden
- (d) Genomic library

Answer: B

Physics

Q790.

Three tubes A, B and C are connected to a horizontal pipe in which liquid is flowing. The radii of pipe at the joints of A, B and C are 2 cm, 1 cm and 2 cm respectively. The height of liquid: A . in A and B is equal

- (b) in A is maximum c . in A and C is same
- (d) is same in all three

Answer: C

Chemistry

Q791.

Which of the following processes is unsuitable for softening water possessing temporary hardness?

- (a) Adding calcium hydroxide
- (b) Filtration
- (c) Boiling
- (d) Distillation

Answer: B

Physics

Q792.

Regarding open organ pipe, which of following is correct?

- (a) Both ends are pressure antinodes
- (b) Both ends are displacement nodes
- (c) Both ends are pressure nodes
- (d) Both (1) and (2)

Answer: C

Q794.

Potential energy is classified into which two energy?

- (a) Gravitational and Elastic Potential Energy
- (b) Kinetic and Elastic Potential Energy
- (c) Mechanical and Elastic Potential Energy
- (d) None

Answer: A

Q795.

The work function of a certain metal is $3.31 \times 10^{-19} \text{ J}$. Then, the maximum kinetic energy of photoelectrons emitted by incident radiation of wavelength $5000 \text{ \AA}$ is (Given $h = 6.62 \times 10^{-34} \text{ J-s}$, $c = 3 \times 10^8 \text{ m s}^{-1}$, $e = 1.6 \times 10^{-19} \text{ C}$)

- (b) 0.41 eV
- (c) 2.07 eV
- (d) 0.82 eV

Answer: D

Chemistry

Q796.

As a lead storage battery is charged.

- (a) Lead dioxide dissolves
- (b) Sulphuric acid is regenerated
- (c) Lead electrode becomes coated with lead sulphate
- (d) The concentration of sulphuric acid decreases

Answer: B

Physics

Q797.

A block of mass ' m ' is kept on the ground as shown in figure. Which of the following statements is/are correct of action-reaction force pair? This question has multiple correct options

- (a) Normal force can never be a reactive force
- (b) Weight of the body is not a part of action reaction pair
- (c) Forces act on different bodies
- (d) None of these

Answer: B,C

Q798.

The weight of 50 kg person at the surface of earth is

- (a) 50 N
- (b) 35 N
- (c) 380 N
- (d) 490 N

Answer: D

Chemistry

Q799.

80% of a first order reaction was completed in 70 min . How much it will take for 90% completion of a reaction?

- (a) 114 min
- (b) 140 min
- (c) 100 min
- (d) 200 min

Answer: A

Q800.

Physical adsorption of a gaseous species may change to chemical adsorption with:

- (a) decrease in temperature

- (b) increase in temperature
- (c) increase in surface area of adsorbent
- (d) decrease in surface area of adsorbent

Answer: B

Q801.

Uncertainty in position of a particle of 25 g in space is 10^{-15} m . Determine uncertainty in velocity (m sec^{-1}) in it. (plank's constant, $h=6.6 \times 10^{-34}$ J s)

- (a) 2.1×10^{-18} ■. 1.2×10^{-18}
- (c) 1.8×10^{-18}
- (d) 2.5×10^{-18}

Answer: B

Physics

Q802.

For the given combination of gates, if the logic states of inputs A, B, C are as follows $A=B=C=0$ and $A=B=1$, $C=0$ then the logic states of output D are $A \cdot 0, 0$

- (b) 0,1
- (c) 1,0
- (d) 1,1

Answer: B

Biology

Q803.

Choose the correct answer from the alternatives given. If a cell is placed in an isotonic solution, what will occur?

- (a) Salts will move into the cell from the surrounding solution
- (b) Water will move into the cell from the surrounding solution
- (c) Salts will move out of the cell into the surrounding solution
- (d) None of the above will occur

Answer: D

Physics

Q804.

Assertion Two balls of different masses are thrown vertically upward with same speed. They will pass through their point of projection in the downward direction with the same speed. Reason The maximum height and downward velocity attained at the point of projection are independent of the mass of the ball.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: A

Q805.

Fraunhofer line of the solar system is an example of:

- (a) line absorption spectrum
- (b) emission of line system
- (c) line emission spectrum
- (d) emission of band spectrum

Answer: A

Biology

Q806.

Which one is a wrong statement?

- (a) Brown algae have chlorophyll a and c and fucoxanthin.
- (b) Archegonia are found in Bryophyta, Pteridophyta and Gymnosperms.
- (c) Mucor has biflagellate zoospores.
- (d) Haploid endosperm is typical feature of gymnosperms.

Answer: C

Physics

Q807.

A spring is kept compressed by a toy car of mass 150 g . On releasing the car it moves with a speed of 0.2 m s^{-1} . So, the elastic potential energy of the spring is A .3 m J

- (b) 3 J
- (c) 1.5 m J
- (d) 4 m J

Answer: A

Q808.

Diodes are used to

- (a) convert DC into AC
- (b) convert AC into DC
- (c) resistance control
- (d) current regulation

Answer: B

Chemistry

Q809.

10 L of hard water required 0.56 g of lime (CaO) for removing hardness. Hence, the temporary hardness in ppm (part per million) of CaCO_3 is :

- (a) 100 ■. 200
- (c) 10
- (d) 20

Answer: B

Q810.

Assertion Ammonia is a base. Reason Ammonia when dissolved in water takes up proton to form ammonium and furnishes OH^- ions.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: A

Physics

Q811.

The transistor are usually made of

- (a) metal oxides with high temperature coefficient of resistivity.
- (b) metals with high temperature coefficient of resistivity.
- (c) metals with low temperature coefficient of resistivity.
- (d) semiconducting materials having low temperature coefficient of resistivity.

Answer: D

Biology

Q812.

A plant with parallel venation is

- (a) castor
- (b) Grass
- (c) colocasia
- (d) Mustard

Answer: B

Physics

Q813.

The temperature at which the domain structure gets destroyed and ferromagnetic substance is converted into paramagnetic substance is called as

- (a) Domain
- (b) Curie Temperature
- (c) Rowland
- (d) Magnetic Field Intensity

Answer: B

Biology

Q814.

Primata and Carnivora are placed together in the same?

- (a) Order
- (b) Division
- (c) Family
- (d) class

Answer: D

Physics

Q815.

Y, k, n represent respectively the young's modulus, bulk modulus and rigidity modulus of a body. If rigidity modulus is twice the bulk modulus, then:

- (a) $Y = 5k / 18$ ■. $Y = 5n / 9$
- (c) $Y = 9k / 5$
- (d) $Y = 18k / 5$

Answer: D

Chemistry

Q816.

The given figure shows the plant for the formation of an important compound X Identify X

- (a) Washing soda
- (b) Common salt
- (c) Baking powder
- (d) Bleaching powder

Answer: D

Q817.

Isotactic polypropylene polymer is one, in which:

- (a) all methyl groups are on one side of the extended chain. It is a highly crystalline, has high melting point and forms strong fibres
- (b) the methyl groups present alternates regularly from one side to the other
- (c) the methyl groups are distributed at random. It is a soft, elastic, and rubbery material
- (d) None of these

Answer: A

Physics

Q818.

A convex mirror of radius of curvature 20 cm forms an image which is half the size of the object. How far is the object from the mirror?

- (a) 5 cm
- (b) 7.5 cm c .-30 cm
- (d) 12.5 cm

Answer: D

Chemistry

Q819.

Which of the following steps is not correct in the mechanism of electrophilic substitution of benzene?

- (a) Generation of electrophile like X^+ , R^+ , RC^+O , NO_2^+ etc.
- (b) Attack of electrophile resulting in the formation of arenium ion in which one of the carbon is sp^3 hybridised.
- (c) Addition of proton on benzene ring to give carbocation.
- (d) Removal of proton from sp^3 carbon atom to restore aromatic character.

Answer: C

Physics

Q820.

The distance of the closest approach of an alpha particle fired at a nucleus with kinetic energy K is r_0 . The distance of the closest approach when the α particle is fired at the same nucleus with kinetic energy 2 K will be

- A. $(r_0)/(2)$
- (b) $4 r_0$
- (c) $(r_0)/(4)$
- (d) $2 r_0$

Answer: A

Biology

Q821.

Sensory crista of ampulla of mammalian ear is connected with

- (a) Sense of balance
- (b) Sense of low hearing
- (c) Sense of loud hearing
- (d) Secretion of ear wax

Answer: A

Physics

Q822.

A body moving with uniform retardation covers 3 km before its speed is reduced to half of its initial value. It comes to rest in another distance of :

- (a) 1 km
- (b) 2 km c .3 km
- (d) 1/2 km

Answer: A

Q823.

Wire of a certain material is stretched slowly by 10%. Its new resistance and specific resistance becomes respectively.

- (a) Both remain same
- (b) 1.1 times in both
- (c) 1.2 times, 1.1 times
- (d) 1.21 times, same

Answer: D

Q824.

The value of $1+1$ is $A \cdot 2$

- (b) 0 c .1
- (d) 10

Answer: A

Chemistry

Q825.

Which of the following order of acidic strength is correct: A · $\text{RCOOH} > \text{ROH} > \text{HOH} > \text{HC} \equiv \text{CH}$

- (b) $\text{RCOOH} > \text{HOH} > \text{ROH} > \text{HC} \equiv \text{CH}$
- (c) $\text{RCOOH} > \text{HOH} > \text{HC} \equiv \text{CH} > \text{ROH}$
- (d) $\text{RCOOH} > \text{HC} \equiv \text{CH} > \text{HOH} > \text{ROH}$

Answer: B

Physics

Q826.

1 atm =

- (a) 100 cm of Hg
- (b) 76 cm of Hg
- (c) 20 cm of Hg

(d) 25 cm of Hg

Answer: B

Chemistry

Q827.

Markovnikov's rule for the addition of hydrogen halides to alkenes states that the incoming hydrogen bonds to the : A · s p^2 carbon with the most hydrogens already

- (b) s p^2 carbon with the fewest hydrogens already
- (c) s p^3 carbon with the most hydrogens already
- (d) s p^3 carbon with the few hydrogens already

Answer: A

Q828.

Which of the following can be used to prepare hydrogen gas in the laboratory?

- (a) Mercuric oxide
- (b) Acid plus zinc
- (c) Potassium chlorate
- (d) Carbon disulfide E. Benzene

Answer: B

Physics

Q829.

For a system to be in equilibrium, the torques acting on it must balance. This is true only if the torques are taken about

- (a) the centre of the system
- (b) the centre of mass of the system
- (c) any point on the system

Answer: C

Chemistry

Q830.

The shape of s p^3 d^2 hybridized molecule with 2 lone pairs will be:

- (a) octahedral
- (b) square pyramidal
- (c) square planar
- (d) tetrahedral

Answer: C

Biology

Q831.

In Opuntia, the function of photosynthesis is carried out by

- (a) cladode
- (b) phyllode
- (c) phyllociade
- (d) stipules

Answer: C

Physics

Q832.

Identify the parameter which measures the time it takes to complete one cycle?

- (a) period
- (b) wavelength
- (c) frequency
- (d) amplitude E. speed

Answer: A

Biology

Q833.

The breakdown of complex components of food into simple ones is called as

- (a) Respiration
- (b) Digestion
- (c) Excretion
- (d) Absorption

Answer: B

Physics

Q834.

A simple pendulum of length 4 m is taken to a height R (radius of the earth) from the earth's surface. The time period of small oscillations of the pendulum is ($g_{\text{surface}} = \pi^2 \text{ m s}^{-2}$)

- (a) 2 s
- (b) 4 s c · 8 s
- (d) 16 s

Answer: C

Q835.

With increase in frequency of an A.C. supply, the inductive reactance

- (a) decreases
- (b) increases directly with frequency
- (c) increases as square of frequency
- (d) decreases inversely with frequency

Answer: B

Q836.

According to Curie's law, the magnetic susceptibility of a substance at an absolute temperature T is proportional to -

- (a) $1 / T \propto T$
- (c) $1 / T^2$
- (d) T^2

Answer: A

Chemistry

Q837.

Assertion The rate of reaction can also increase w.r.t its product if one of the product act as catalyst. Reason A catalyst lowers the activation energy of reactions. Read the above assertion and reason and choose the correct option regarding it.

- (a) Both Assertion and Reason are true Reason is the correct explanation to Assertion
- (b) Both Assertion and Reason are true but Reason is not the correct explanation to Assertion
- (c) Assertion is true but Reason is false
- (d) Both Assertion and Reason are false

Answer: A

Biology

Q838.

Identify the given figures A, B, C, D and E. E

- (a) A: Marginal, B: Axile, C: Free central, D: Parietal, E: Basal
- (b) A: Marginal, B: Parietal, C: Free central, D: Axile, E: Basal
- (c) A: Marginal, B: Axile, C: Parietal, D: Free central, E: Basal
- (d) A: Marginal, B: Axile, C: Parietal, D: Basal, E: Free central

Answer: A

Physics

Q839.

"I is smooth wall, as shown in the diagram, making a variable angle θ with the horizontal floor. There is no friction with the vertical wall. At which angle θ will the normal force exerted by the floor upwards be at its maximum? Do not consider any motion by the plank (consider the forces exerted when the plank is perfectly still for a given angle θ)

- (a) When $\theta=0.00^{\circ}$ (the plank is horizontal)
- (b) When $\theta=45.0^{\circ}$
- (c) When $\theta=90.0^{\circ}$ (the plank is vertical)
- (d) The angle θ makes no difference E. More information is needed

Answer: D

Q840.

A fly strikes the windshield of a moving truck as shown. What can you say about the forces: the force on the fly and the force on the truck.

- (a) Force on the fly > Force on the truck
- (b) Force on the fly < Force on the truck
- (c) Force on the fly = Force on the truck
- (d) None of these

Answer: C

Q841.

The work done by an external agent to shift a point mass from infinity to the centre of the earth is W. Then choose the correct relation.

- (a) $w=0$
- (b) $w>0$ c · $w<0$
- (d) $w \leq 0$

Answer: C

Chemistry

Q842.

What is the name for the number of protons in an atom?

- (a) Atomic identity
- (b) Atomic mass
- (c) Atomic weight
- (d) Atomic number

Answer: D

Q843.

Violet coloured complex obtained in the detection of sulphur is : A · N a₂[F e(N O)(C N)₅]

- (b) N a₃[Fe(ONSNa)(CN)₅].
- (c) N a₄[F e(C N)₅ N O S].
- (d) Both (B) and (C)

Answer: D

Biology

Q844.

Assertion Coenzymes can be vitamins or minerals. Reason NAD is an example of a coenzyme.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion.
- (c) Assertion is correct but Reason is incorrect.
- (d) Assertion is incorrect but Reason is correct.

Answer: D

Chemistry

Q845.

A company which makes the cases for mobile phones requires a material that should be opaque, light, a bad conductor of heat and electricity, and which can be moulded into material would the company prefer?

- (a) Mol ■. ■■■ c . ■■4
- (d) ■■■

Answer: *(plastic/polymer)*

Physics

Q846.

Calculate the electrostatic potential energy at the instant when the alpha particle stops?

- (a) 36.3 MeV
- (b) 45.0 MeV
- (c) 3.63 MeV
- (d) 40.0 Mev

Answer: C

Q847.

As shown in the figure, force of 105 N each are applied in opposite directions, on the upper and lower force of a cube of side 10 cm, shifting the upper face parallel to itself by 0.5 cm . If the side of another cube of the same material is 20 cm, then under similar condition as shown as above, the displacement will be: A · 0.25 cm

- (b) 0.37 cm
- (c) 0.75 cm

(d) 1.00 cm

Answer: A

Q848.

Assertion: By roughening the surface of a glass sheet its transparency can be reduced. Reason: Glass sheet with rough surface absorbs more light.

- (a) Both assertion and reason are true and reason is the correct explanation of assertion
- (b) Both assertion and reason are true but the reason is not the correct explanation of assertion
- (c) Assertion is true but reason is false
- (d) Both the assertion and reason are false E. Reason is true but assertion is false

Answer: C

Q849.

If a magnetic material is having magnetic susceptibility (χ)=-1, then the relative magnetic permeability (μ_r) and type of magnetic material is?

- (a) 0, diamagnetic
- (b) 2, ferromagnetic
- (c) 1, paramagnetic
- (d) -1, diamagnetic

Answer: A

Q850.

Which of the following is not a characteristic of the standard unit?

- (a) It should be of convenient size
- (b) It should change with respect to space and time
- (c) It should not be perishable
- (d) It should be eerily reproducible

Answer: B

Q851.

The inertia of a moving body makes it:

- (a) stop due to friction
- (b) increase its speed
- (c) decrease its speed
- (d) resist a change in its state of motion

Answer: D

Biology

Q852.

Which of the following chemical cause the thinning of layer P?

- (a) Carbon monoxide
- (b) sulphur dioxide
- (c) chlorofluorocarbons
- (d) Both A and C

Answer: C

Chemistry

Q853.

The least count of burette used normally is x (in mL). The value of 10 x is: A

- (b) 1.5

- (c) 0.10
(d) none of these

Answer: C

Physics

Q854.

Two electric charges of $9 \mu\text{C}$ and $-3 \mu\text{C}$ are placed 0.16 m apart in air. There will be a point P at which electric potential is zero on the line joining the two charges and in between them. The distance of P from $9 \mu\text{C}$ charge is: A .0 .14 m

- (b) 0.12 m
(c) 0.08 m
(d) 0.06 m

Answer: B

Q855.

Is a circular ring made of a uniform wire. Current enters and leaves the ring through straight conductors which, if produced, would have passed through the centre C of the ring. The magnetic field at C: This question has multiple correct options

- (a) due to the straight conductors is zero
(b) due to the loop is zero
(c) due to the loop is proportional to θ
(d) due to the loop is proportional to $(\pi - \theta)$

Answer: A,D

Chemistry

Q856.

From Bohr's model, radius of the orbit and energy of the hydrogen atom are calculated as: This question has multiple correct options

- (a) the radius of the 1st orbit is $0.529 \text{ \AA}^{\text{circ}}$
(b) the energy of the 1st orbit is -13.6 eV
(c) the energy of the 1st orbit is $-21.72 \times 10^{-19} \text{ J}$
(d) the radius of the 1st orbit is $1.06 \text{ \AA}^{\text{circ}}$

Answer: A,B,C

Physics

Q857.

The path followed by a projectile is called its:

- (a) trajectory
(b) range
(c) amplitude
(d) none of these

Answer: A

Q858.

In secondary rainbow what is the order of colours observed from earth?

- (a) Violet innermost, Red outermost
(b) Red innermost, Violet outermost
(c) Random

(d) White and dark alternatively

Answer: B

Q859.

The force primarily responsible for the existence of the solar system is the

- (a) force of friction
- (b) gravitational force
- (c) electrostatic force
- (d) magnetic force

Answer: B

Q860.

How many seconds are there in a week?

- (a) 60480
- (b) 604800
- (c) 6048000
- (d) 6048

Answer: B

Biology

Q861.

The enzyme which has polymerising activity in $5' \rightarrow 3'$ direction but exonuclease activity in $3' \rightarrow 5'$ direction only is :

- (a) RNA polymerase III
- (b) DNA polymerase II
- (c) DNA polymerase
- (d) All of these

Answer: B

Chemistry

Q862.

Two particles A and B are in motion. If the wavelength associated with particle A is 5×10^{-8} m. Calculate the wavelength associated with particle B if its momentum is half of A

- (a) 5×10^{-8} m
- (b) 10^{-5} cm
- (c) 10^{-6} cm
- (d) 5×10^{-8} cm

Answer: B

Q863.

Ratio of solubilities of gases N_2 & O_2 in water from air at $25^\circ C$ and 1 atm will be if air is 20 % by volume of O_2 and 80 % by volume of N_2 is: Given : $K_H(N_2) = 2 \times 10^4$ atm and $K_H(O_2) = 1 \times 10^4$ atm

- (a) 1: 8
- (b) 2: 1
- (c) 1: 2
- (d) 8: 1

Answer: D

Q864.

Chief source of soil and water pollution is:

- (a) mining
- (b) agro industry

- (c) thermal power plant
- (d) all the three

Answer: D

Physics

Q865.

Which of the following property shows the property of ferromagnetic substances?

- (a) The ferromagnetic property depends on temperature
- (b) The ferromagnetic property does not depend on temperature.
- (c) At high enough temperature ferromagnet becomes a diamagnet
- (d) At low temperature ferromagnet becomes a paramagnet

Answer: A

Chemistry

Q866.

J J Thomson Model could not able to explain the following?

- (a) Stability of Atom
- (b) electrical neutrality of atom
- (c) stability of nucleus
- (d) all of these

Answer: A

Q867.

If 9 g of H_2O is electrolysed completely with the current of 50 % efficiency then :

- (a) 96500 charge is required
- (b) 2×96500 C charge is required
- (c) 5.6 L of O_2 at STP will be formed
- (d) 11.2 L of O_2 at STP will be formed

Answer: B

Biology

Q868.

Drinking aerated drink with low levels (0.02 ppm) of pesticide for long period would cause

- (a) cancer of intestine
- (b) Accumulation in the body
- (c) Leukaemia
- (d) Immunity against mosquito

Answer: B

Q869.

Which of the following system of classification is based on gross morphology of plants?

- (a) Natural system
- (b) Artificial system
- (c) Practical system
- (d) All system

Answer: B

Chemistry

Q870.

Which of the following statements is not correct?

- (a) In a chemical change, color of the object may change.
- (b) In a chemical change, heat will always be released.
- (c) A chemical change always involves a physical change.
- (d) Rusting of iron becomes faster when humidity is high.

Answer: B

Physics

Q871.

A body is suspended from a spring balance kept in a satellite. The reading of the balance is W_1 when the satellite goes in an orbit of radius R and is W_2 when it goes in an orbit of radius $2R$. Then

- (a) $W_1 = W_2$
- (b) $W_1 < W_2$ c · $W_1 > W_2$
- (d) $W_1 \neq W_2$

Answer: A

Biology

Q872.

In the lac operon, the structural genes are switched off when repressor binds to the

- (a) Operator
- (b) Promoter
- (c) Regulator
- (d) Inducer E. Allolactose

Answer: A

Chemistry

Q873.

Sulphur dioxide levels in the atmosphere can be reduced by using:

- (a) catalytic converters in industry
- (b) Static electricity to attract it in Factory chimneys
- (c) More efficient car-engines.
- (d) Low-sulphur fuels

Answer: D

Biology

Q874.

Water transport from roots to leaves is explained by

- (a) The pressure flow theory
- (b) Differences in source and sink solute concentrations
- (c) The pumping force of xylem vessels
- (d) The cohesion-tension theory

Answer: D

Physics

Q875.

When all the inputs of a NAND gate are connected together, the resulting circuit is:

- (a) a NOT gate
- (b) an AND gate
- (c) an OR gate
- (d) a NOR gate

Answer: A

Chemistry

Q876.

Give the number of neutrons in an atom of the isotope ${}_{47}^{107}\text{Ag}$

- (a) 59
- (b) 60
- (c) 61
- (d) 58

Answer: B

Q877.

The equilibrium constant for a reaction is K, and the reaction quotient is Q. For a reaction mixture, the ratio K/Q is 0.33. This means that:

- (a) the reaction mixture will equilibrate to form more reactant species
- (b) the reaction mixture will equilibrate to form more product species
- (c) the equilibrium ratio of reactant to product concentrations will be 3
- (d) the equilibrium ratio of reactant to product concentrations will be 0.33

Answer: A

Physics

Q878.

A ball is thrown from the top of a tower with an initial velocity of 10 m/s at an angle of 30° above the horizontal. It hits the ground at a distance of 17.3 m from the base of the tower. The height of the tower ($g = 10 \text{ m/s}^2$) will be

- (a) 10 m
- (b) 110 m
- (c) 12 m
- (d) 100 m

Answer: A

Q879.

The number of degrees of freedom for a gas system is

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: B

Q880.

Suppose your writing desk is tilted a little, a book kept on it starts sliding down. The direction of frictional force acting on it is

- (a) upwards

- (b) downwards
- (c) no direction
- (d) cant say

Answer: A

Q881.

An ideal monoatomic gas at 27°C is compressed adiabatically to $8/27$ times of its present volume. The increase in temperature of the gas is: A $.375^{\circ}\text{C}$

- (b) 402°C
- (c) 175°C
- (d) 475°C

Answer: D

Chemistry

Q882.

Define homologous series.

- (a) Compound with same melting point
- (b) Compound which are stereo isomers
- (c) Compounds which have different parameters
- (d) Compound with same general formula

Answer: D

Q883.

In nucleic acids, the sequence is:

- (a) phosphate-sugar-base
- (b) sugar-acid-phosphate
- (c) sugar-base-phosphate
- (d) base-phosphate-sugar

Answer: A

Q884.

Atomic number of strontium (Sr) is equal to:

- (a) 28
- (b) 38
- (c) 42
- (d) 34

Answer: B

Q885.

Carbon and hydrogen are estimated in organic compounds by:

- (a) Kjeldahl's method
- (b) Duma's method
- (c) Liebig's method
- (d) Carius method

Answer: C

Q886.

The vapour pressure is least for?

- (a) pure water
- (b) 0.1 m aqueous urea
- (c) 0.2 m aqueous urea
- (d) 0.3m aqueous urea

Answer: D

Q887.

Given below is a redox reaction. Which of the following types the reaction belong to? $\text{Cu}^{+2} + \text{SO}_4^{2-} + \text{Zn} \rightarrow \text{Cu}_{(s)} + \text{ZnSO}_{4(aq)}$

- (a) Combination reaction
- (b) Decomposition reaction
- (c) Metal displacement reaction
- (d) Non-metal displacement reaction

Answer: C

Q888.

Diethyl amine when treated with nitrous acid yields:

- (a) diethyl ammonium nitrite
- (b) ethyl alcoho
- (c) N-nitroso diethyl amine
- (d) triethyl ammonium nitrite

Answer: C

Q889.

Complete the correlation. Silk worm: Cocoon : : Wool:

- (a) Hair of rabbitt
- (b) Cotton plant
- (c) Fleece of sheep
- (d) Jute of hemp

Answer: C

Q890.

The reactivities of methanol (I), 1- propanol (II), 2 - butanol (III) and 2 methyl-2- propanol (IV) towards sodium metal follow the order: A . > > > > V

- (b) IV > III > II > I
- (c) I > IV > II > III
- (d) IV > II > III > I

Answer: A

Q891.

How many grams of NaOH are needed to make 100 g of a 5 % solution? A . 2

- (b) 5
- (c) 20
- (d) 40 E . 95

Answer: B

Biology

Q892.

The plasma membrane of animal cells is elastic due to the presence of

- (a) Lipids
- (b) Carbohydrates
- (c) Proteins
- (d) water

Answer: A

Chemistry

Q893.

Elements with high melting point are present in:

- (a) p-block elements
- (b) s-block elements
- (c) d-block elements
- (d) None of these

Answer: C

Q894.

Assertion: Thallium (III) hydroxide is highly soluble in water. Reason: Thallium (III) hydroxide is the most basic among 13th group hydroxides.

- (a) Assertion is True, Reason is True, Reason is a correct explanation of Assertion
- (b) Assertion is True, Reason is True, Reason is not a correct explanation of Assertion
- (c) Assertion is True, Reason is False
- (d) Assertion is False, Reason is True

Answer: D

Physics

Q895.

A telescope gives an (inverted erect) and (enlarged/diminished) image of a distant object. Choose the correct alternative.

- (a) Inverted, diminished
- (b) Inverted, enlarged
- (c) Erect, diminished
- (d) Erect, enlarged

Answer: B

Chemistry

Q896.

In a mixture of acetic acid and sodium acetate, the ratio of concentrations of the salt to the acid is increased ten times. Then the p H of the solution:

- (a) Increases by one
- (b) Decreases by one
- (c) Decreases ten fold
- (d) Increases ten fold

Answer: A

Q897.

Which of the following is an elastomer?

- (a) Vulcanized rubber
- (b) Dacron
- (c) Polystyrene
- (d) Melamine

Answer: A

Q898.

The position in the periodic table to which F e and Z n elements belong to

- (a) Alkali metal
- (b) Alkaline earth metal
- (c) d-block

(d) None of these

Answer: C

Physics

Q899.

Assertion For a particle performing SHM, its speed decreases as it goes away from the mean position. Reason In SHM, the acceleration is always opposite to the velocity of the particle.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: D

Chemistry

Q900.

Primary battery is a battery:

- (a) which can be recharged.
- (b) which cannot be reconditioned by replacing chemical.
- (c) which cannot be reused.
- (d) which cannot be recharged.

Answer: D

Q901.

Bohr modified the adding that the electrons move in orbits of fixed size and energy.

- (a) Rutherford's model
- (b) Dalton's model
- (c) chadwick's model
- (d) Goldstein's model

Answer: A

Biology

Q902.

Stomata having pores bounded by a single ring shaped guard cell are found in the

- (a) Leaf of ferns
- (b) Pinnule of Cycas.
- (c) capsule of Funaria
- (d) All of the above

Answer: D

Physics

Q903.

A diverging electron beam enters a region of varying potentials. Intersection of equispaced equipotential plane surfaces with the plane of the paper are shown, with their potentials mentioned in the figure. If before entering this region, x and y components of the velocity of electrons are $3\sqrt{(2eV)/(m)}$ and $\sqrt{(eV)/(m)}$, respectively, where m is the mass of an electron and e is the electronic charge, then : This question has multiple correct options

- (a) the electron beam is converged
- (b) the electron beam is diverged
- (c) the electric field is uniform and is directed towards negative x-axis
- (d) the electric field is uniform and is directed towards positive x-axis

Answer: A,C

Q904.

A drop of liquid having radius 2 mm has a terminal velocity 20 cm s^{-1} , the terminal velocity of a drop 1mm radius will be:

- (a) 40 cm s^{-1}
- (b) 20 cm s^{-1}
- (c) 10 cm s^{-1} D · 5 cm s^{-1}

Answer: D

Chemistry

Q905.

A reaction $A+B \rightarrow C$ is second order with respect to A and independent of B The rate expression for the reaction is:

- (a) $\text{rate} = K[A][B]$
- (b) $\text{rate} = K[A]^2[B]$
- (c) $\text{rate} = K[A]^2[B]^2$
- (d) $\text{rate} = K[A]^2$

Answer: D

Biology

Q906.

Multiple Choice Questions Depletion of which gas in the atmosphere can lead to an increased incidence of skin cancers.

- (a) Methane
- (b) Nitrous oxide
- (c) Ozone
- (d) Ammonia

Answer: C

Physics

Q907.

Human body temperature is well measured by

- (a) A clinical thermometer
- (b) A laboratory thermometer
- (c) Thermostat
- (d) Both A & B can be used

Answer: A

Q908.

A machine gun fires 10 bullets/s, each of mass 10 g , the speed of each bullet is 20 c m / s, then force of recoil is

- (a) 200 dyne
- (b) 2000 dyne

(c) 20 dyne

(d) None

Answer: A

Chemistry

Q909.

Which of the following options is not correct?

(b) 126,000 has 3 significant figures

(c) $15.15 \mu s = 1.515 \times 10^{-5} s$

(d) $0.0048 = 48 \times 10^{-3}$

Answer: B

Q910.

Which of the following compounds will not undergo Friedel-Craft reaction easily?

(a) cumene

(b) xylene

(c) nitrobenzene

(d) toluene

Answer: C

Q911.

Assertion A small amount of acid or alkali is added before electrolysis of water. Reason Pure water is weak electrolyte.

(a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion

(b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion

(c) Assertion is correct but Reason is incorrect

(d) Assertion is incorrect but Reason is correct.

Answer: A

Q912.

Which one given below is a nonreducing sugar?

(a) Maltose

(b) Lactose

(c) Glucose

(d) sucrose

Answer: D

Q913.

Read the following passage and answer the questions at the end of it. Dilute solution of alkali metals in liquied NH_3 is blue. It is the ammoniated electron which is responsible for the blue color of the solution and the electrical conductivity is due to the ammoniated cation, $[M(NH_3)_x]^+$ as well as the ammoniated electron, $[e(NH_3)_y]^-$ values of x and y depend on the extent of solvation (by NH_3). Dilute solutions are paramagnetic due to free electrons. What happens if alkali metal is allowed to react with concentrated ammonia solution?

(a) Paramagnetic character of solvated electrons is retained

(b) Solvated electrons associate to form electron-pairs and paramagnetic character decreases

(c) Reducing character is increased

(d) Reducing character is not affected

Answer: B

Q914.

A compound of sodium does not give CO_2 when heated but it gives CO_2 when treated with dilute acids. A crystalline compound is found to have 37.1% Na and 14.52% H_2O . Hence, the compound is :

- (a) $\text{NaHCO}_3 \cdot 10\text{H}_2\text{O}$
- (b) $\text{NaHCO}_3 \cdot 5\text{H}_2\text{O}$
- (c) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
- (d) $\text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O}$

Answer: C

Physics

Q915.

Two equal charges q are kept fixed at a and $+a$ along the x -axis. A particle of mass m and charge $q/2$ is brought to the origin and given a small displacement along the x -axis, then:

- (a) the particle executes oscillatory motion
- (b) the particle remains stationary
- (c) the particle executes, SHM along x -axis
- (d) the particle executes SHM along y -axis

Answer: C

Chemistry

Q916.

$E_n = -3.13 \times 10^{-18} / n^2 \text{ k cal/mol}$. If the value of $E = -3.4 \times 10^{-18} \text{ k cal/mol}$, to which value does 'n' correspond? A . 4

- (b) 3
- (c) 2
- (d)

Answer: B

Q917.

Which one of the following undergoes nucleophilic substitution exclusively by S_N1 mechanism?

- (a) Benzyl chloride
- (b) Ethyl chloride
- (c) chlorobenzene
- (d) Isopropyl chloride

Answer: A

Physics

Q918.

When an object is placed between optical centre and focus of lens:

- (a) Inverted, real and enlarged image is formed
- (b) Virtual, erect and enlarged image is formed
- (c) Inverted, erect, diminished image is formed
- (d) Virtual, real, diminished image is formed

Answer: B

Biology

Q919.

Catalytic converter used for controlling pollution is used in

- (a) Thermal plants
- (b) Chimneys

- (c) Brick kilns
- (d) Motor vehicles

Answer: D

Physics

Q920.

In a LR circuit of 3 m H inductance and 4 Ω resistance, emf $E=4$ (1000 t) volt is applied. The amplitude of current is: A · 0.8 A

- (b) 4/7
- (c) 1.0 A
- (d) $(4)/(\sqrt{7})$ A

Answer: D

Q922.

Which of the following pair of quantities do not have the same dimensions?

- (a) Potential gradient, electric field
- (b) Torque, kinetic energy
- (c) Light year, time period
- (d) Impedance, reactance

Answer: C

Chemistry

Q923.

Identify a metalloid from the following list of elements.

- (a) carbon
- (b) Neon
- (c) sodium
- (d) Tellurium

Answer: D

Q924.

What is not true about polymers?

- (a) Polymers do not carry any charge.
- (b) Polymers have high viscosity.
- (c) Polymers scatter light.
- (d) Polymers have low molecular weight.

Answer: D

Biology

Q925.

Enzyme renin is secreted by

- (a) Cells of stomach
- (b) Cells of intestine
- (c) The cortical cells of kidney
- (d) The cells of juxtaglomerular apparatus of kidney

Answer: A

Chemistry

Q926.

The value of K_c for the reaction $3\text{O}_2(\text{g}) \rightleftharpoons 2\text{O}_3(\text{g})$ is 1.7×10^{-56} at 25°C . Do you expect pure air at 25°C to contain much O_3 (ozone) when O_2 and O_3 are in equilibrium? If the equilibrium concentration of O_2 in air at 25°C is $8 \times 10^{-3} \text{ M}$, what is the equilibrium concentration of O_3 ? A · Yes, $3 \times 10^{-32} \text{ mol / L}$

- (b) Yes, $6 \times 10^{-32} \text{ mol / L}$
- (c) Yes, $9 \times 10^{-32} \text{ mol / L}$
- (d) None of these

Answer: A

Q927.

Which of the following compounds can be purified by vacuum distillation? This question has multiple correct options

- (a) Glycerine
- (b) Glycerol
- (c) Propane-1,2, 3-triol
- (d) Ethanol

Answer: A,B,C

Q928.

The correct IUPAC name of the given compound is:

- (a) N-dimethyl 1-methyl-1-ethyl propane
- (b) 3-methyl-3(N, N-dimethyl amino) pentane
- (c) N-dimethyl amino-3-methyl pentane
- (d) 3, N, N - trimethyl-3-pentanamine

Answer: D

Biology

Q929.

Absorption of mineral elements from the soil occurs through

- (a) Diffusion
- (b) Active transport
- (c) osmosis
- (d) All of the above

Answer: B

Q930.

In humans, blood vascular system has

- (a) Double circulation
- (b) Venous heart
- (c) Arterial heart
- (d) Arteriovenous heart

Answer: A

Chemistry

Q931.

Which of the following statements is correct about the equilibrium constant?

- (a) Its value increases by increase in temperature
- (b) Its value decreases by decrease in temperature

- (c) Its value may increase or decrease with increase in temperature
 (d) Its value is constant at all temperature

Answer: C

Q932.

Assertion Sodium metal is stored under Kerosene. Reason Metallic sodium melts when exposed to air.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
 (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
 (c) Assertion is correct but Reason is incorrect
 (d) Assertion is incorrect but Reason is correct

Answer: C

Q933.

According to Bohr's model, the angular momentum of an electron in 4th orbit is: $A \cdot h / 3$

- (b) $h / 2$
 (c) $2 h / \pi$
 (d) none of these

Answer: C

Physics

Q934.

A steady current flows in a metallic conductor of non-uniform cross-section. The quantity/quantities constant along the length of the conductor is /are

- (a) current, electric current and drift velocity
 (b) drift speed only
 (c) current and drift speed only
 (d) current only

Answer: D

Q935.

The electric field at a distance $(3R)/2$ from the centre of a charged conducting spherical shell of radius R is E . The electric field at a distance R/2 from the centre of the sphere is :

- (a) zero
 (b) E
 (c) $E/2$
 (d) $E/3$

Answer: A

Chemistry

Q936.

Consider the following salts: NaCl, HgCl₂, Hg₂Cl₂, CuCl₂, CuCl and AgCl . Identify the correct set of insoluble salts in water.

- (a) Hg₂Cl₂, CuCl, AgCl, HgCl₂, CuCl, AgCl
 (c) Hg₂Cl₂, CuCl₂, AgCl
 (d) Hg₂Cl₂, CuCl, NaCl

Answer: A

Q937.

$A \cdot O_{1/2} A_2 + (1/2) O_2$; $K = 5 \times 10^5$ $B \cdot O_{1/2} B_2 + (1/2) O_2$; $K = 1.10 \times 10^1$ $C \cdot O_{1/2} C_2 + (1/2) O_2$; $K = 2.3 \times 10^{18}$ $D \cdot O_{1/2} D_2 + (1/2) O_2$; $K = 1.4 \times 10^{21}$ Which oxide is most stable? A . A O

- (b) B O
- (c) C O
- (d) D O

Answer: D

Q938.

The geometry of XeF_6 is:

- (a) planar hexagon
- (b) regular octahedron
- (c) distorted octahedron
- (d) square bipyramid

Answer: C

Physics

Q939.

The motion of centre of mass depends on

- (a) total external forces
- (b) total internal forces
- (c) sum of (a) and (b)
- (d) None of these

Answer: A

Biology

Q940.

How many nodes are found in heart of human?

- (a) one
- (b) Two
- (c) many
- (d) None of the above

Answer: B

Physics

Q941.

In a semiconductor, which of the following statement is correct?

- (a) At 0 K, Si is a super conductor.
- (b) In a p-type semiconductor the acceptor level lies near the conduction band.
- (c) Each donor atom contributes one hole.
- (d) p-n junction is electrically neutral

Answer: D

Chemistry

Q942.

An oxidation process involves:

- (a) increase in oxidation number
- (b) Decrease in oxidation number
- (c) Addition of electrons

(d) Removal of electronegative element

Answer: A

Biology

Q943.

In Spirogyra, asexual reproduction takes place by

- (a) Breaking up of filaments into smaller bits.
- (b) Division of a cell into two cells.
- (c) Division of a cell into many cells.
- (d) Formation of young cells from older cells.

Answer: A

Chemistry

Q944.

The chelating ligand used to remove excess of copper and iron in chelate therapy is:

- (a) D-penicill amine
- (b) oxalate ion
- (c) EDTA
- (d) ethylenediamine E. dimanyl glyoxime

Answer: A

Biology

Q945.

Which one from the following is associated with occupational hazard?

- (a) Aurorosis
- (b) Pneumoconiosis
- (c) silicosis
- (d) Both B and C

Answer: D

Physics

Q946.

Consider the following pairs of quantities: 1. Young's modulus; pressure 2. Torque; energy 3. Linear momentum; work 4. Solar day; light year In which cases are the dimensions, within a pair, same?

- (a) 1 and 3
- (b) 1 and 4 c · 1 and 2
- (d) 2 and 4

Answer: C

Q947.

The value of g at a particular point is 9.8 m s^{-2} . Suppose the earth suddenly shrinks uniformly to half its present size without losing any mass. The value of g at the same point (distance of the point from the centre of earth does not change) will now be

- (a) 9.8 m s^{-2}
- (b) 4.9 m s^{-2} c · 19.6 ms^{-2}
- (d) 39.2 ms^{-2}

Answer: A

Chemistry

Q948.

Which of the following pair consists of ore of the same metal? This question has multiple correct options

- (a) Magnesite, Cerussite
- (b) Chalcocite, Copper pyrites
- (c) Bauxite, Corundum
- (d) Anglesitc, Cerussite

Answer: B,C,D

Q949.

What is the density of wet air with 75 % relative humidity at 1 atm and 300 K ? Given : vapour pressure of H_2O is 30 torr and average molar mass of air is 29 g / m o l

- (a) 1.614 g / L ■. 0.96 g / L
- (c) 1.06 g / L
- (d) 1.164 g / L

Answer: D

Biology

Q950.

Carnassial teeth are modified for

- (a) Crushing
- (b) Tearing
- (c) Grinding
- (d) cutting

Answer: D

Q951.

Which one of the following shows bilateral symmetry?

- (a) Virus
- (b) Bacteria
- (c) Amoeba
- (d) Planaria E. sponge

Answer: D

Physics

Q952.

Fill in the blank: In order to increase the capacity of a parallel plate condenser, one should introduce a sheet of between the plates (assume that the space is completely filled).

- (a) Mica
- (b) Tin
- (c) copper
- (d) Stainless Steel

Answer: A

Q953.

A thick shell with inner radius R and outer radius $3R$ has a uniform charge density $\sigma \text{ C/m}^3$. It has a spherical cavity of radius R as shown in the figure. The electric field at the centre of the cavity is

- (a) zero
- (b) $2\sigma R / \epsilon_0$
- (c) $3\sigma R / 4\epsilon_0$
- (d) $7\sigma R / 12\epsilon_0$

Answer: B

Q954.

A current is induced in a coil by electromagnetic induction.

- (a) Only when coil moves in the magnetic field
- (b) Only when a magnetic is moved into a coil
- (c) Only when the coil and magnet move with respect to each other
- (d) None of the above

Answer: C

Biology

Q955.

Reflex action is

- (a) Stimulus $\rightarrow$ sensory neuron $\rightarrow$ spinal cord $\rightarrow$ motor neuron $\rightarrow$ response
- (b) Stimulus $\rightarrow$ motor neuron $\rightarrow$ sensory neuron $\rightarrow$ response
- (c) Reception $\rightarrow$ motor neuron $\rightarrow$ sensory neuron $\rightarrow$ response
- (d) Sensory neuron $\rightarrow$ stimulus $\rightarrow$ motor neuron $\rightarrow$ response

Answer: A

Chemistry

Q956.

The first inorganic pesticide to be used commercially was

- (a) Bordeaux mixture
- (b) Burgandy mixture
- (c) D.D.T.
- (d) 2-4 D

Answer: A

Physics

Q957.

A needle of length 5 cm, placed 45 cm from a lens forms an image on a screen placed 90 cm on the other side of the lens. The type of lens and its focal length are :

- (a) convex, 30 cm
- (b) concave, 30 cm
- (c) convex, 60 cm
- (d) concave, 60 cm

Answer: A

Q958.

The co-ordinates of a moving particle at any time t are given by $x = \alpha t^3$ and $y = \beta t^3$. The speed of the particle at time t is given by $A \cdot t^2 \sqrt{(\alpha^2 + \beta^2)}$

- (b) $3 t^2 \sqrt{(\alpha^2 + \beta^2)}$

- (c) $3 \sqrt{\alpha^2 + \beta^2}$
(d) $\sqrt{\alpha^2 + \beta^2}$

Answer: B

Q959.

A stick is thrown in the air and lands on the ground at some distance from the thrower. The centre of mass of the stick will move along a parabolic path

- (a) in all cases
(b) only if the stick is uniform
(c) only if the stick has linear motion but no rotational motion
(d) only if the stick has a shape such that its centre of

Answer: A

Q960.

A substance contains 35.0 g nitrogen, 5.05 g hydrogen, and 60.0 g oxygen. How many grams of oxygen are there in a 200 g sample of the substance?

- (a) 10.1 g
(b) 60.0 g
(c) 70.0 g
(d) 120.9 g

Answer: D

Biology

Q961.

The thylakoids occur inside

- (a) Mitochondria
(b) Chloroplast
(c) Golgi apparatus
(d) Endoplasmic reticulum

Answer: B

Q962.

In pachytene stage of meiosis, the chromosomes appear

- (a) single stranded
(b) Double stranded
(c) Three stranded
(d) Four stranded

Answer: D

Chemistry

Q963.

Which of the following statements is not true?

- (a) The ratio of the mean speed of the RMS speed is independent of the temperature
(b) The square of the mean speed of the molecules is equal to the mean square speed at a certain temperature
(c) Mean kinetic energy of the gas-molecules at any given temperature is independent of the mean speed
(d) The difference between RMS speed and mean speed at any temperature for different gases diminishes as larger and yet larger molar masses are considered

Answer: B

Physics

Q964.

The displacement-time graph of a particle executing SHM is shown in the figure. Then This question has multiple correct options

- (a) the velocity is maximum at $t = T / 2$
- (b) the acceleration is maximum at $t = T$
- (c) the force is zero at $t = 3 T / 4$
- (d) the potential energy equals the oscillation at $t = T/2$

Answer: B,C,D

Q965.

The height at which the value of acceleration due to gravity becomes 50 % of that at the surface of the earth. (radius of the earth = 6 4 0 0 k m) is

- (a) 2650 ■. 2430
- (c) 2250
- (d) 2350

Answer: A

Biology

Q966.

Dictyostelium is

- (a) Cellular slime mould
- (b) Plasmodial slime mould
- (c) Water mould
- (d) None of the above

Answer: A

Physics

Q967.

When air is replaced by a dielectric medium of force constant K , the maximum force of attraction between two charges, separated by a distance :

- (a) decreases K times
- (b) increases K times
- (c) remains unchanged
- (d) becomes $(1)/(K^2)$ times

Answer: A

Q968.

Assertion When current through a bulb decreases by 0.5 %, the glow of bulb decreases by 1 % Reason Glow (Power) which is directly proportional to square of current.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion
- (c) Assertion is correct but Reason is incorrect
- (d) Both Assertion and Reason are incorrect

Answer: A

Q969.

The transformer voltage induced in the secondary coil of a transformer is mainly due to:

- (a) a varying electric field

- (b) a varying magnetic field
- (c) the vibrations of the primary coil
- (d) the iron core of the transformer

Answer: B

Q970.

A substance is placed in a non uniform magnetic field. It experiences weak force towards the strong field. The substance is _____ type.

- (a) Ferromagnetic
- (c) Paramagnetic
- (d) None of these

Answer: C

Biology

Q971.

Which one of the following has plantings life cycle?

- (a) Funaria
- (b) Polytrichum
- (c) ustilago
- (d) Triticum

Answer: D

Q972.

The colour of lymph is A · Red

- (b) Pale yellow
- (c) colourless
- (d) Milky

Answer: B

Physics

Q973.

Person who use boolean algebra for describing the operation of logic gates first was

- (a) Boole
- (b) Shannon
- (c) schottky
- (d) zener

Answer: B

Q974.

Two identical conductors of copper and aluminium are placed in an identical electric field. The magnitude of induced charge in the aluminium will be :

- (a) zero
- (b) Greater than in copper
- (c) less than in copper
- (d) Equal to that of copper

Answer: D

Q975.

The lens used to correct long sightedness or hypermetropla.

- (a) Convex lens

- (b) Concave lens
- (c) simple lens
- (d) Cylindrical lens

Answer: A

Chemistry

Q976.

The solubility of water decreases rapidly with the increase in the size of groups attached to carbonyl group.

- (a) alkyl
- (b) aryl
- (c) phenyl
- (d) benzyl

Answer: A

Q977.

5 mol of liquid water is compressed from 1 bar to 10 bar at constant temperature. Change in Gibb's energy (ΔG) in Joule is: (Density of water = 1000 kg / m^3)

- (a) 18
- (b) $225 \text{ c} \cdot 450$
- (d) 900

Answer: C

Physics

Q978.

Nuclear fusion produces: A · argon

- (b) deuterium
- (c) helium
- (d) krypton

Answer: C

Q979.

Magnitude of e.m.f. produced in a coil when a magnet is inserted into it does not depend upon:

- (a) Number of turns in the coil
- (b) Speed of the magnet
- (c) Magnetic strength of magnet
- (d) Temperature of coil

Answer: D

Q980.

With what speed should a bus travel so that it can cover a distance of 10 km in 5 ? A · 60 km / hr

- (b) 10 km/hr
- (c) 12 km / hr
- (d) 120 km/hr

Answer: D

Chemistry

Q981.

Organic compounds are numerous since:

- (a) carbon has high catenation ability
- (b) tetravalency of carbon
- (c) isomerism of organic compounds
- (d) All of these

Answer: D

Q982.

Assertion Nylon wire is stronger than steel wire. Reason Nylon is a fibre while steel is an alloy.

- (a) Both assertion and reason are correct and reason is the correct explanation for assertion
- (b) Both assertion and reason are correct but reason is not the correct explanation for assertion
- (c) Assertion is correct but reason is incorrect
- (d) Both assertion and reason are incorrect

Answer: B

Physics

Q983.

Assertion Total current entering a circuit is equal to leaving the circuit by Kirchhoff's law. Reason It is based on conservation of energy.

- (a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion.
- (c) Assertion is correct but Reason is incorrect.
- (d) Both Assertion and Reason are incorrect.

Answer: C

Chemistry

Q984.

The conductivity of an electrolytic solution decreases on dilution due to:

- (a) decrease in number of ions per unit volume
- (b) increase in ionic mobility of ions
- (c) increase in percentage ionisation
- (d) increase in number of ions per unit volume

Answer: A

Physics

Q985.

Which of the following statements is NOT true for friction force?

- (a) Friction force is produced when two objects rub against each other
- (b) A lighter object will experience a smaller frictional force
- (c) Friction produces more wear and tear
- (d) Friction is harmful

Answer: B

Chemistry

Q986.

Aluminothermy used for the spot welding of large iron structures is based upon the fact that: This question has multiple correct options

- (a) As compared to iron, aluminium has greater affinity for oxygen.
- (b) As compared to aluminium, iron has greater affinity for oxygen.
- (c) Reaction between aluminium and oxygen is endothermic.
- (d) Reaction between iron oxide and aluminium is exothermic.

Answer: A,D

Physics

Q987.

If there are N nuclear particles in a nucleus of radius R then the number of nuclear particles in radius $3R$ will be

- (a) N
- (b) $2N$
- (c) $27N$
- (d) $(27)^{1/3}N$

Answer: C

Q988.

An electron with energy 12.09 eV strikes hydrogen atom in ground state and gives its all energy to the hydrogen atom. Therefore hydrogen atom is excited to

- (a) Fourth
- (b) Third
- (c) second
- (d) First

Answer: B

Chemistry

Q989.

Ether is formed when alkyl halide is treated with sodium alkoxide. This method is known as:

- (a) Hoffmann reaction
- (b) Williamson synthesis
- (c) Wurtz synthesis
- (d) Kolbe's reaction

Answer: B

Physics

Q990.

A particle is in SHM. Then the graph of its acceleration as a function of displacement is a

- (a) circle
- (b) hyperbola
- (c) straight line with negative slope
- (d) straight line with positive slope

Answer: C

Q991.

There are two charges $+2\text{ }\mu\text{C}$ and $-3\text{ }\mu\text{C}$. The ratio of forces acting on them will be

- (a) 2: 3
- (b) 1: 1
- (c) 3: 2
- (d) 4: 9

Answer: B

Q992.

A wire suspended vertically from one of its ends is stretched by attaching a weight of 200 N to the lower end. The weight stretches the wire by 1 mm. Then the energy stored in the wire is

- (a) 0.1
- (b) 0.2
- (c) 10
- (d) 20

Answer: A

Q993.

A body of mass $1/4$ kg is in S.H.M and its displacement is given by the relation $y = 0.05 \sin(20t + \pi/2)$ m. If t is in seconds, the maximum force acting on the particle is:

- (a) 5 N
- (b) 2.5 N
- (c) 10 N
- (d) 0.25 N

Answer: A

Biology

Q994.

Which one of the following is the correct description of a certain part of a normal human skeleton?

- (a) Parietal bone and the temporal bone and the skull are joined by fibrous joint
- (b) First vertebrae is axis, which articulates with the occipital condyles.
- (c) The 11th and 10th pairs of ribs are called the floating ribs
- (d) Glenoid cavity is a depression to which the thigh bone articulates

Answer: A

Q995.

Which of the following state of an ecosystem helps it to avoid changes?

- (a) Homeostasis
- (b) Shortage of components
- (c) Deficiency of light
- (d) Imbalance

Answer: A

Physics

Q996.

A candle is placed in front of an optical instrument, and an image is obtained as inverted, real, and on the same side of the device. Identify the device that would produce the types of images described above.

- (a) Concave mirror
- (b) Convex mirror
- (c) concave lens
- (d) Convex lens E. Flat mirror

Answer: A

Biology

Q997.

Green house effect is due to

- (a) Accumulation of O_3 and depletion of O_2
- (b) Accumulation of O_3 and CO_2
- (c) Depletion of CO_2 and O_3
- (d) Accumulation of green plants

Answer: B

Chemistry

Q998.

The force which is responsible for circular motion of electrons around the nucleus is

- (a) Van der Waals
- (b) Electrostatic forces
- (c) centrifugal and Centripetal
- (d) Intermolecular forces

Answer: B

Q999.

Which of the following mixtures is used in the preparation of tips of match sticks? A . S+K

- (b) $Sb_2S_3 + S$
- (c) $K_2Cr_2O_7 + S + P$
- (d) P+K

Answer: B

Physics

Q1000.

Two identical rods each of moment of inertia I about a normal axis through centre are arranged in the form of a cross. The M . I . of the system about an axis through centre and perpendicular to the plane of system is:

- (a) $2I$
- (b) I
- (c) $2.5I$
- (d) $6I$

Answer: A